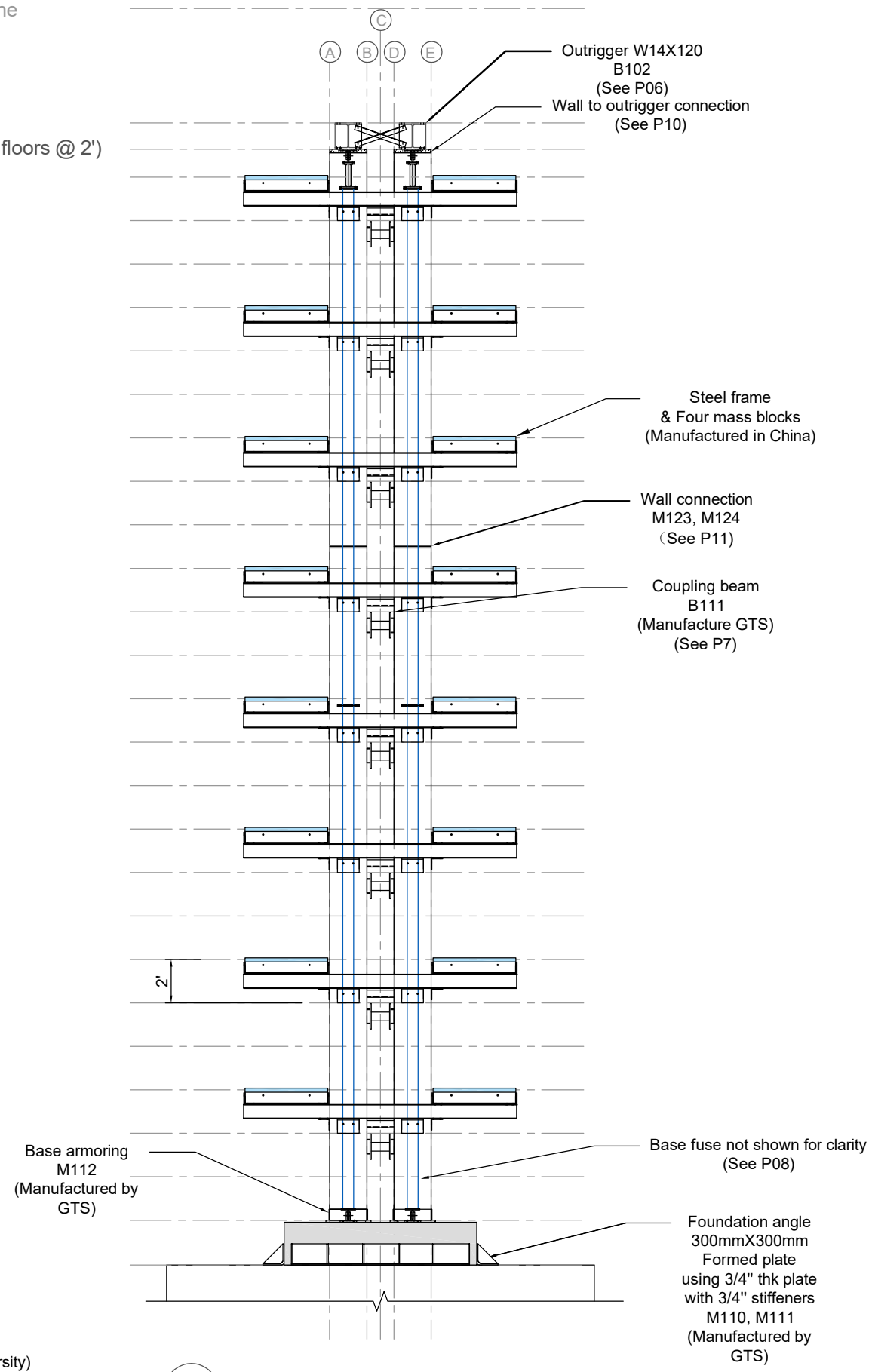
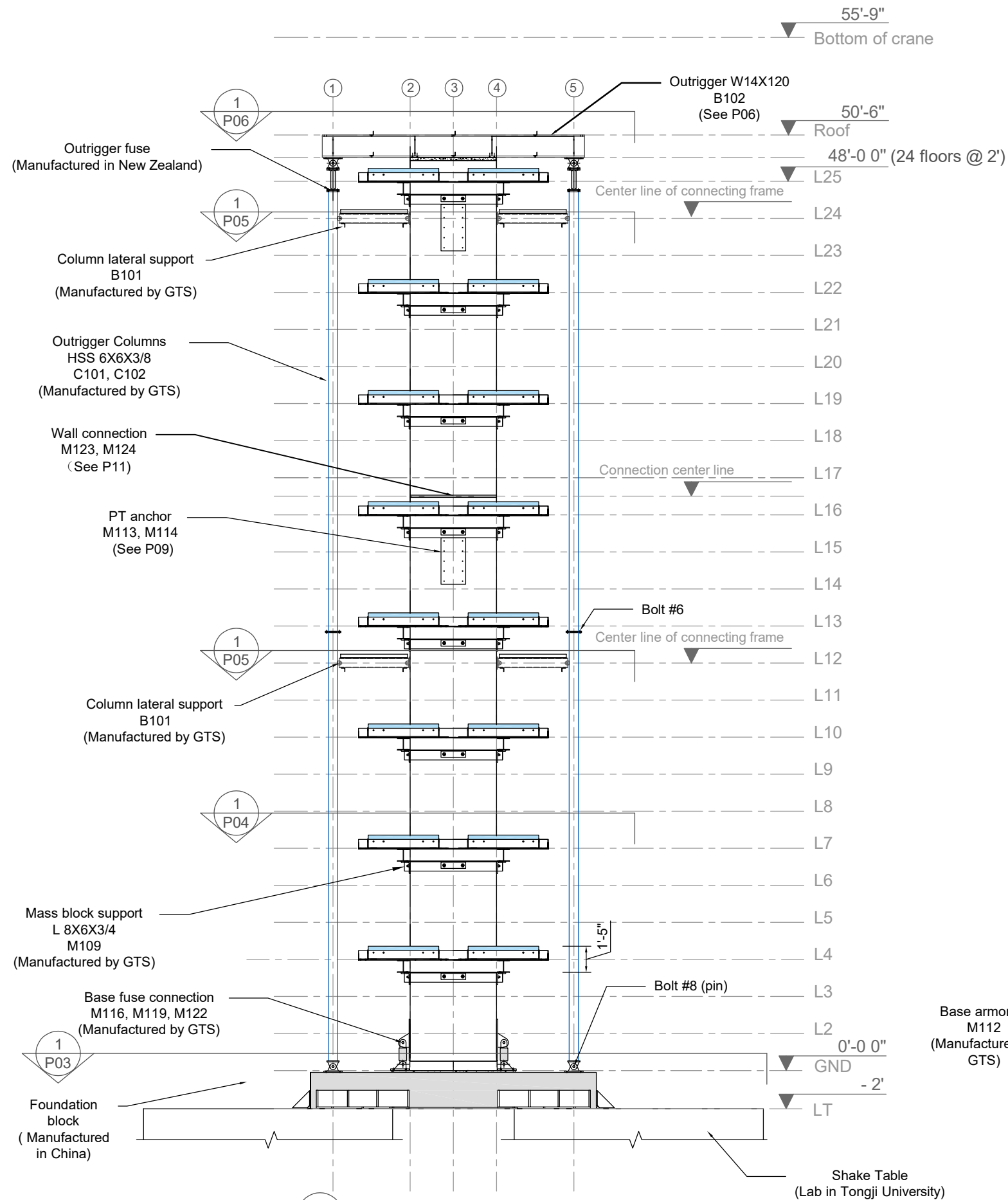
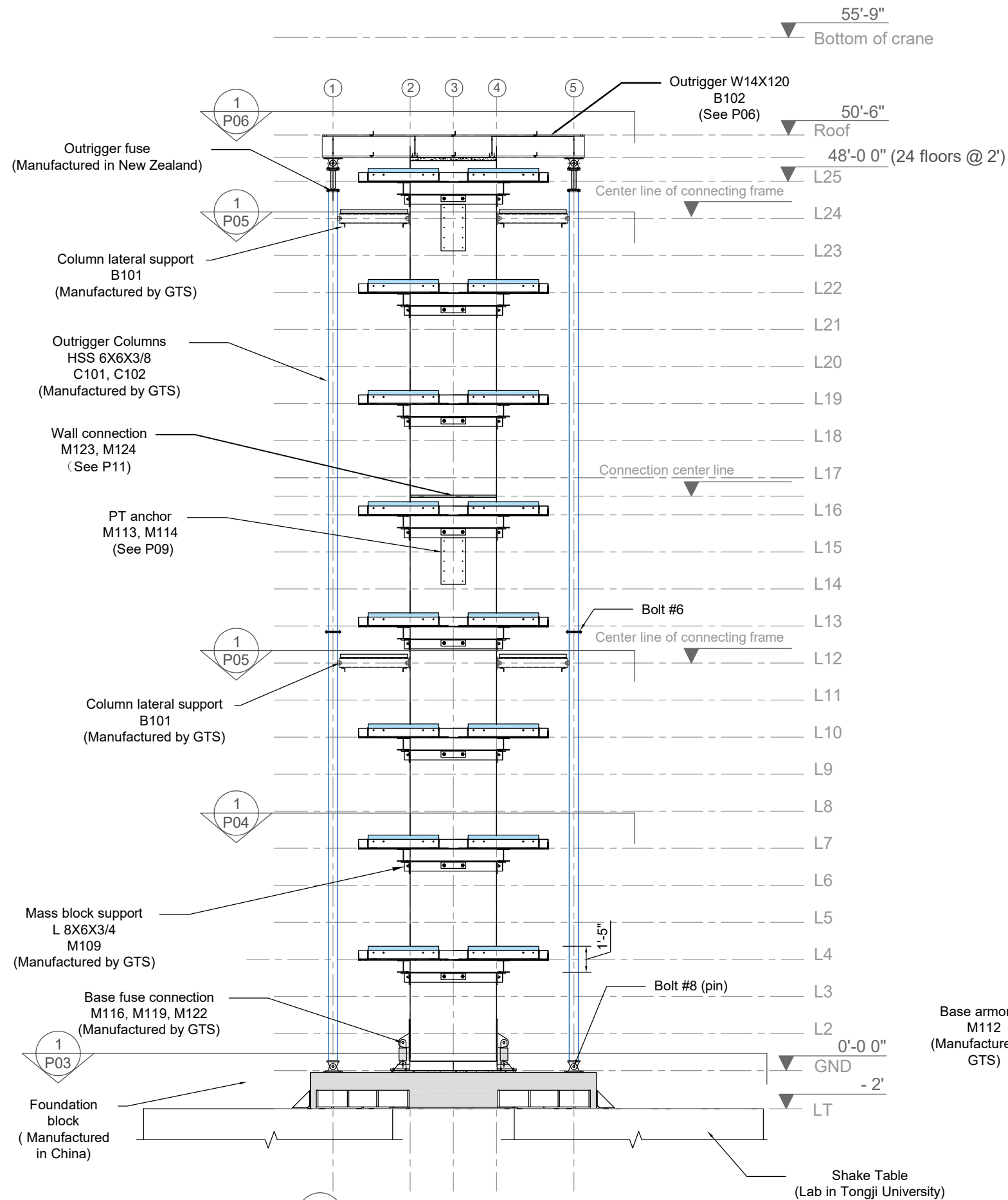
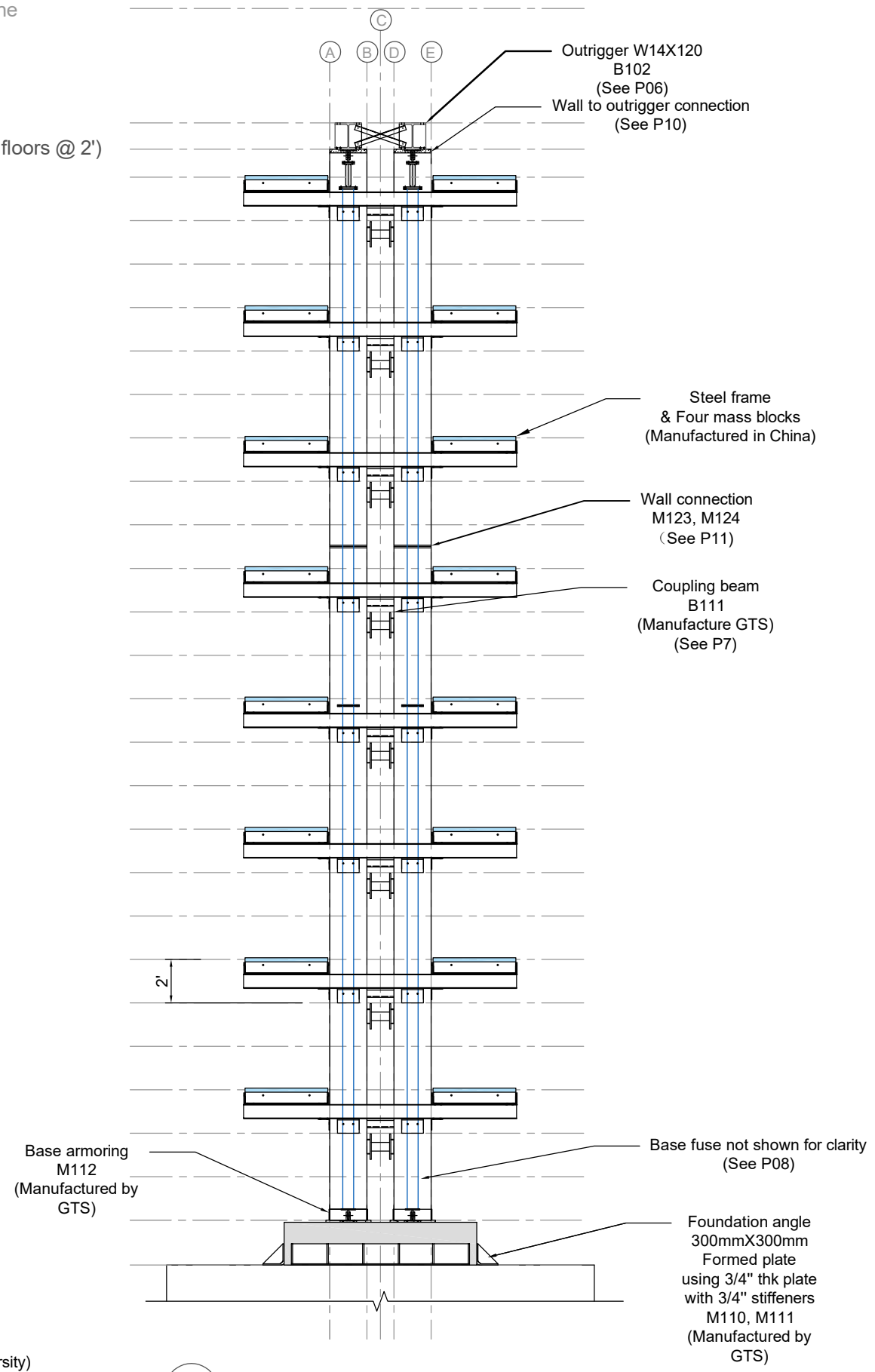


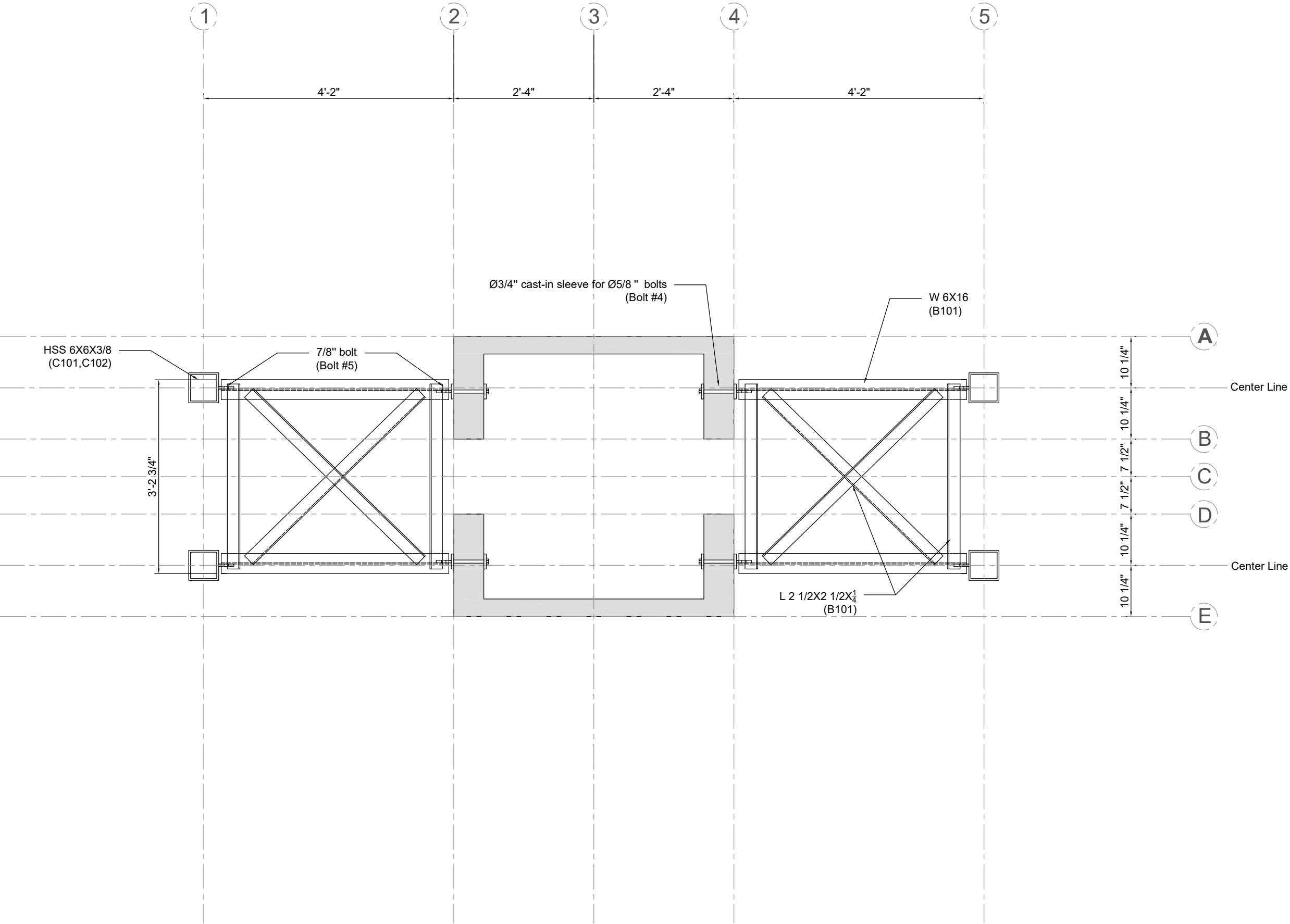




1 P02 **Test set-up: front view**



2 P02 **Test set-up: side view**



Note for elevation of lateral support:
240" (L11) & 552" (L24)



ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

PlanView
(Lateral Support Level)

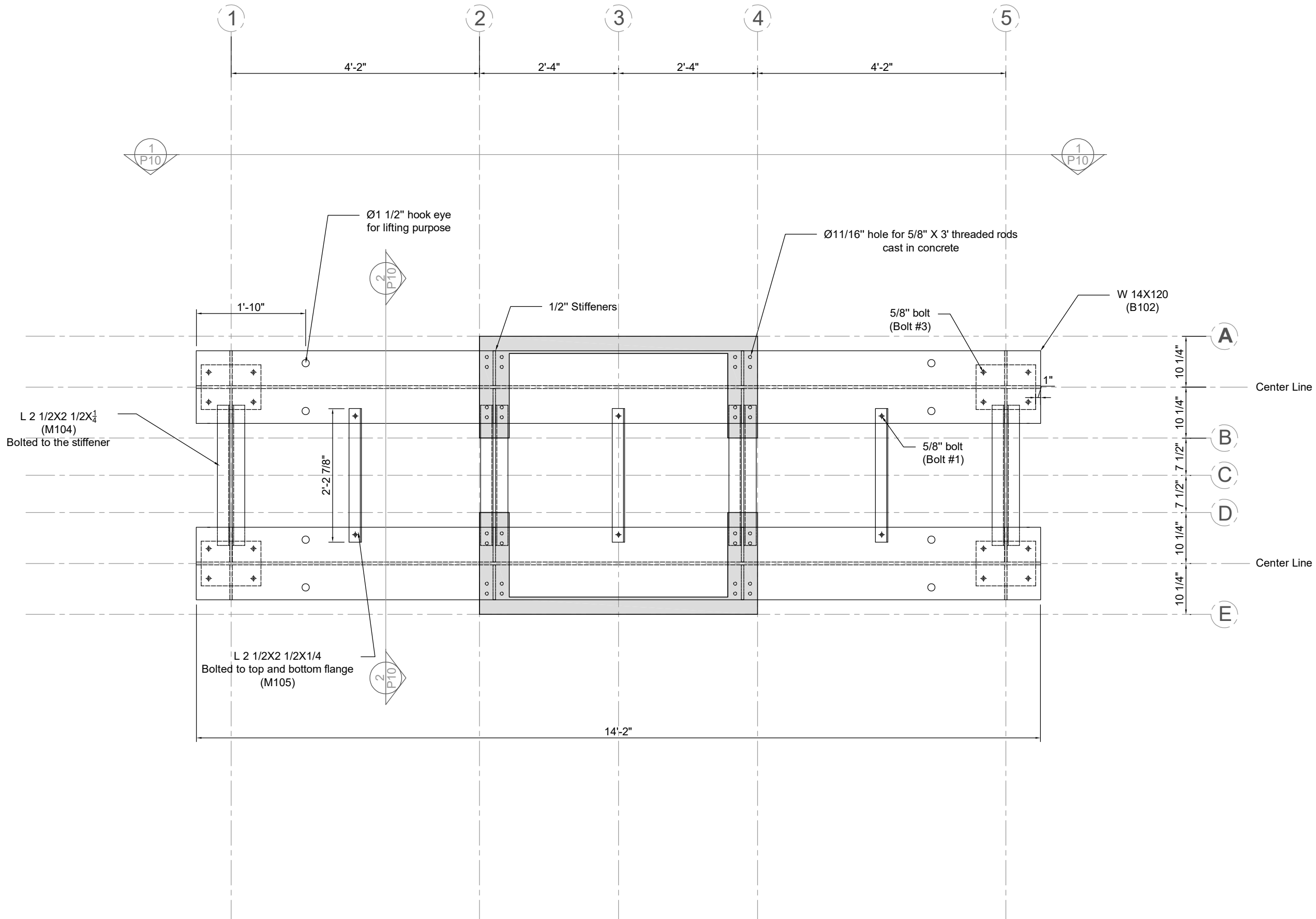
SHEET NO.

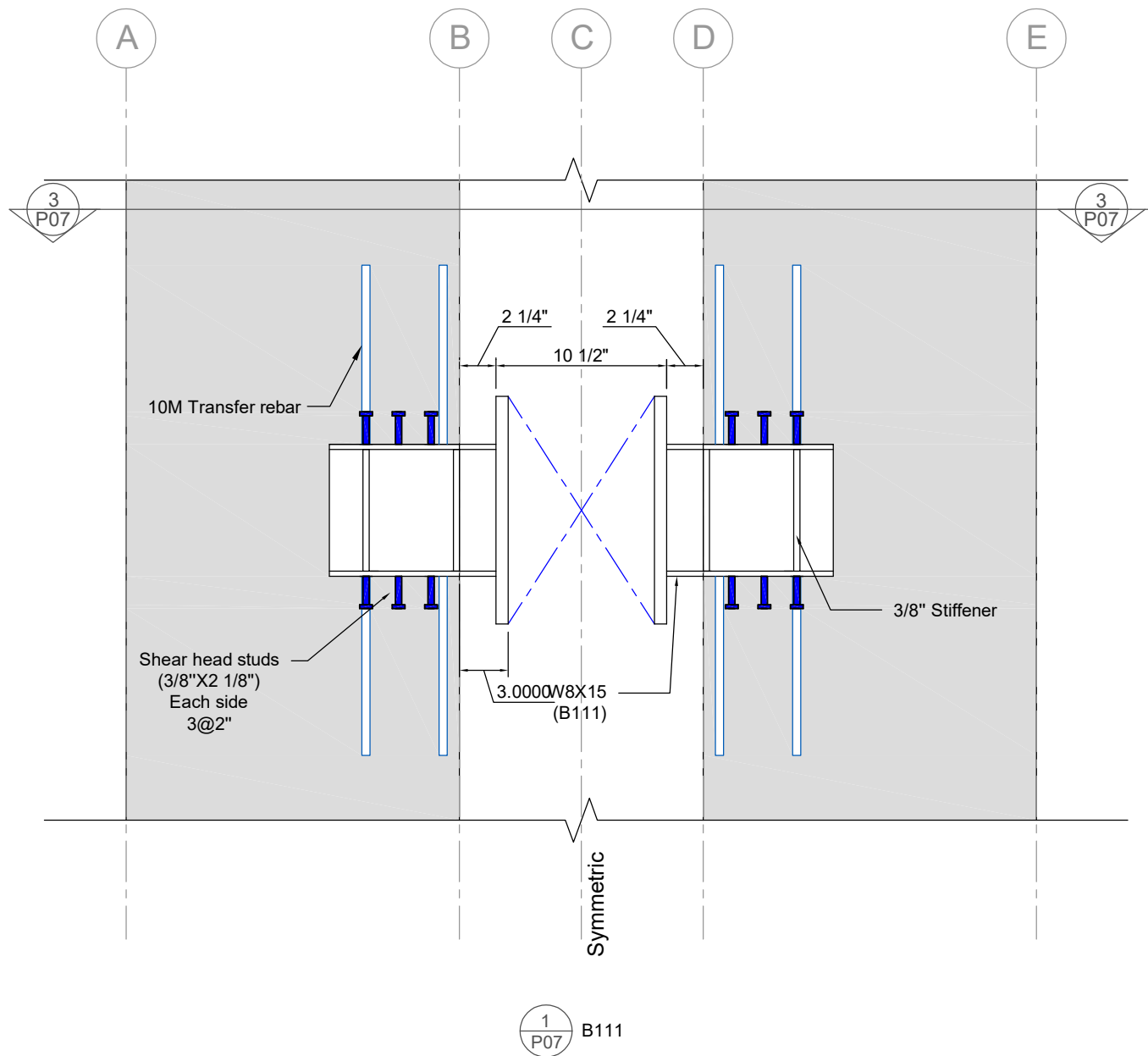
P05

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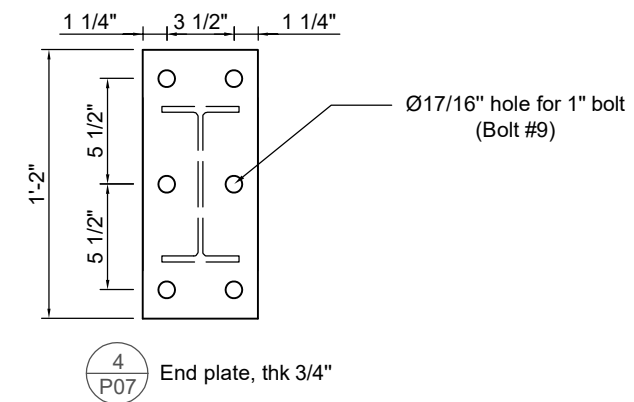
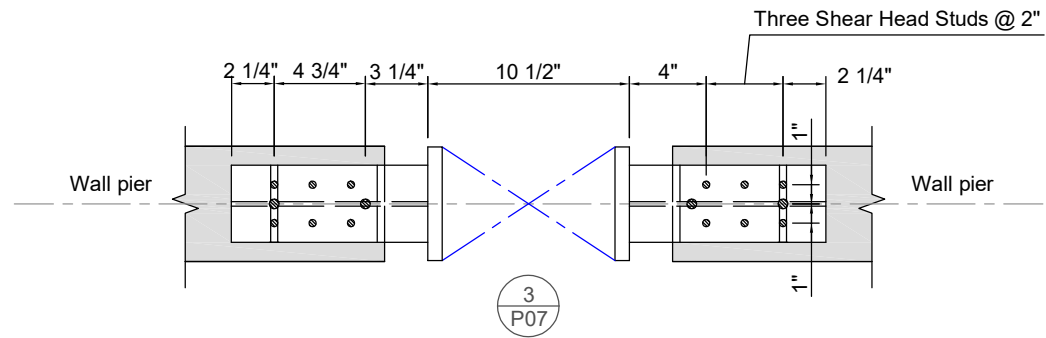
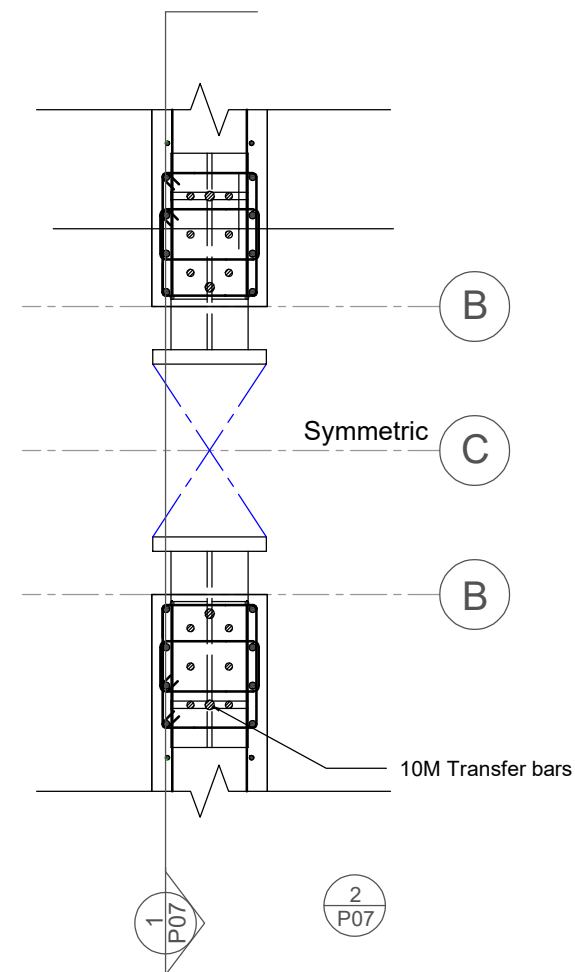
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2020/02/17

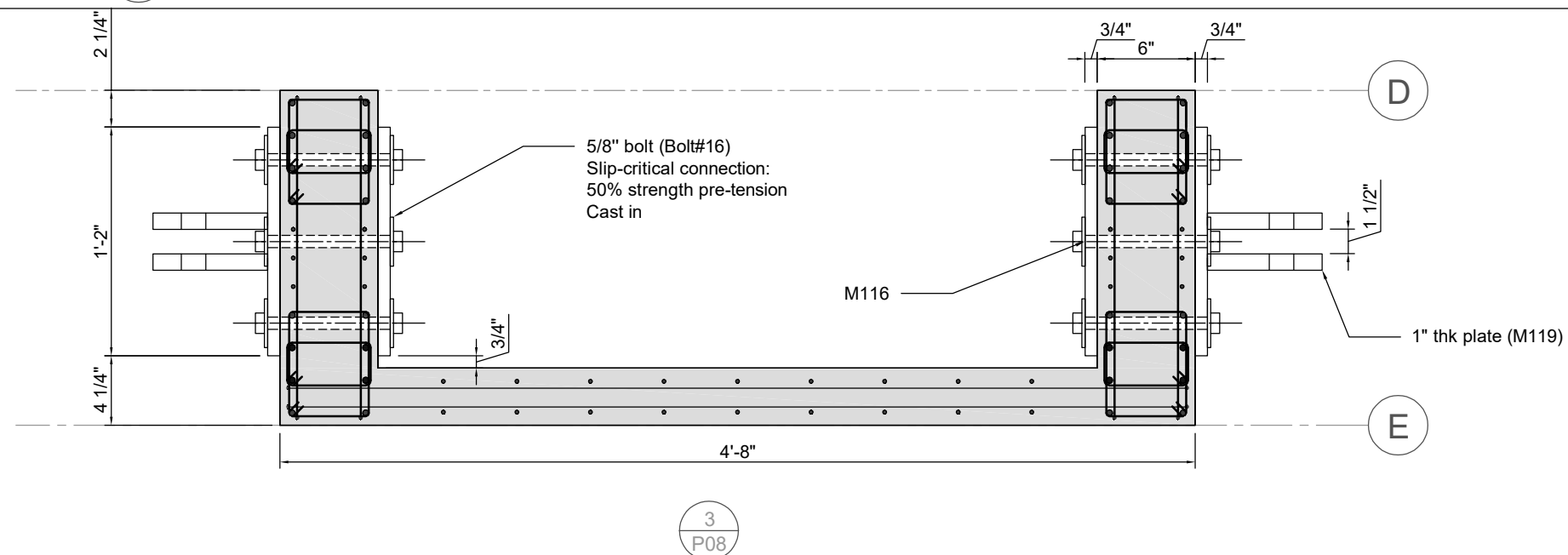
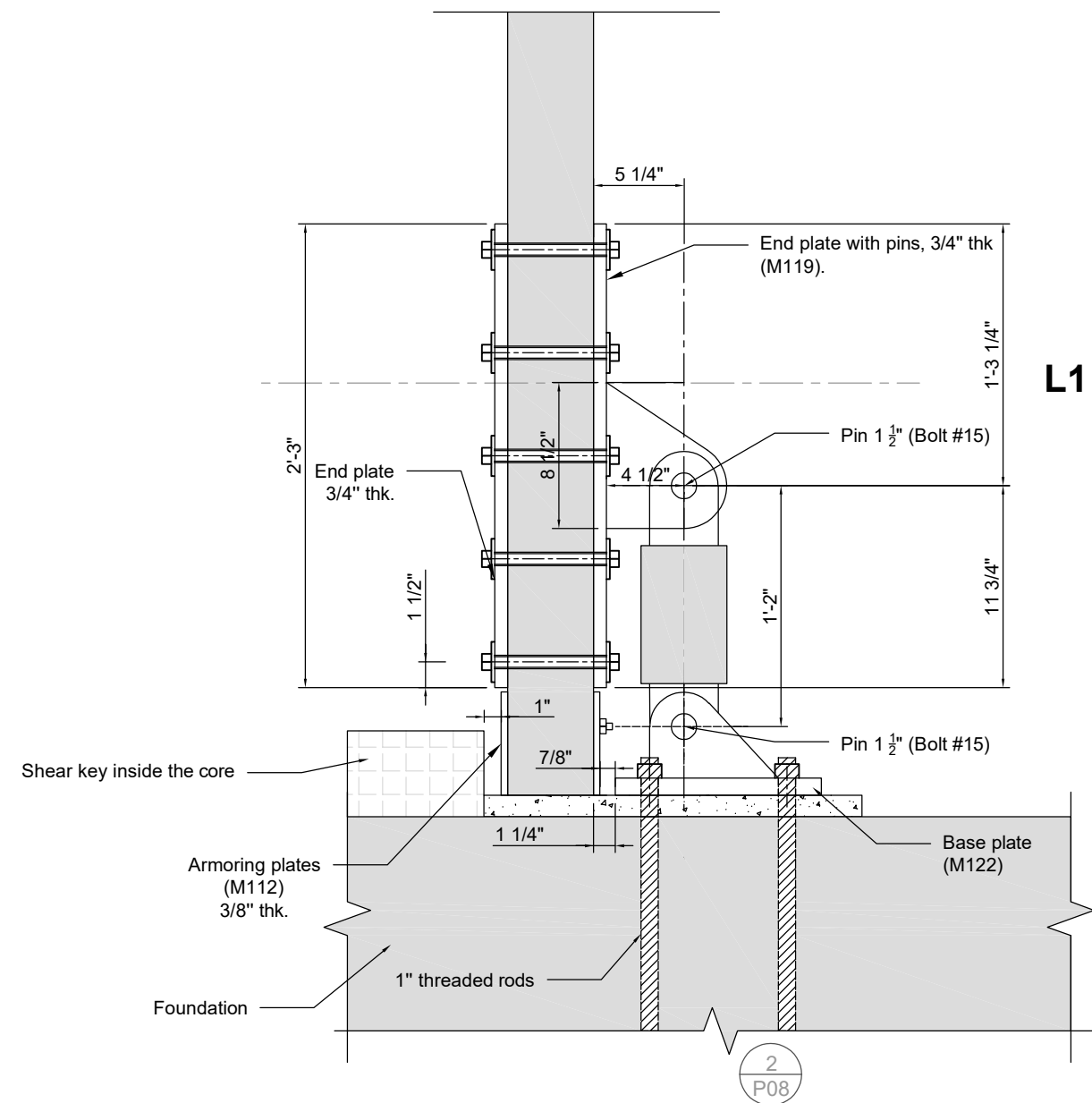
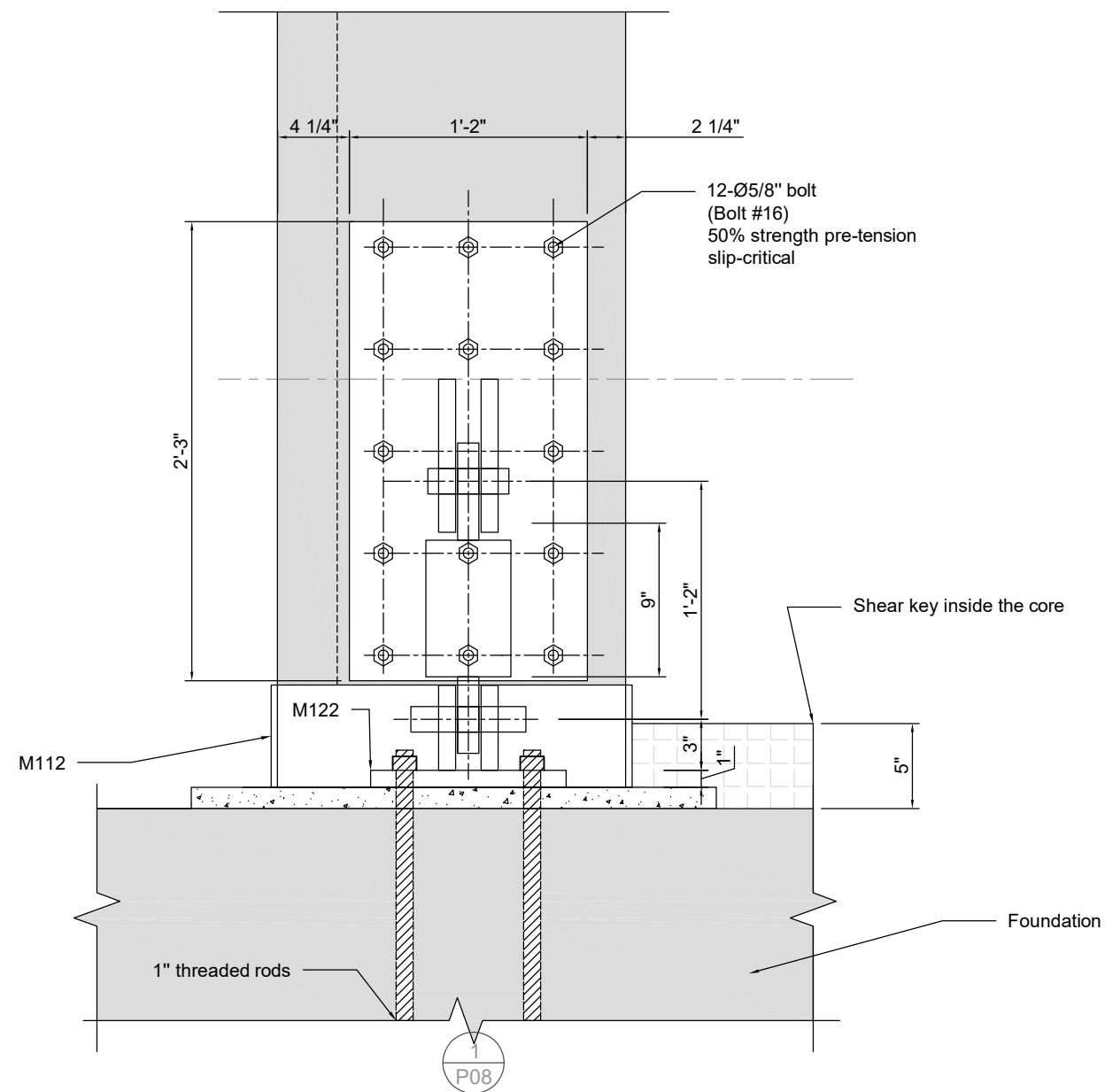


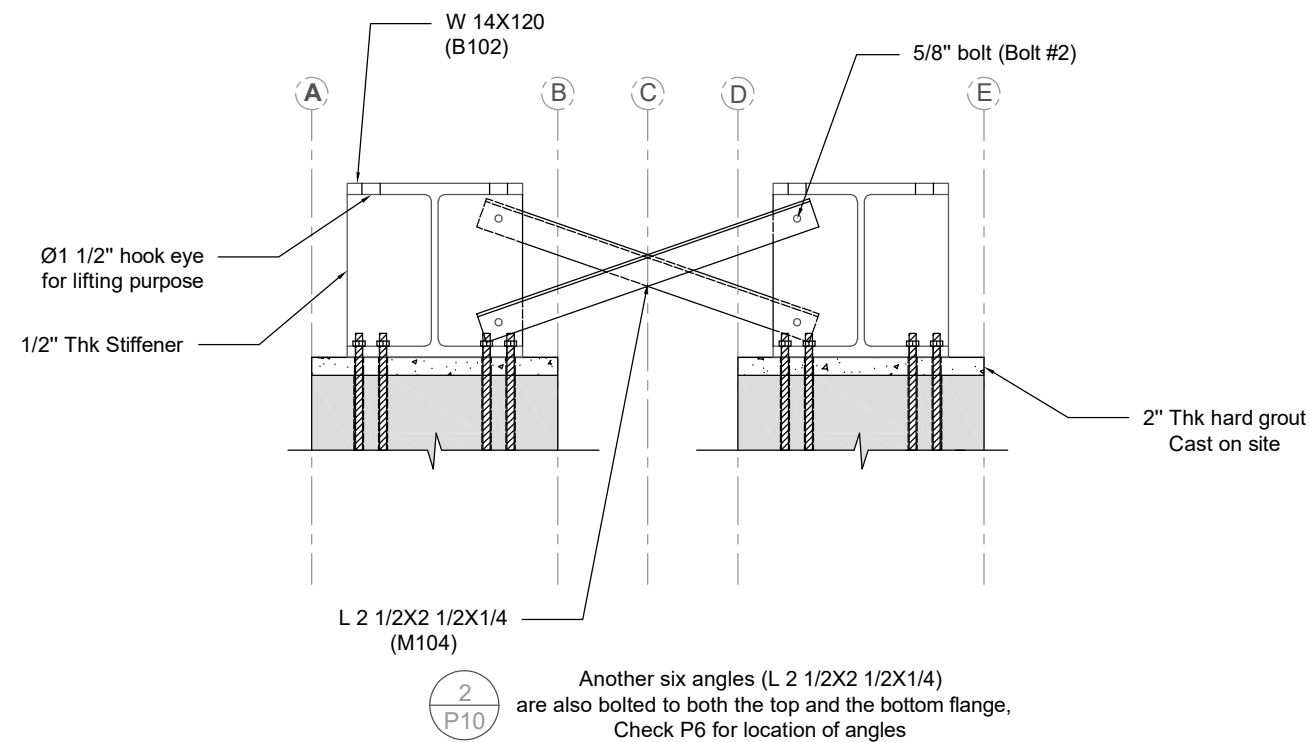
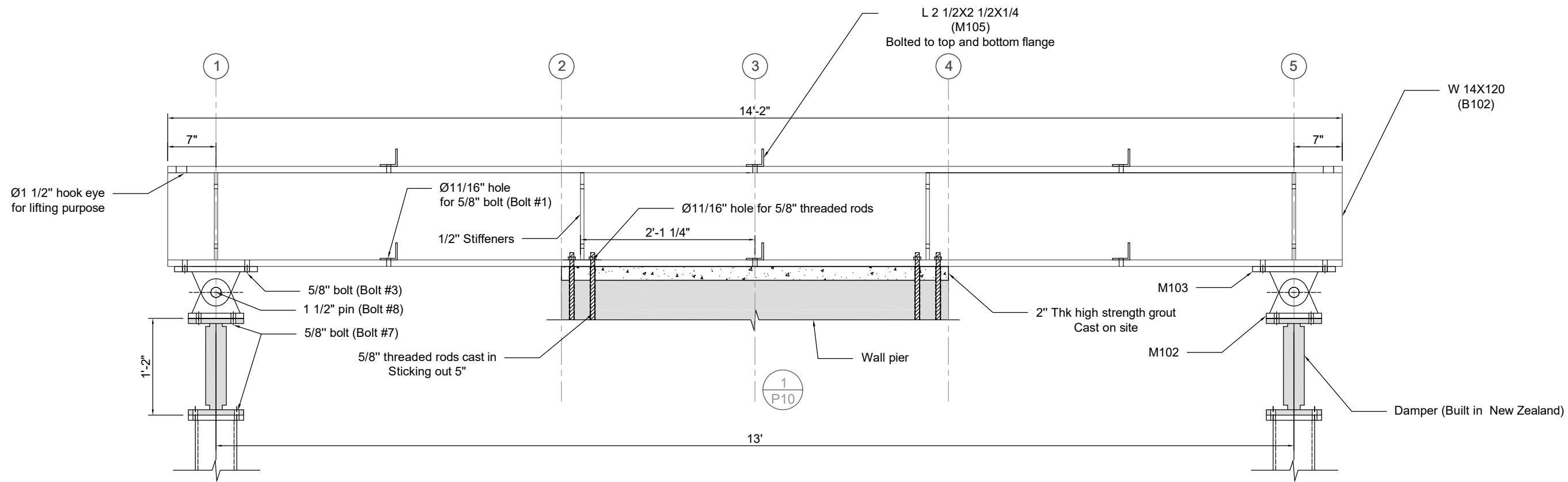


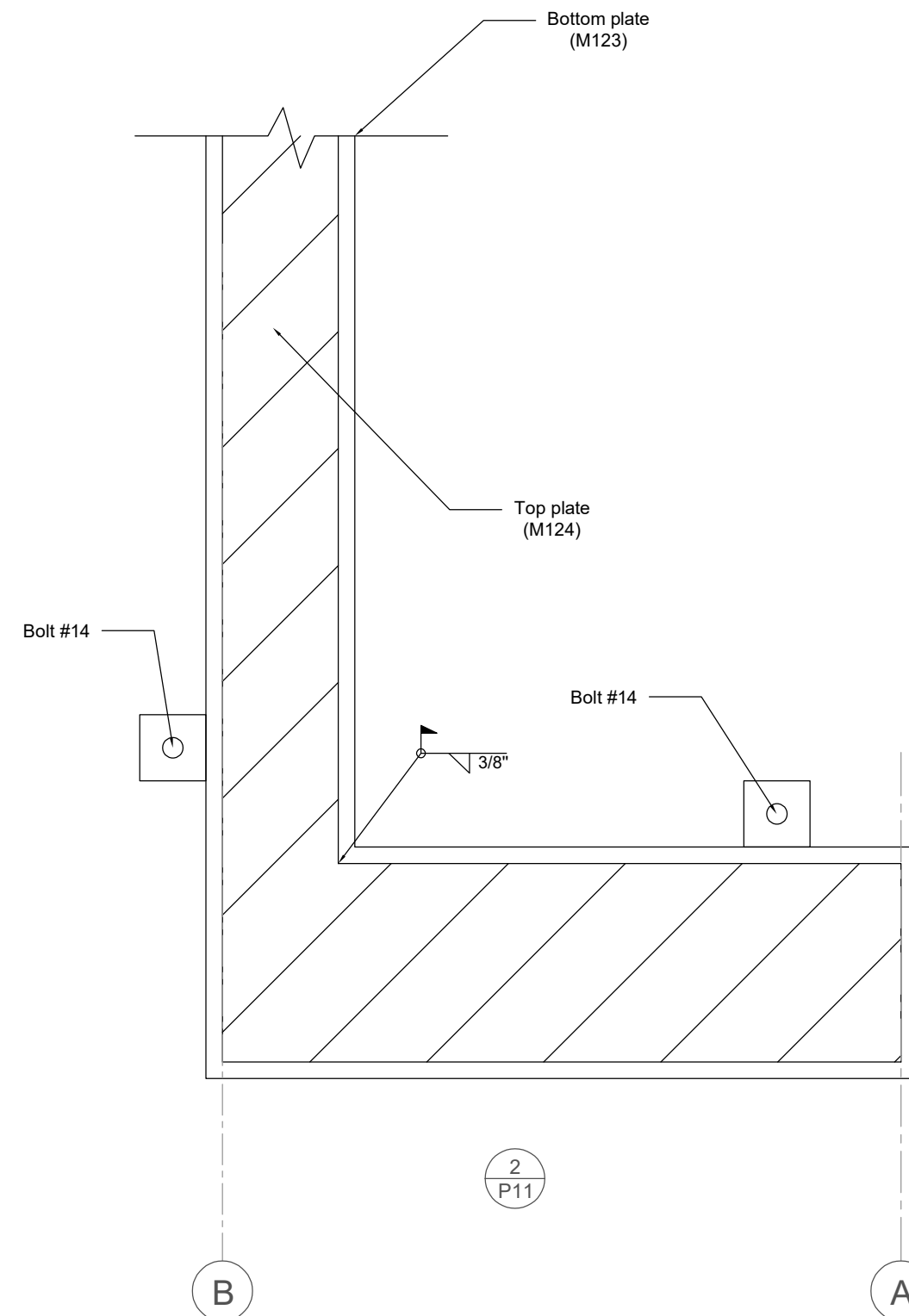
1 P07 B111

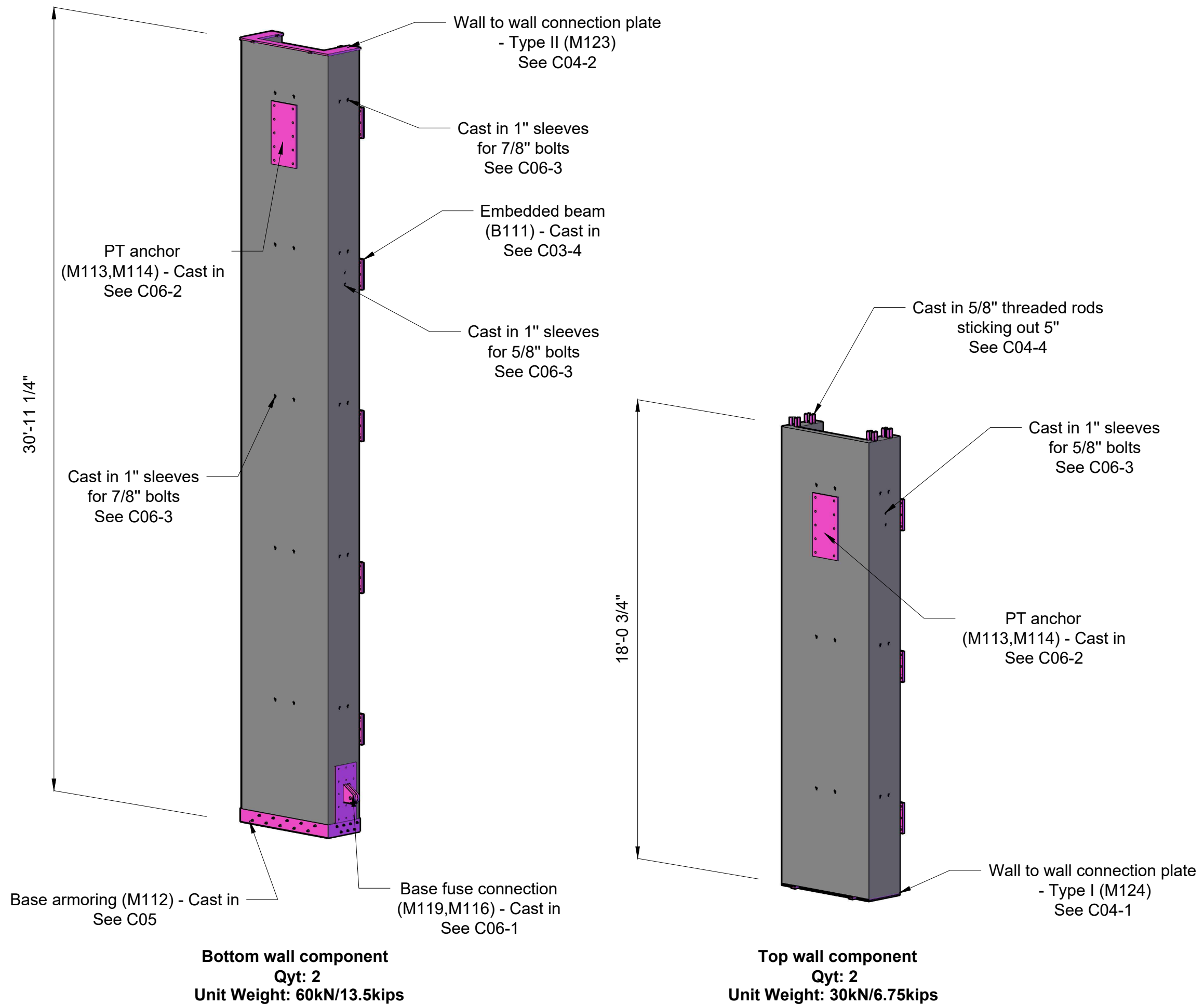


4 P07 End plate, thk 3/4"

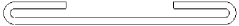
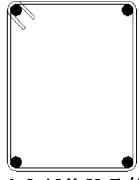
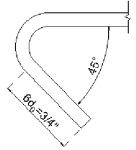
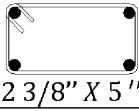








Rebar Information

Name	Shape	Size	Anchorage Type	Unit Length	Qty	Supplier
Vertical zone reinforcement -Top wall		#3 3/8"	/	18'	64	LMS (UBC bends all the ties, LMS bends the horizontal rebar)
Vertical zone reinforcement -Bottom wall				31'	64	
Vertical distributed reinforcement - Top wall				18'	48	
Vertical distributed reinforcement - Bottom wall				31'	48	
Horizontal distributed reinforcement at flanges		#3 3/8"		2' - 1"	784	
Horizontal distributed reinforcement at web				5' - 1"	392	
Ties in zone (undeformed wire) Type I	 6 3/8" X 5"			2'	1622	
Ties in zone (undeformed wire) Type II	 2 3/8" X 5"			1' - 4"	1622	
Additional wires at flanges	 4 7/8"			7 1/2"	300	
Additional wires at webs	 1 3/8"			5"	430	
Threaded Dywidag bar		Ø5/8"	/	2' - 3"	16	Dywidag
Material Summary						
Type	Total length	Volume [inch3]		Weight [kg]		
#3 deformed rebar	9114 ft	12073		1543		
11ga undeformed wire	5773 ft	783		100		
Threaded Dywidag bar	36 ft	132		17		

Sleeve Information

Inner Diameter	Unit Length	Qty	Total length
Ø1"	6"	64	32 ft
Ø3/4"	6"	16	8 ft
Ø1"	3 1/2"	32	9.3 ft

Material Properties

Unless otherwise specified in the drawings, materials for all components should be:

Rebar and Wire:
#3 rebar : ASTM 706 Grade 60

11ga wire: Grade ASTM A1064

Concrete:
f 'c = 45 Mpa, from 28 day concrete cylinder tests

Clear cover is 5/16"

Max. aggregate size is 1/4"

Cast in Steel Components

M112 - Base armoring
M113, M114 - PT anchor
M116, M119 - Base fuse connection
M123 - Wall to wall connection plate Type II
M124 - Wall to wall connection plate Type I
B111 - Embedded beam

Volume of Wall Piers

7.704 m^3 (Four pieces together)

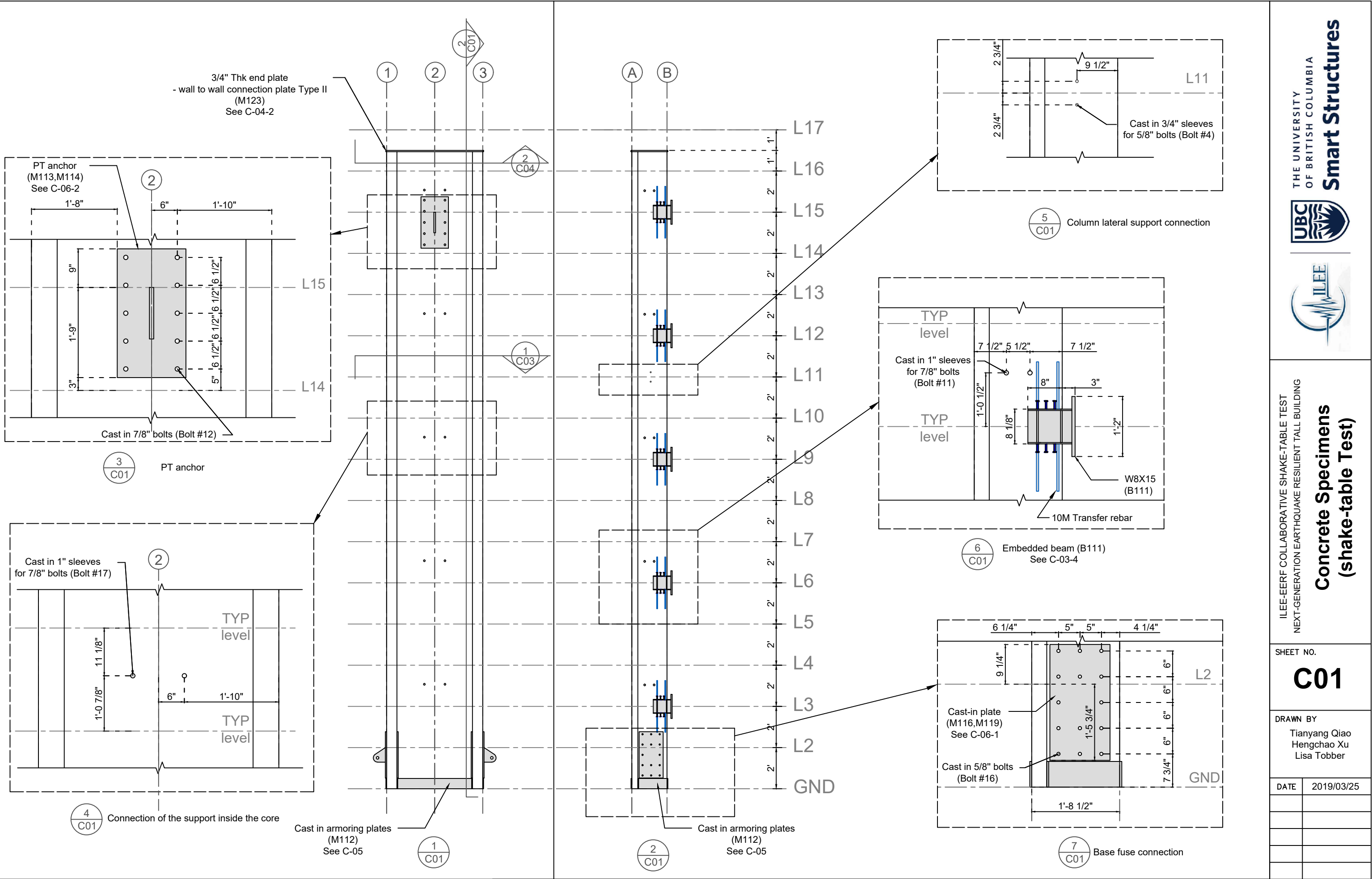


SHEET NO.

Info

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DATE 2019/03/25



ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

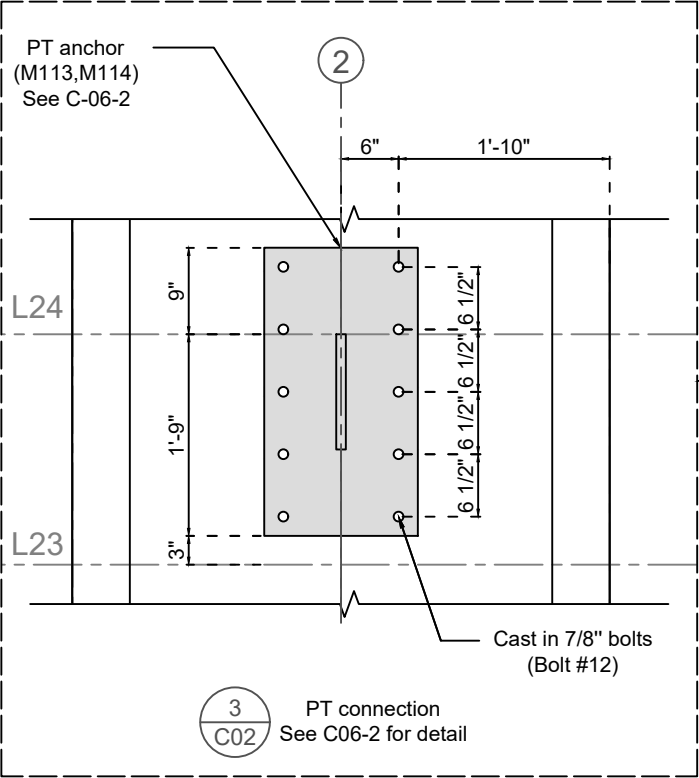
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C01

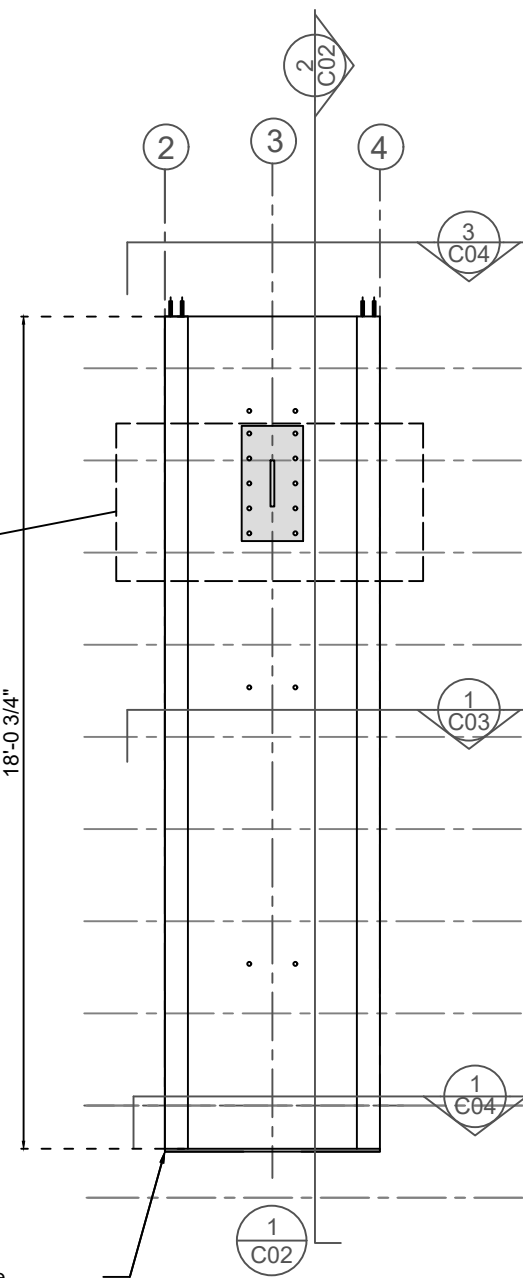
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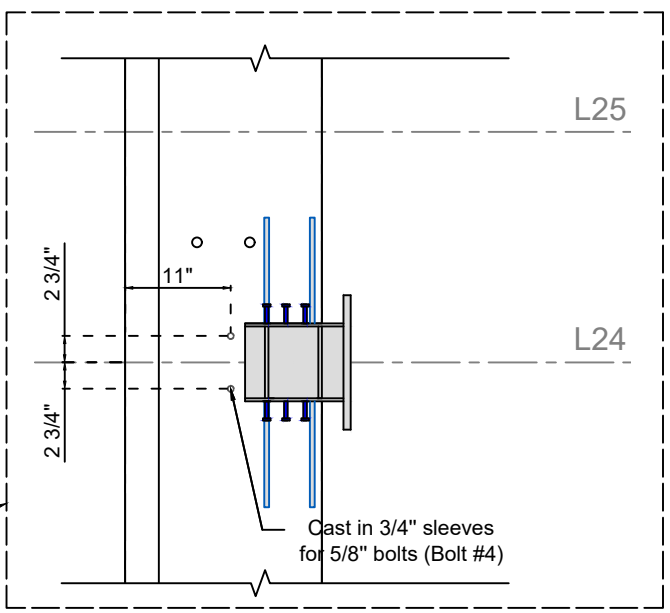
DATE 2019/03/25



3/4" Thk end plate
- wall to wall connection plate Type I
(M124)
See C-04-1



Cast in Ø 5/8" threaded rod
See C-04-4



4
C02

Column lateral support connection

ILee-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Concrete Specimens
(shake-table Test)

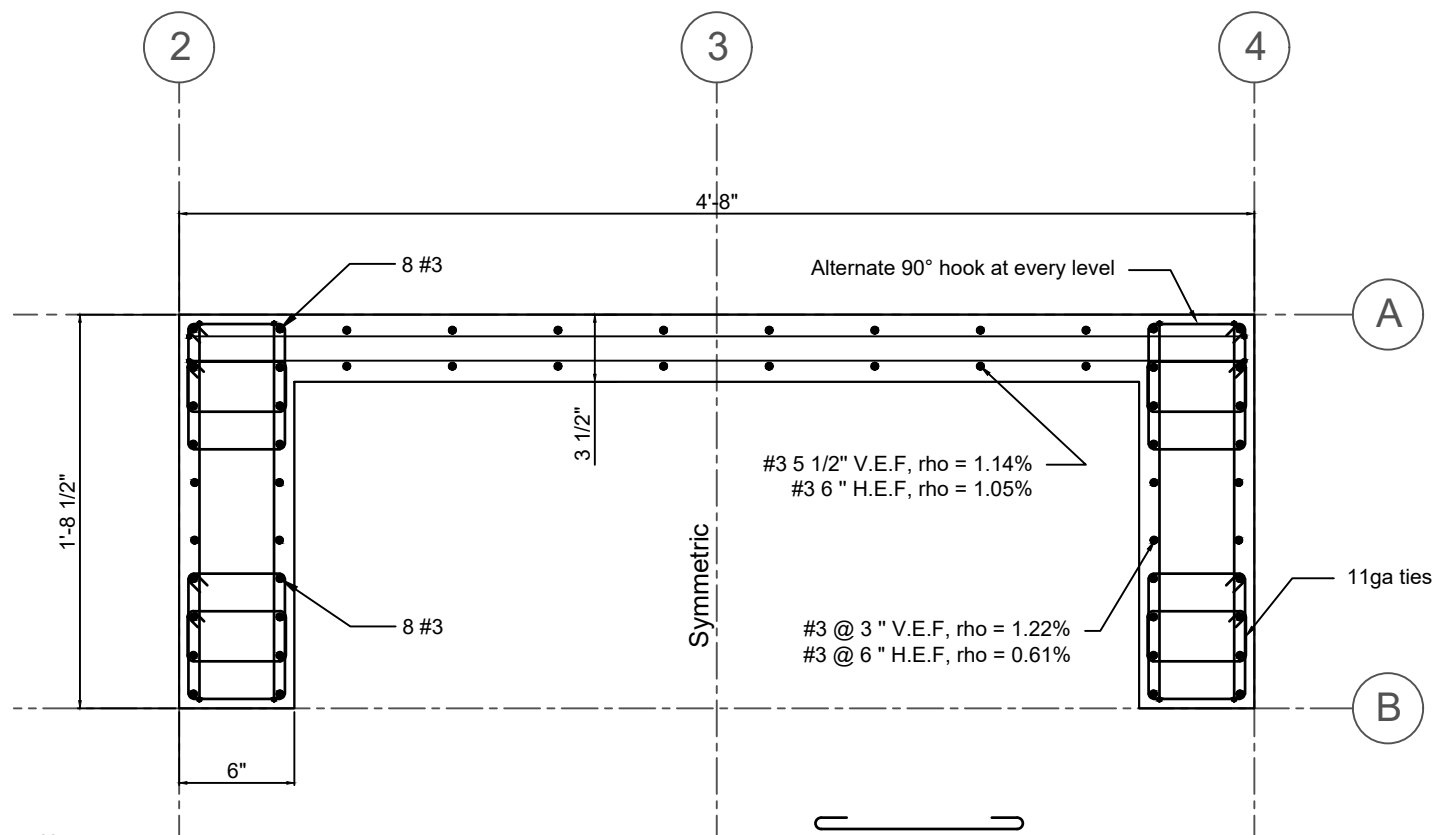
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C02

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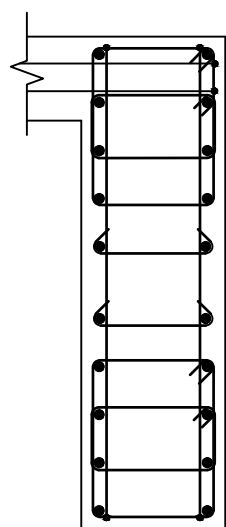
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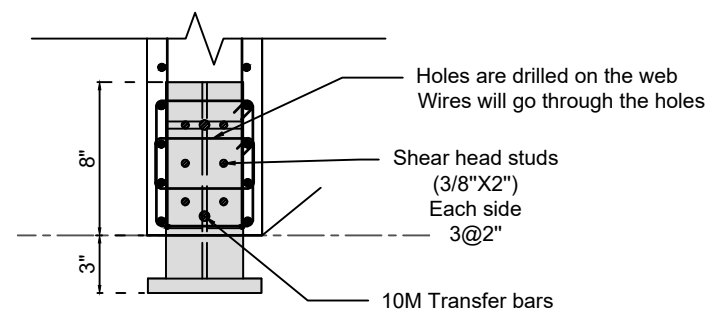


Note:
#3 : dia = 3/8"
11ga : dia = 0.12"
Minimum clear thickness 5/16"

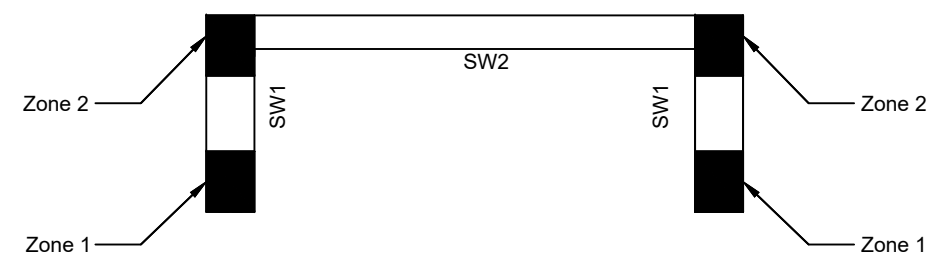
Hook horizontal bars 180° both ends
Hook tight to end of zone at end of wall
(Alternate 90° hook if necessary)



Additional hoops @ 2 1/4"
between level GND and L2



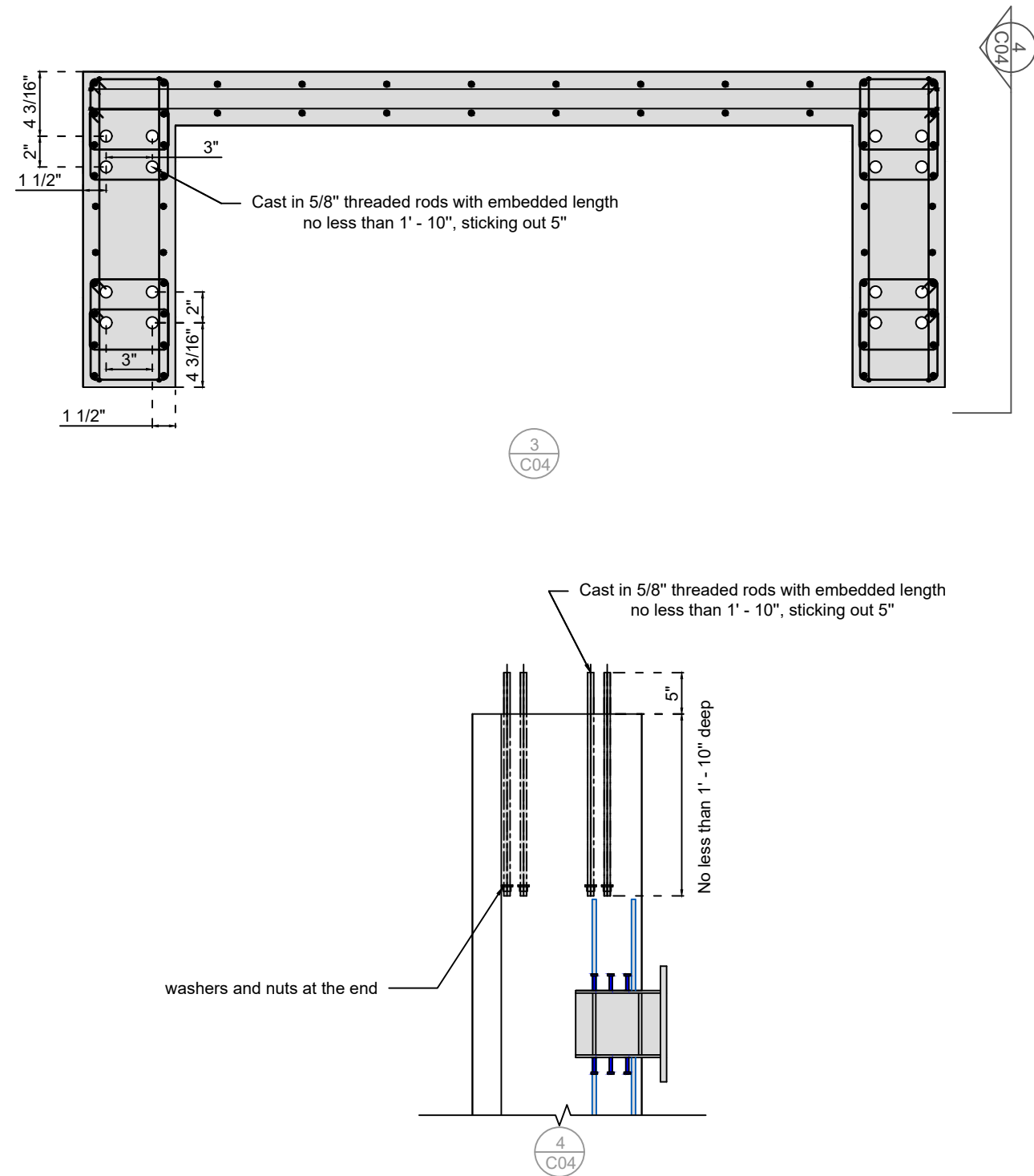
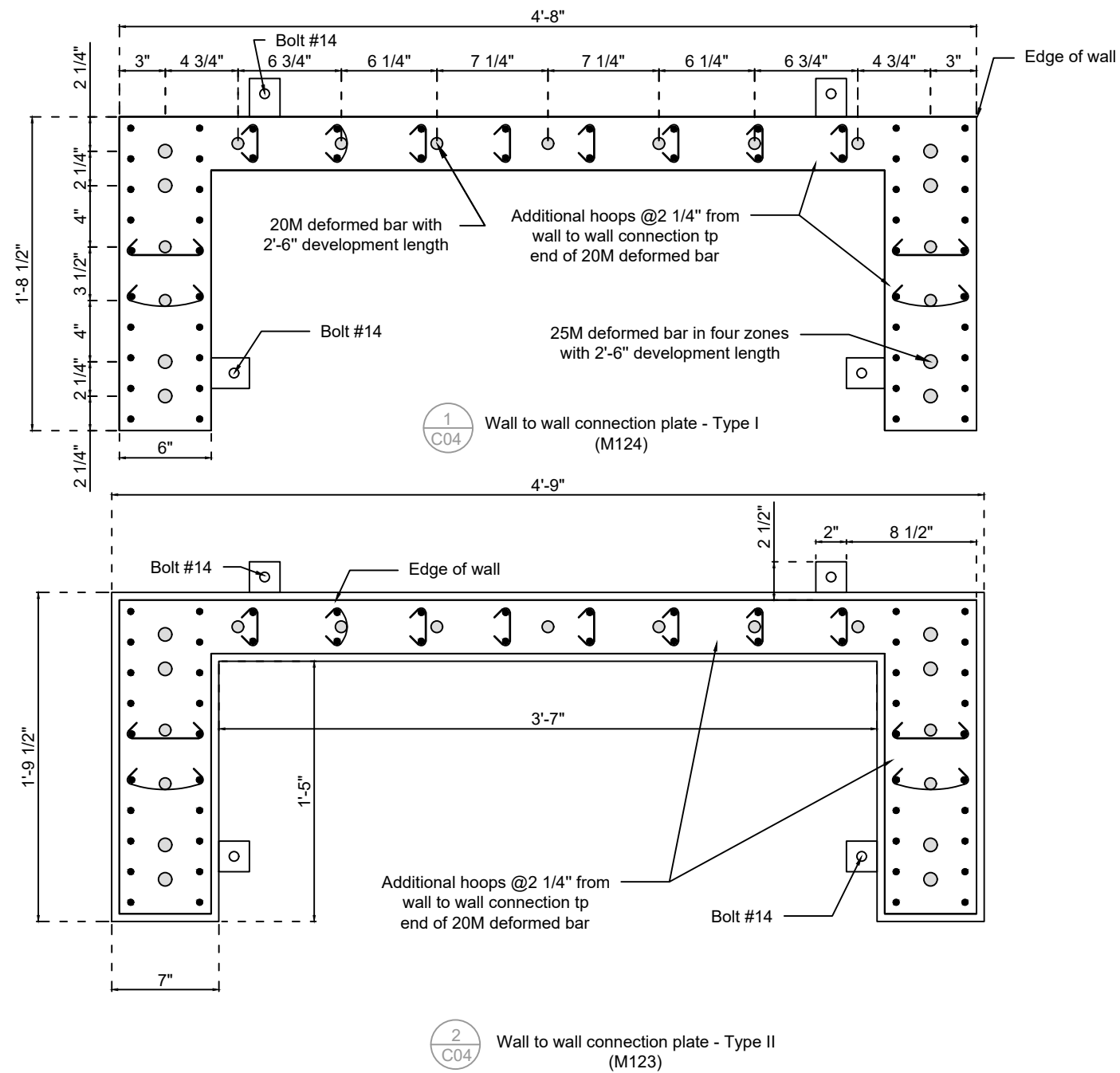
Special case with embed beam
(B111)

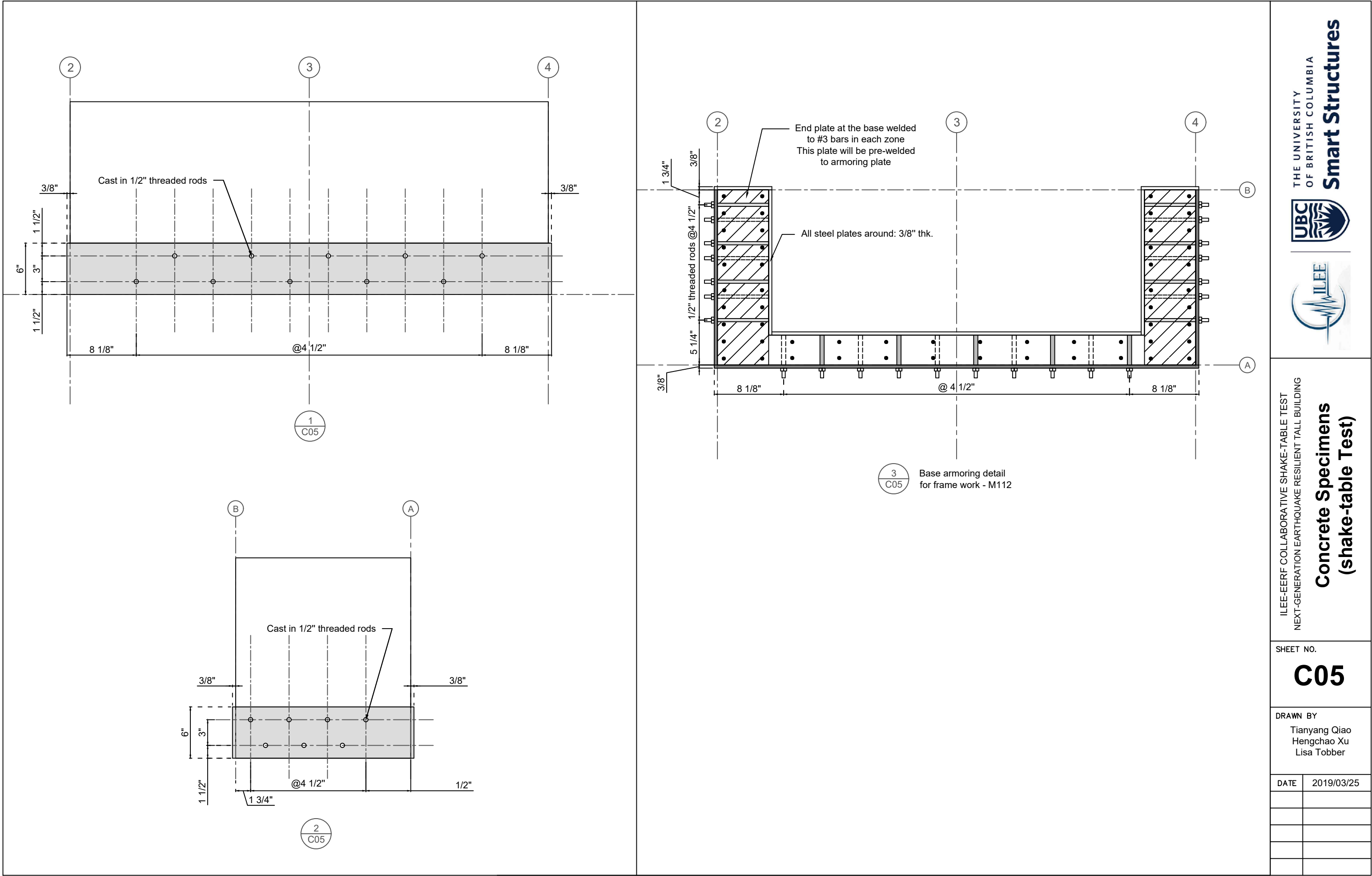


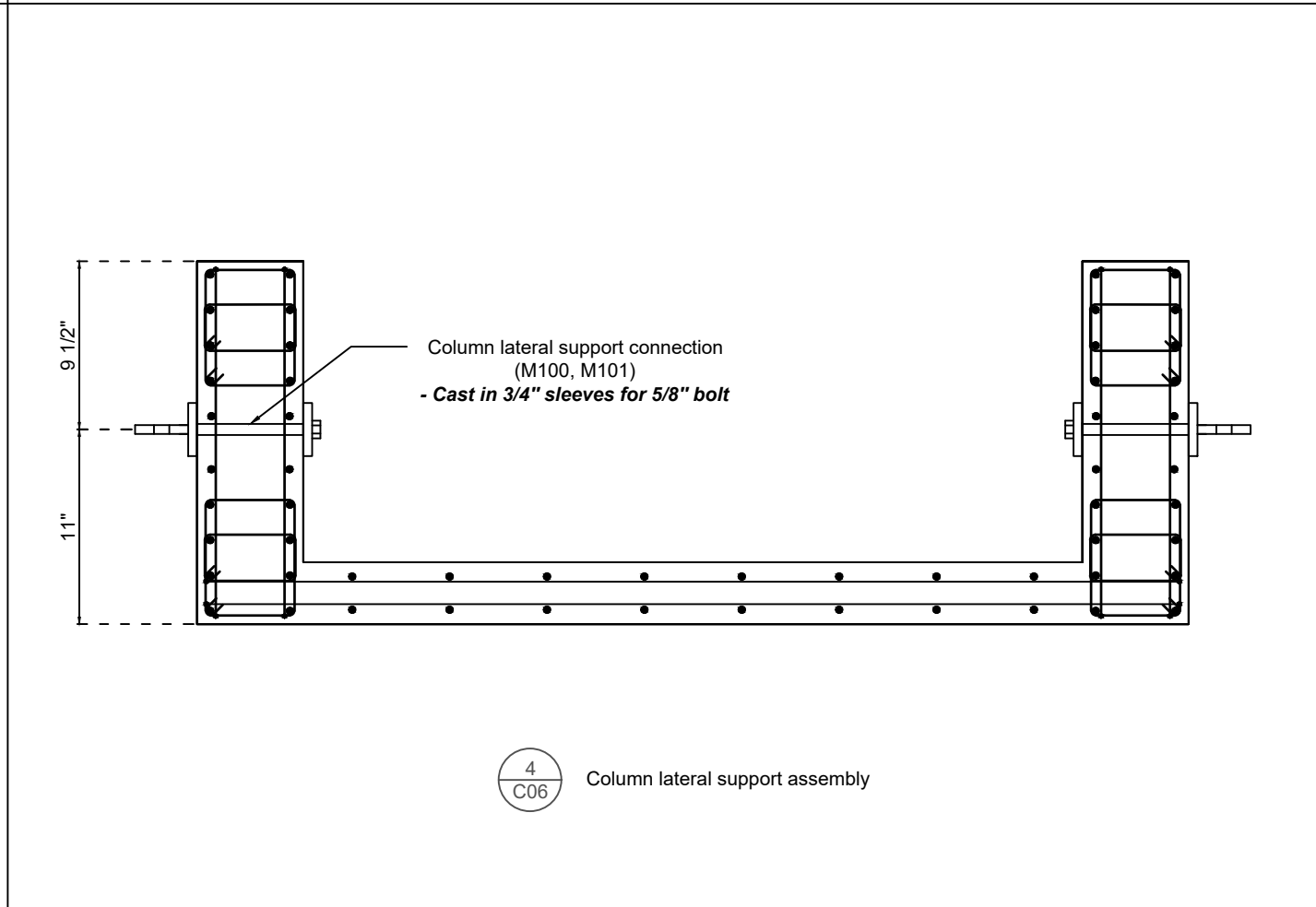
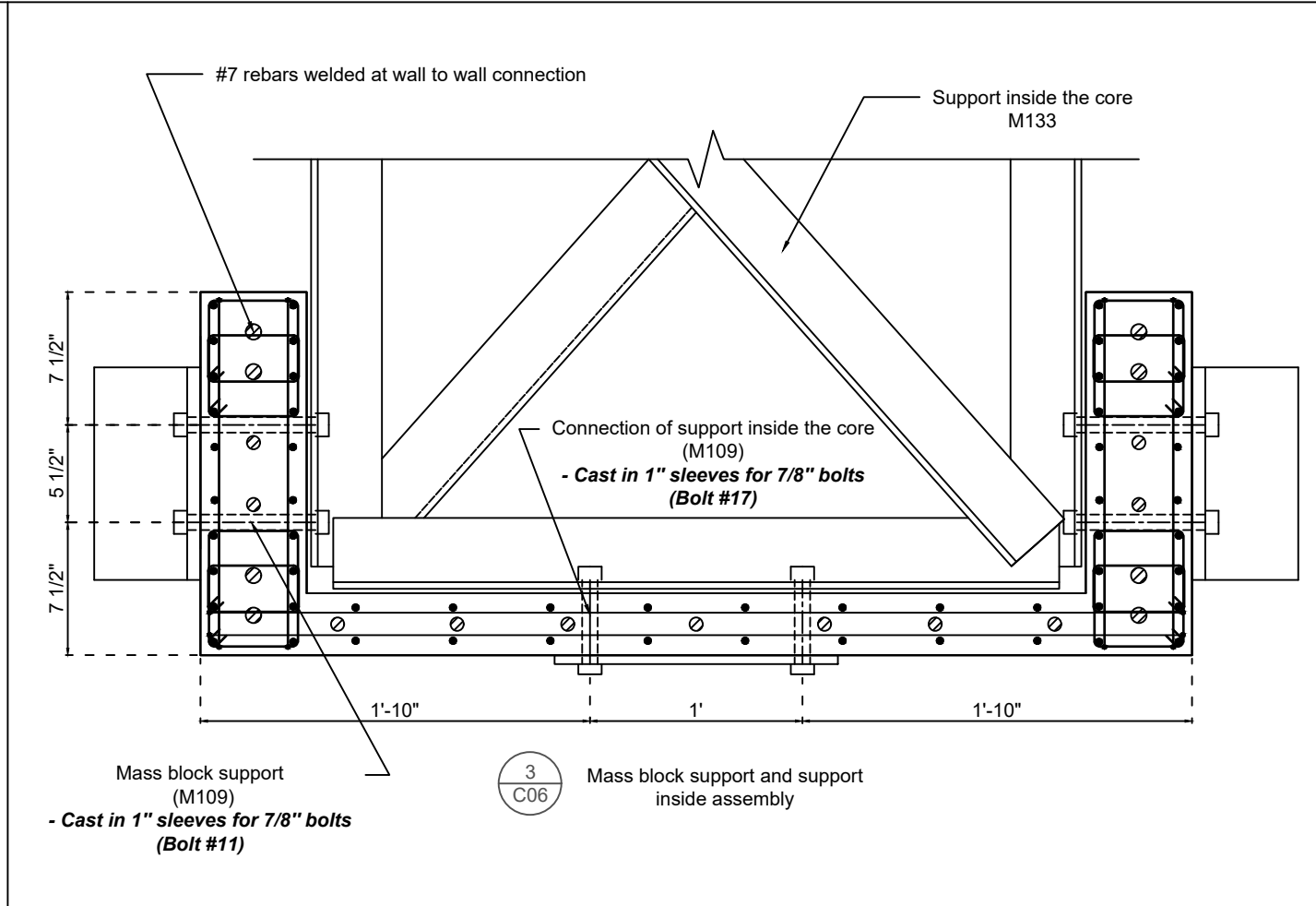
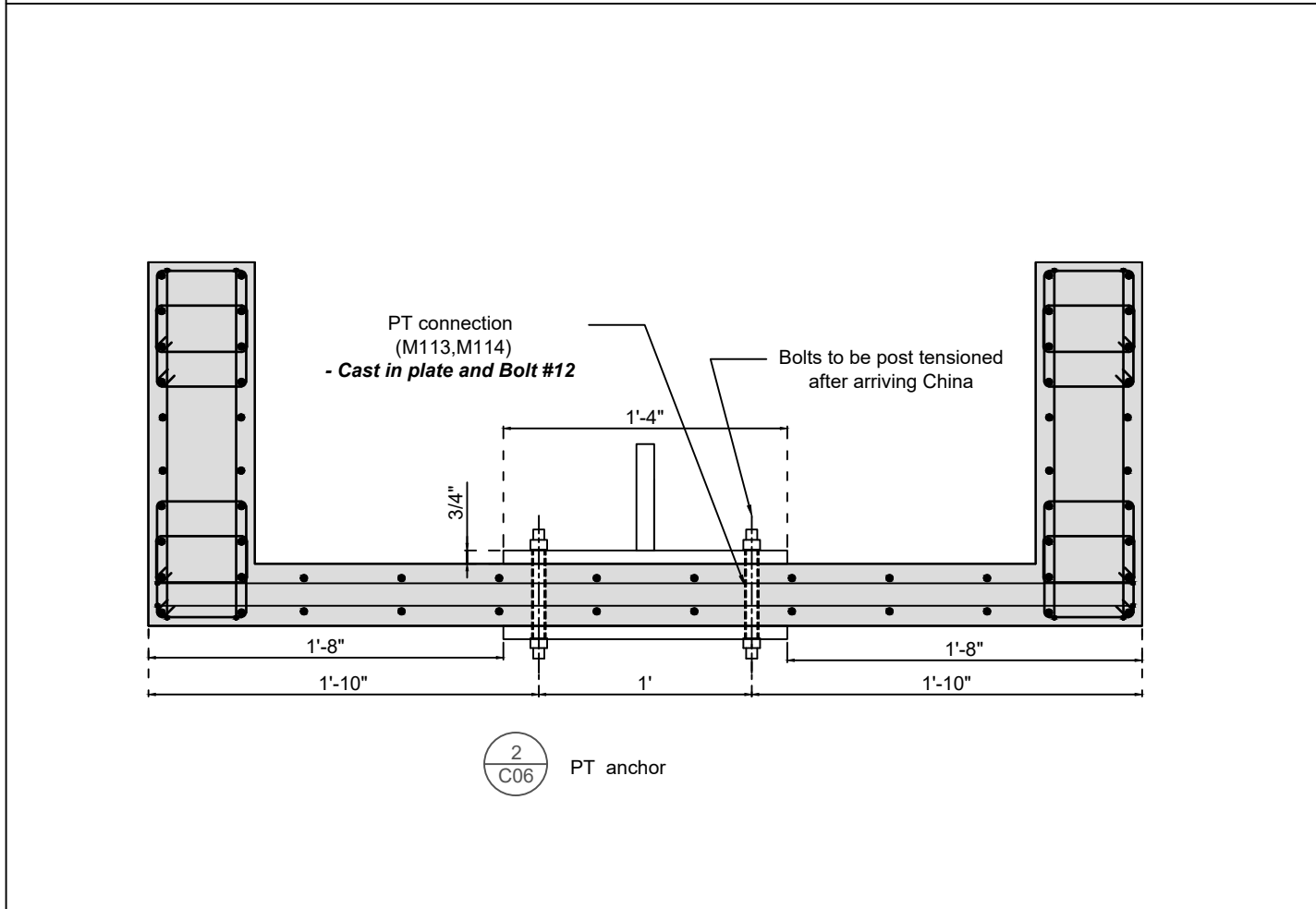
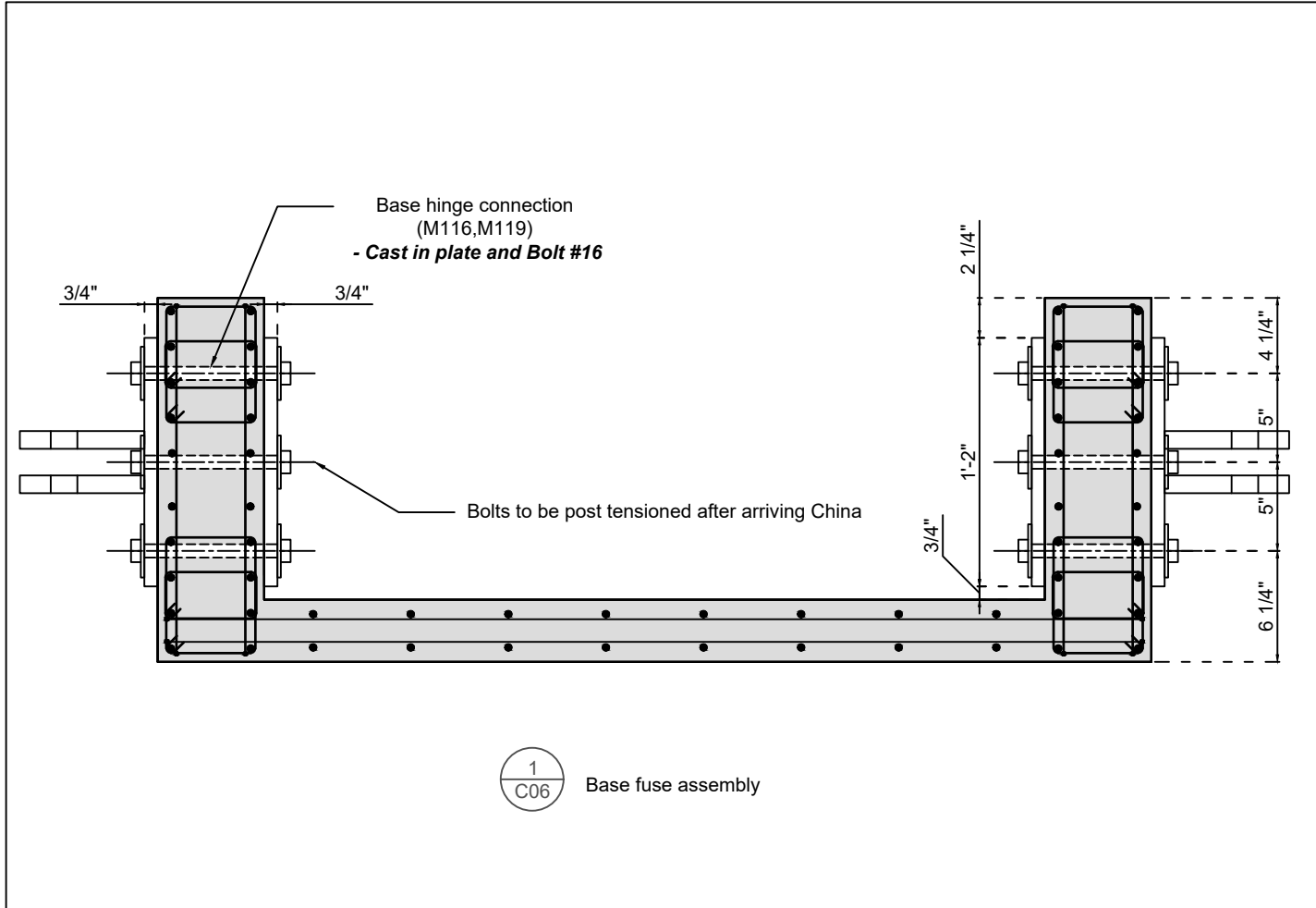
2
C03 Location of four zones, two flanges
and one web

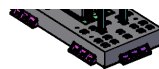
LVL	Zone reinforcing		Distribution steel		f'c
	Z1	Z2	SW1 (flange)	SW2 (web)	
Roof					
L24					45MPa
L23					
L22					
L21		8-#3 vert 11ga ties@2 1/4"			
L20					
L19					
L18					
L17					
L16					
L15					
L14					
L13					
L12					
L11					
L10					
L9					
L8					
L7					
L6		8-#3 vert 11ga ties@6"			
L5					
L4					
L3					
L2					
L1	8-#3 vert 11ga ties@2 1/4"	8-#3 vert 11ga ties@2 1/4"	6in wall #3 @ 3" VEF #3 @ 6" HEF	3.5in wall #3 @ 5 1/2" VEF #3 @ 6" HEF	

5
C03 Detailed information of reinforcement in elevation









Components

Num	Component	Drawing	Qty	Location of Fabrication
1	Outrigger beam	P13	2	GTS Canada
2	Outrigger fuse and connectors	P16	4	GTS Canada
3	Column lateral support	P15	4	GTS Canada
4	Mass block	P04,P07	8	China
5	Mass block angle	P04,P07	32	GTS Canada
6	Outrigger column and connections	P16,P17	4	GTS Canada
7	PT channels inside the wall	P12	4	GTS Canada
8	Wall to wall connection	P14	2	Canada
9	Coupling beam	P09	32	GTS Canada
10	Base fuse and connectors	P10	4	GTS Canada
11	Foundation	P03	1	China
12	Foundation anchors	P03	12	GTS Canada
13	Base armoring	P11	2	GTS Canada
14	Foundation angle	P03	6	GTS Canada

Material Properties

Unless otherwise specified in the drawings, materials for all components should be:

Steel:

ASTM A36 for steel plates;
G40.21 300W for angles;
ASTM A572- Gr50 for hot-rolled W, M, HP, and channel sections
ASTM A500 - Gr50 for HSS sections

Threaded Rods:

ASTM F1554-Gr105 for threaded rods (no welding required);
ASTM F1554-Gr55 for threaded rods (welding required);

Bolts:

ASTM A490 for bolts to connect coupling beams
ASTM A325 for other bolts

Rebars and PT:

Gr60 for rebars;
ASTM A416-Gr270 for strands in PT tendons

Concrete:

f'c = 45 Mpa, from 28 day concrete cylinder tests

For the concrete core wall channels:

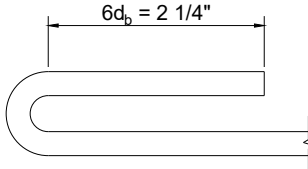
the clear cover is 5/16"; the max. aggregate size is 1/4".

For all mass blocks and the foundation :

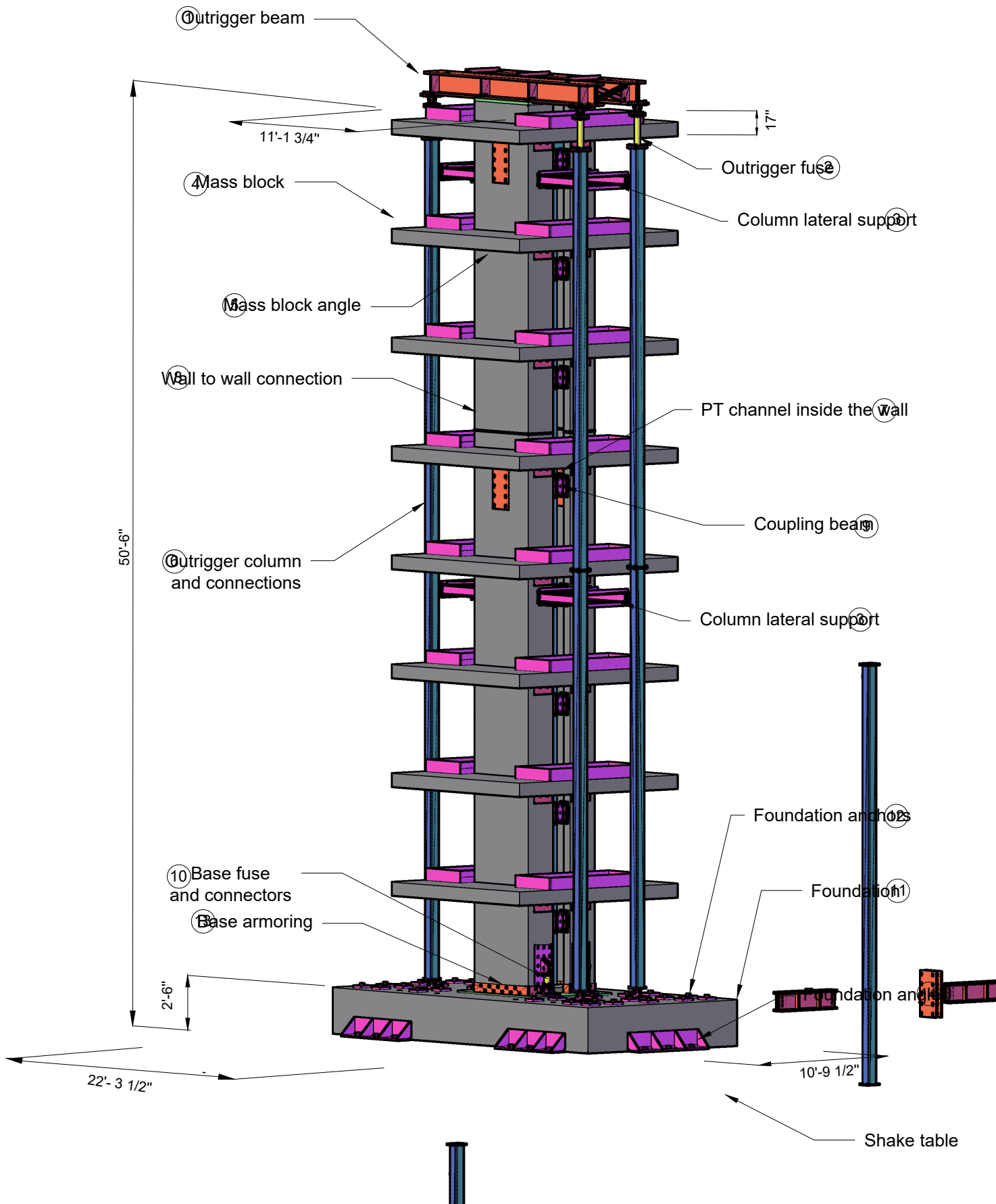
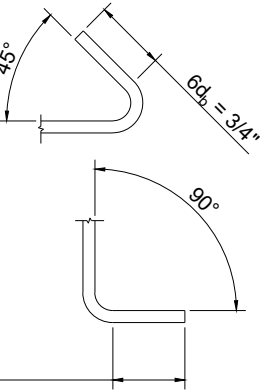
the clear cover is 1 3/16" and the max. aggregate size is 3/4".

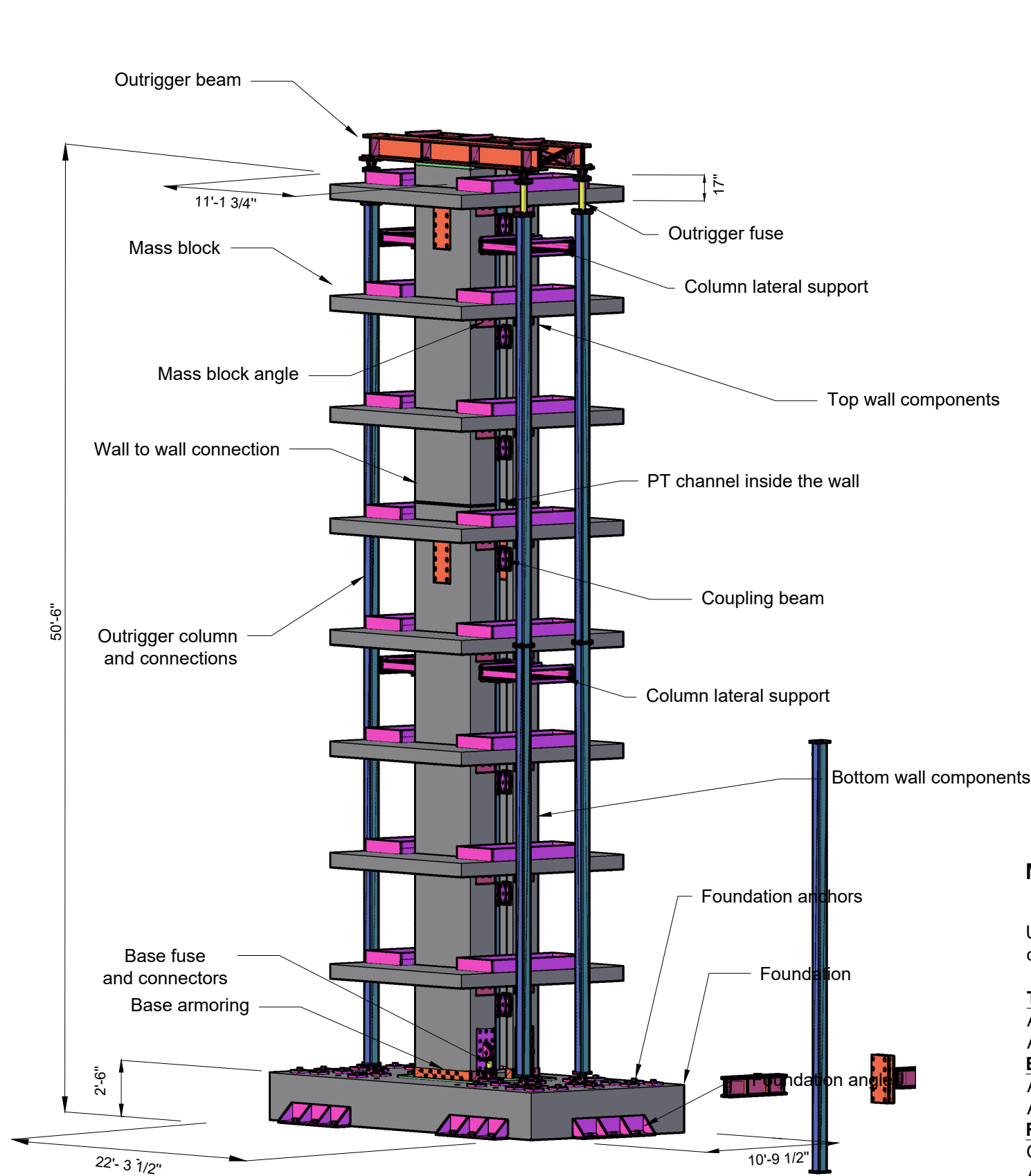
Typical Development Length of Reinforcement

#2 Rebars:

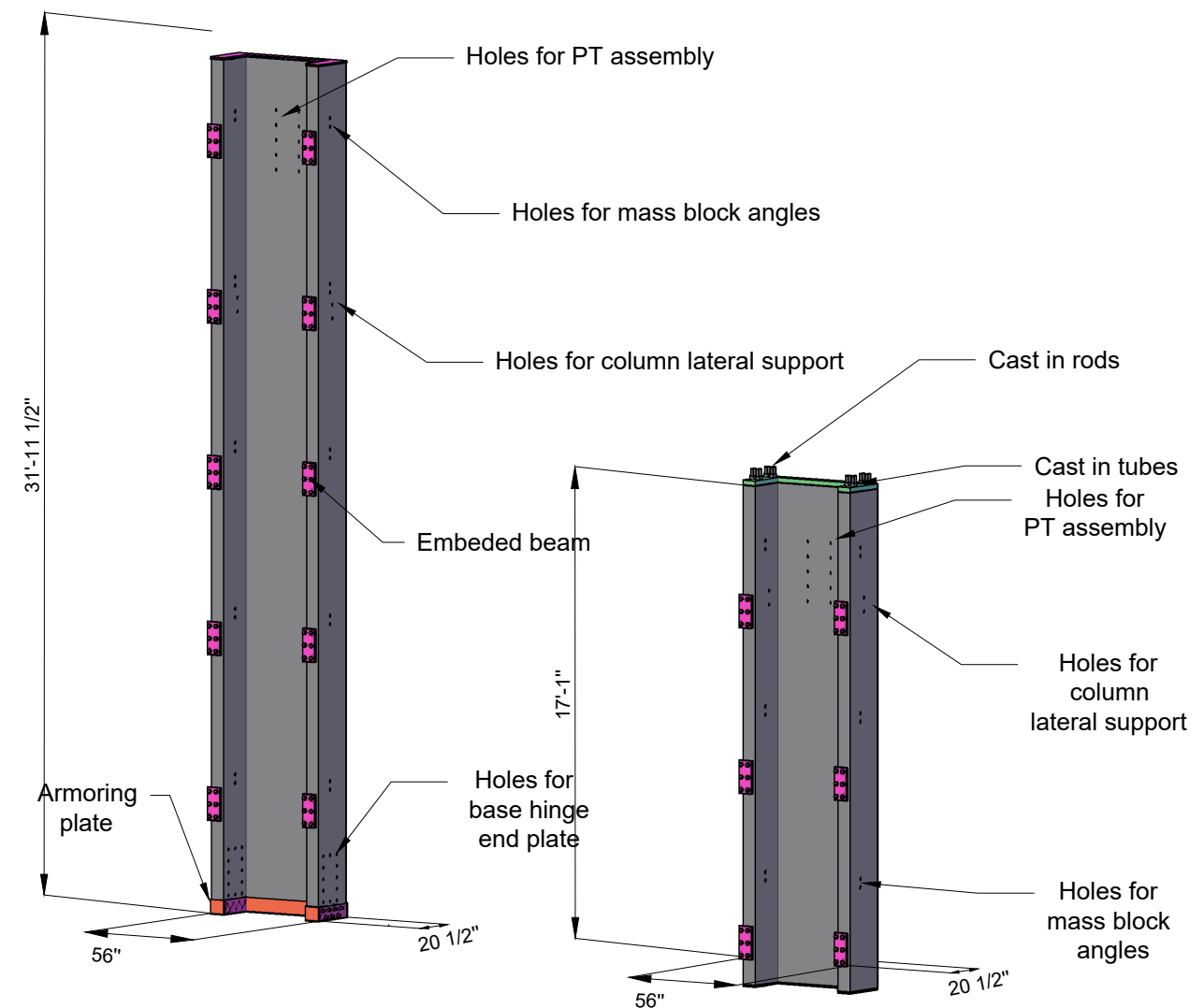


1/8" Wires:





Shake table test



Bottom wall component
Qty: 2
Unit Weight: 60kN/13.5kips

Top wall component
Qty: 2
Unit Weight: 30kN/6.75kips

Material Properties

Unless otherwise specified in the drawings, materials for all components should be:

Threaded Rods:

ASTM F1554-Gr105 for threaded rods (no welding required);
ASTM F1554-Gr55 for threaded rods (welding required);

Bolts:

ASTM A490 for bolts to connect coupling beams
ASTM A325 for other bolts

Rebars and PT:

Gr60 for rebars;
ASTM A416-Gr270 for strands in PT tendons

Concrete:

$f'_c = 45$ Mpa, from 28 day concrete cylinder tests

For the concrete core wall channels:

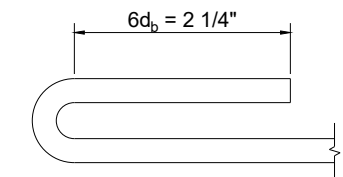
the clear cover is 5/16"; the max. aggregate size is 1/4".

For all mass blocks and the foundation :

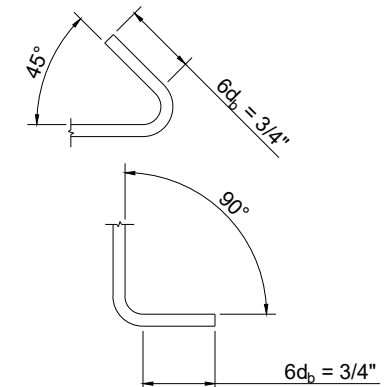
the clear cover is 1 3/16" and the max. aggregate size is 3/4".

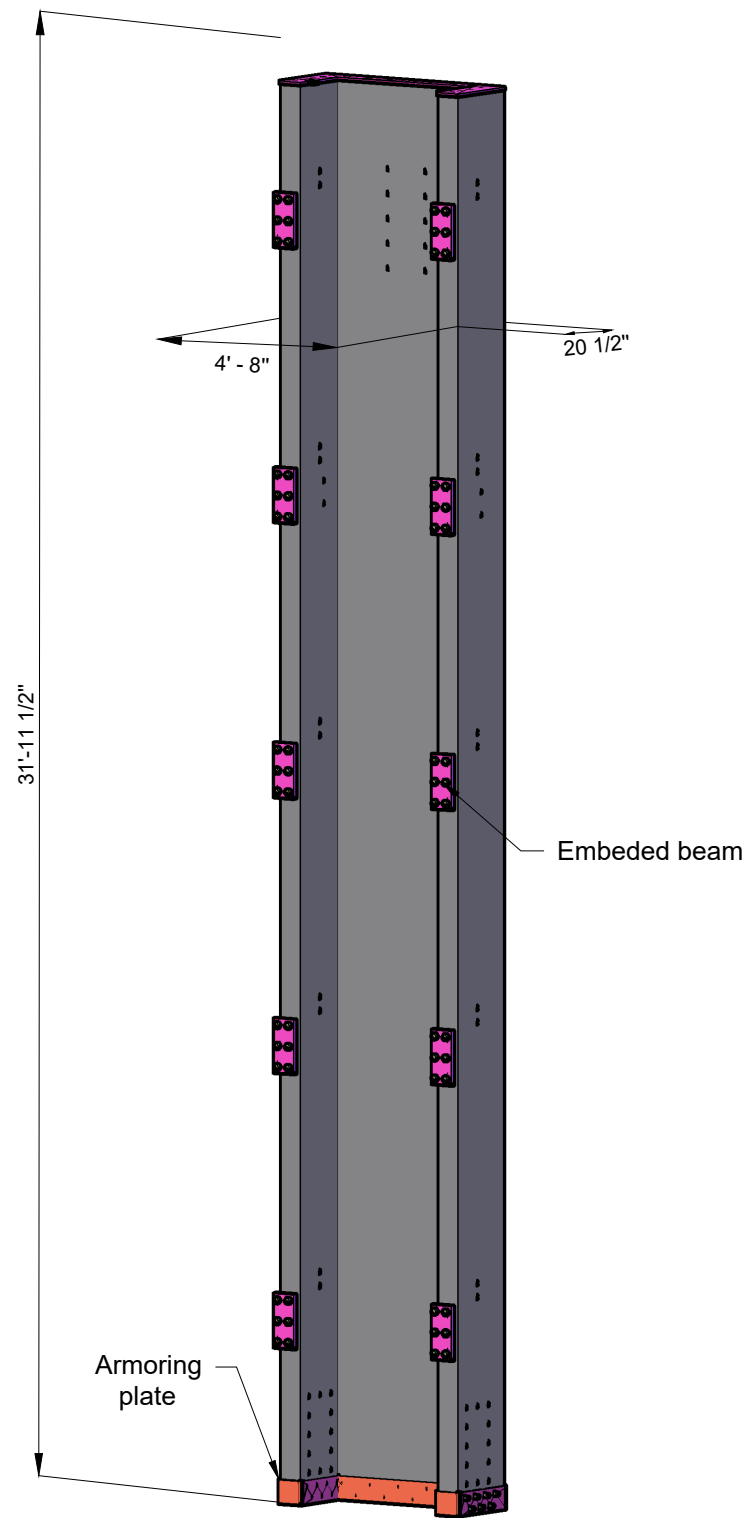
Typical Development Length of Reinforcement

#2 Rebars:

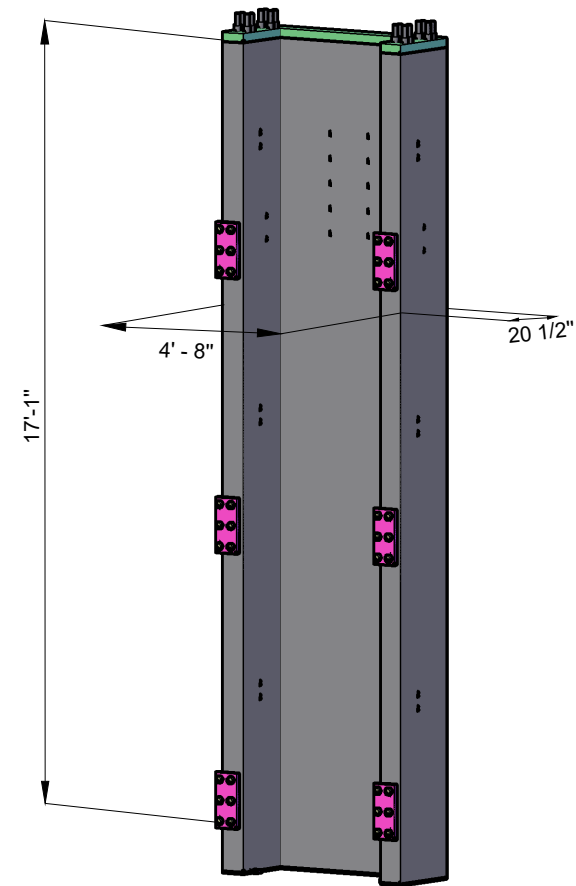


1/8" Wires:



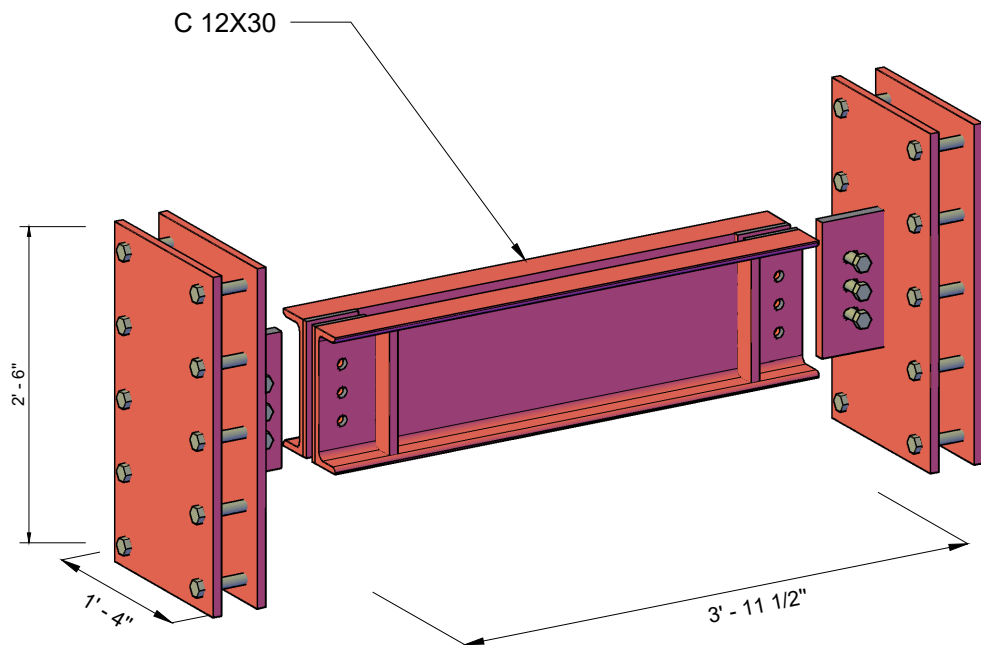


Bottom wall component
Qty: 2
Unit Weight: 60kN/13.5kips

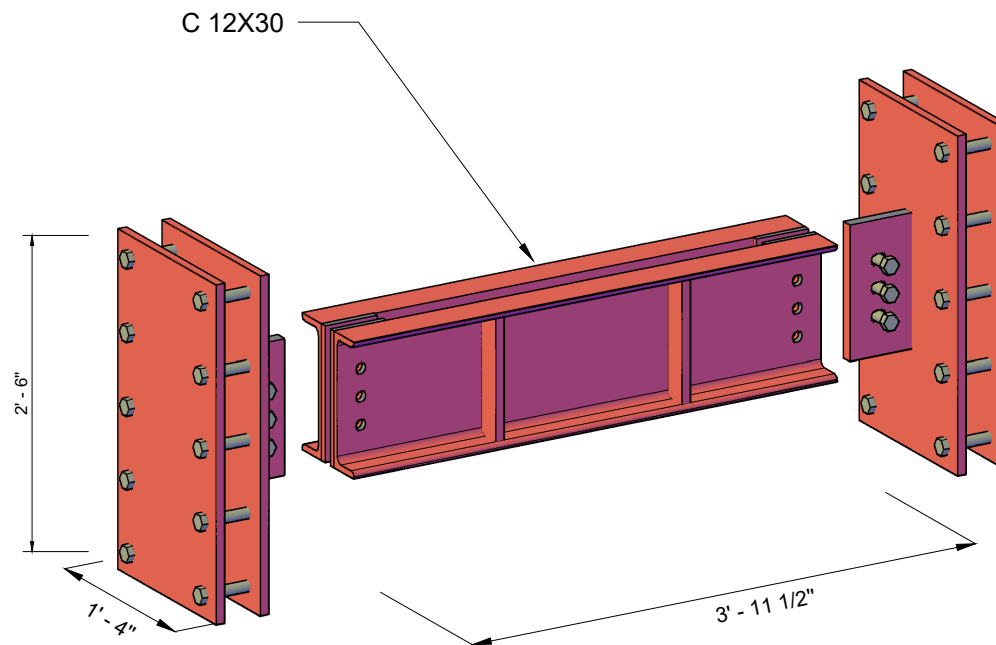


Top wall component
Qty: 2
Unit Weight: 30kN/6.75kips

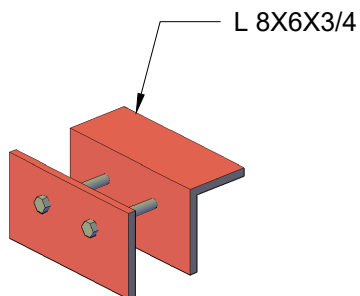




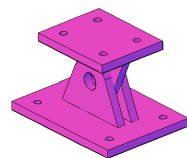
PT anchor - Type I
Qty: 1
Unit Weight: 3.5kN/0.8kips



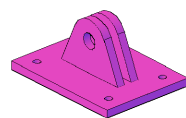
PT anchor - Type II
Qty: 1
Unit Weight: 3.5kN/0.8kips



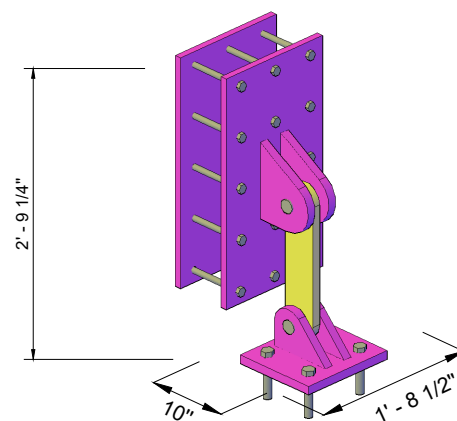
Mass block support
Qty: 32
Unit Weight: 0.3kN/0.065kips



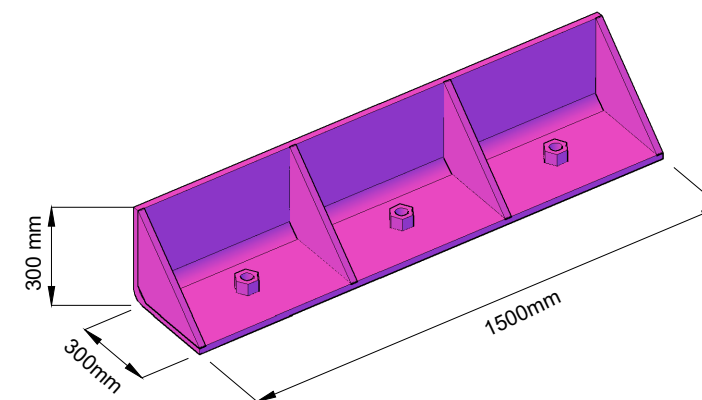
Outrigger end connection
Qty: 4
Unit Weight: 0.2kN/0.04kips



Outrigger end connection
Qty: 4
Unit Weight: 0.14kN/0.03kips

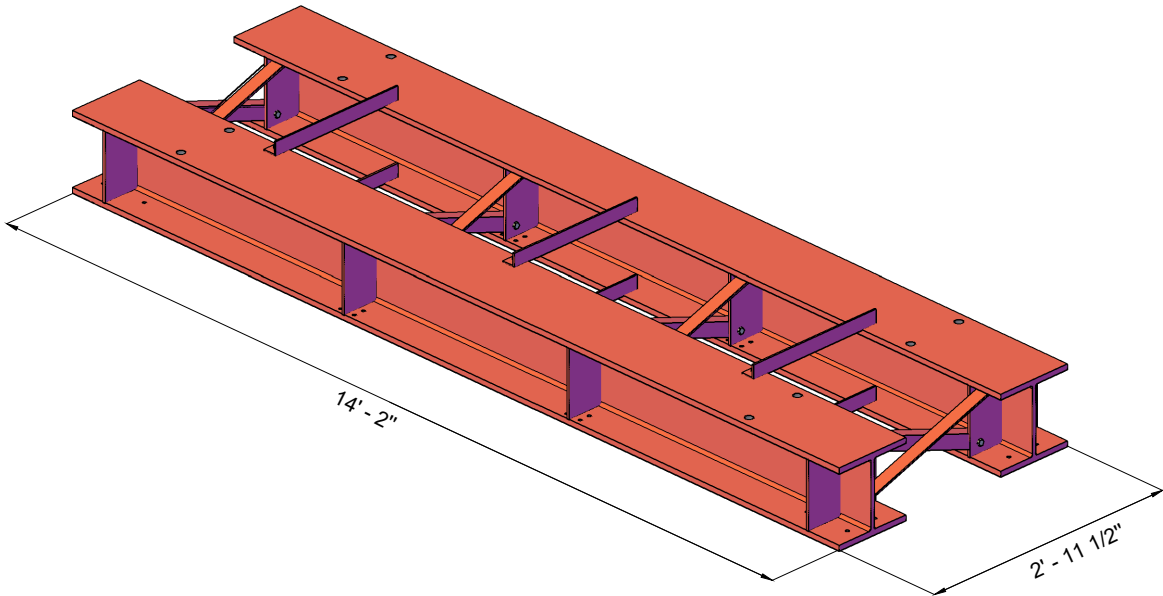


Base fuse
Qty: 4
Unit Weight (without fuse): 1.1kN/0.25kips

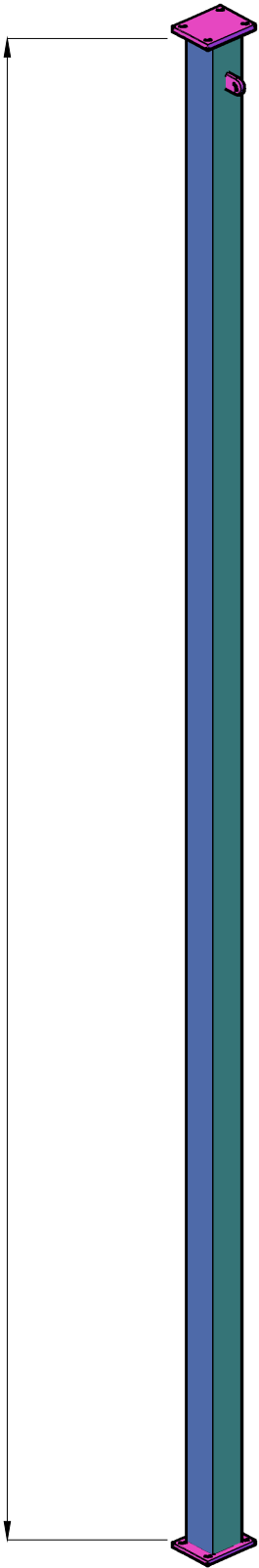


Foundation shear angle (Formed by 3/4" Thk plate)
Qty: 6
Unit Weight: 2.4kN/0.54kips

Outrigger
Qty: 1
Unit Weight: 16.8kN/3.8kips

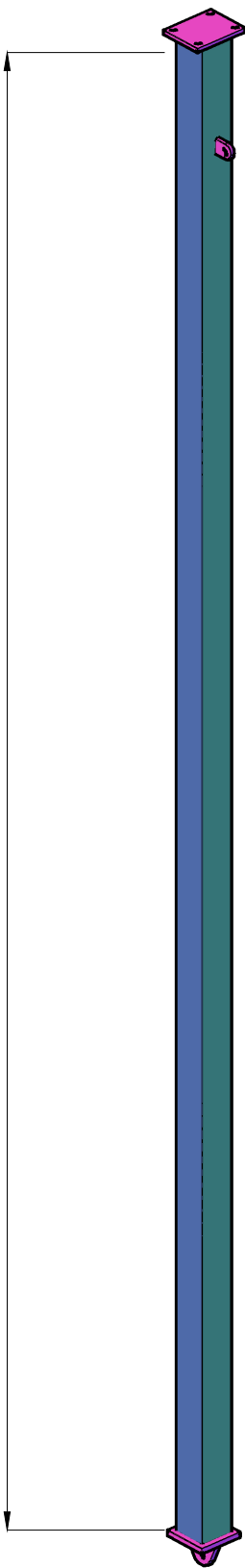


22' - 11 3/8"



Outrigger column - Type I
Qty: 4
Unit Weight: 3.1kN/0.7kips

22' - 11 3/8"



Outrigger column - Type II
Qty: 4
Unit Weight: 3.1kN/0.7kips

ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING



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Smart Structures

Components in 3D View

SHEET NO.

3

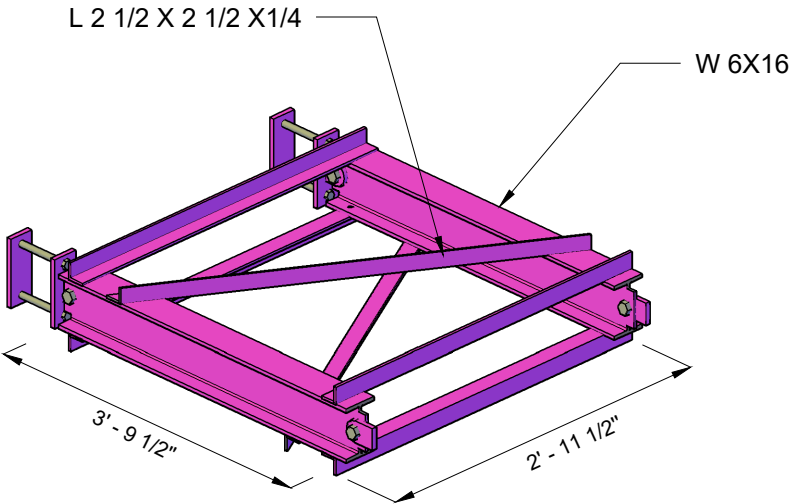
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Hengchao Xu
Lisa Tobber

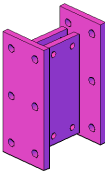
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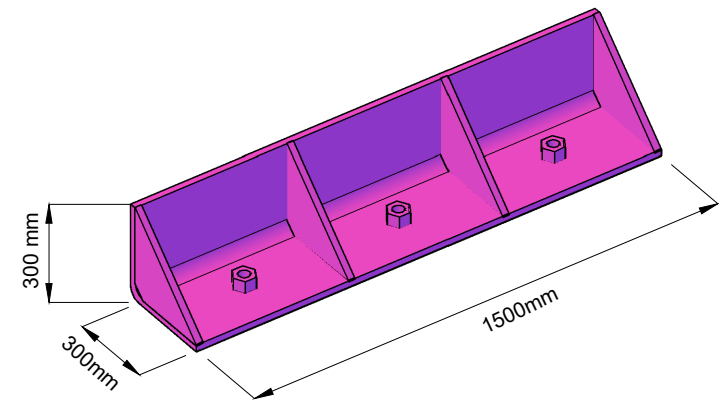
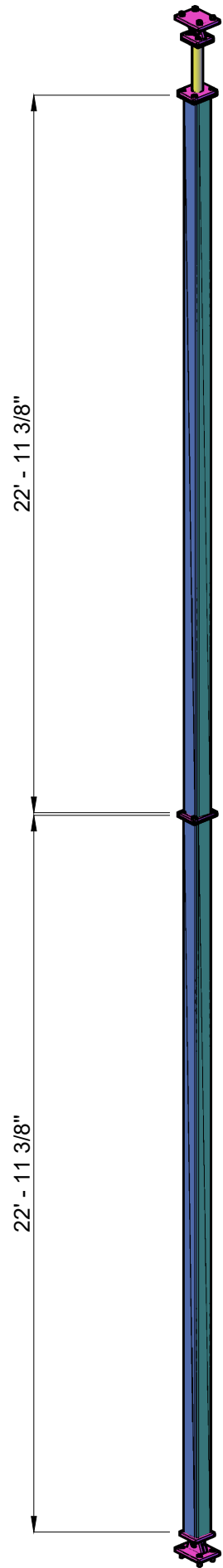
2023/10/08

Column lateral support
Qty: 4
Unit Weight: 0.75kN/0.17kips



Friction Device
Qty: 16
Unit Weight: 0.2kN/0.04kips





Foundation shear angle (Formed by 3/4" Thk plate)
Qty: 6
Unit Weight: 2.4kN/0.54kips

ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

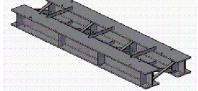
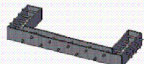
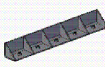
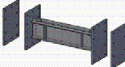
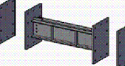
Components in 3D View

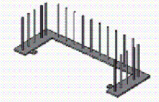
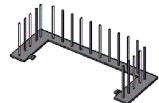
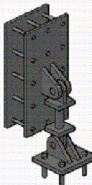
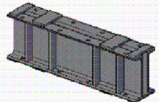
SHEET NO.	
4	
DRAWN BY	
Tianyang Qiao Hengchao Xu Lisa Tobber	
DATE	2023/10/08

Steel Components List

Manufactured by GTS

Ready for Fabrication

Num	Component	Figure	Qty	Drawing
1	Outrigger		1	S1-1
2	Outrigger column lateral support		4	S2
3	Outrigger columnc Type I		4	S3-1
4	Outrigger columnc Type II		4	S3-1
5	Outrigger end connection Type I		8	S3-2
6	Outrigger end connection Type II		4	S3-2
7	Wall base armoring		2	S4-1/S4-2
8	Steel embedded beam		32	S5
9	Foundation anchor Type I		4	S6-1
10	Foundation anchor Type II		2	S6-2
11	Mass block support		32	S7
12	PT anchor Type I		1	S8-1/S8-3
13	PT anchor Type II		1	S8-2/S8-3

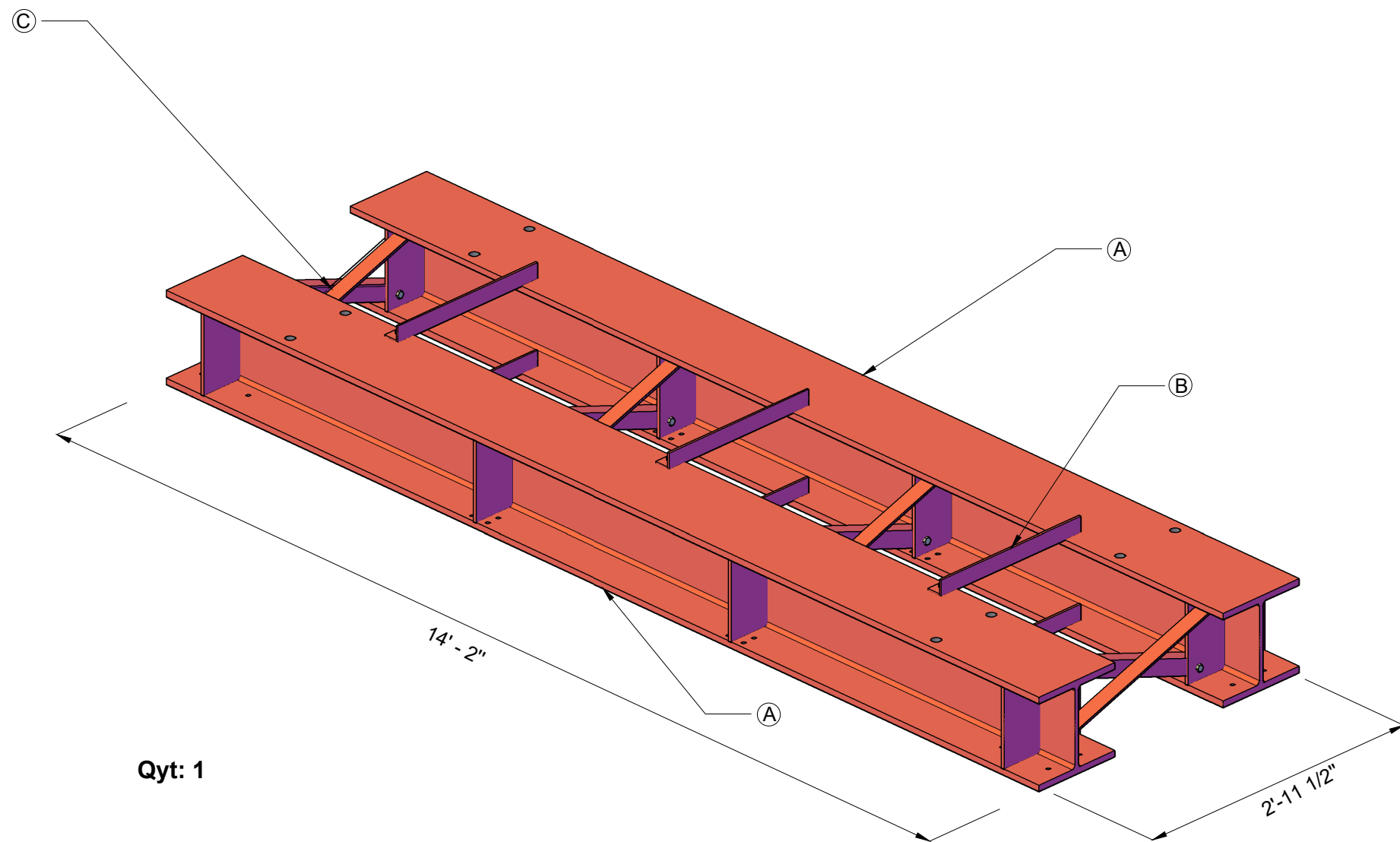
Num	Component	Figure	Qty	Drawing
14	Wall to wall connection Type I		2	S9
15	Wall to wall connection Type II		2	S9
16	Base fuse connection		4	S10
17	PT base anchor plate		1	S11
18	Friction damper		16	S12 UnderTest

Standard Section Used

Num	Section Type	Total Length
1	W 14X120	28'-2"
2	W 8X15	24'-8"
3	W 6X16	30'-4"
4	L 8X6X3/4	32'
5	L 2.5X2.5X1/4	61'-3 3/4"
6	HSS 6X6X3/8	45'-10 3/4"
7	C12X30	22'-3"

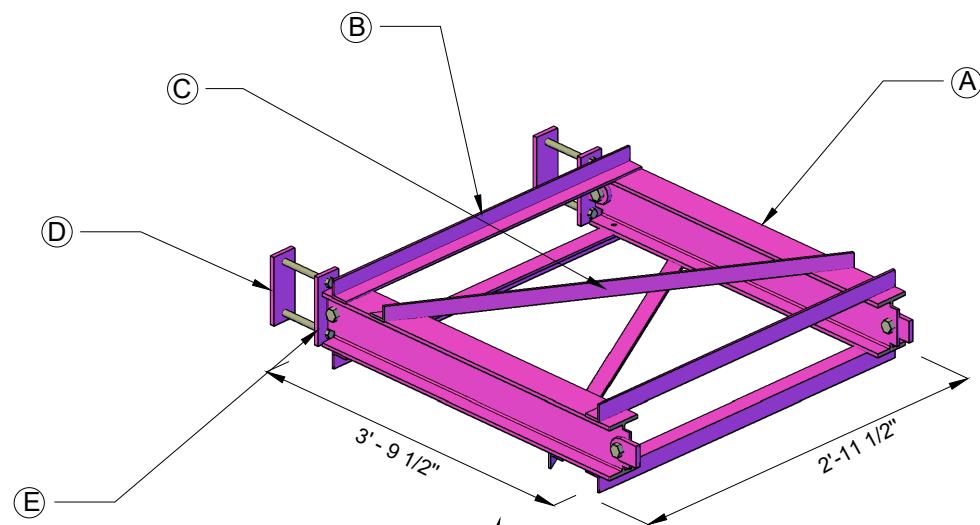
Material Properties

Steel:
ASTM A36 for steel plates;
G40.21 300W for angles;
ASTM A572- Gr50 for hot-rolled W, M, HP, and channel sections
ASTM A500 - Gr50 for HSS sections
Threaded Rods:
ASTM F1554-Gr105 for threaded rods (no welding required);
ASTM F1554-Gr55 for threaded rods (welding required);
Bolts:
ASTM A325 for bolts



Qty: 1

* Ø11/16" hole for 5/8" bolt

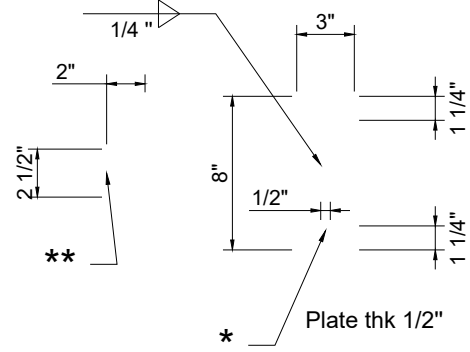
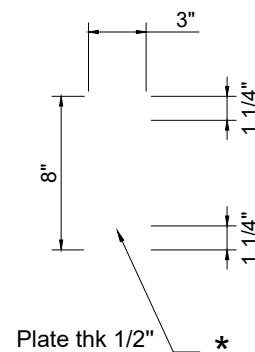
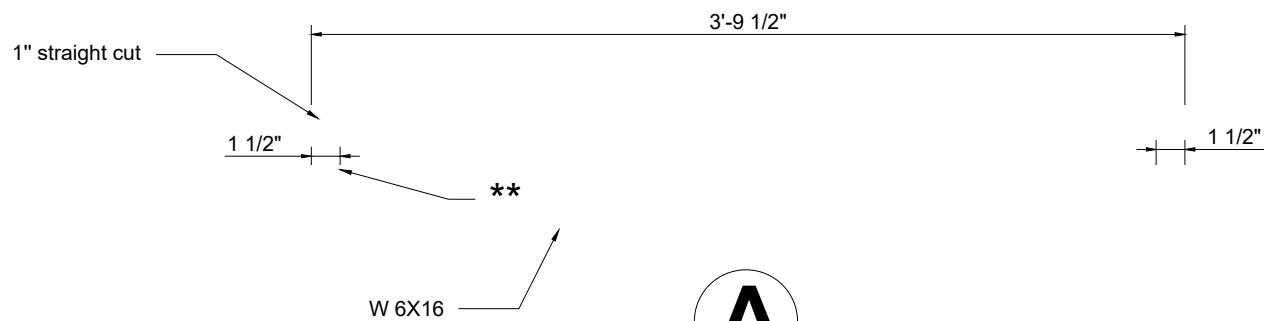


Qyt: 4

Notes for GTS:

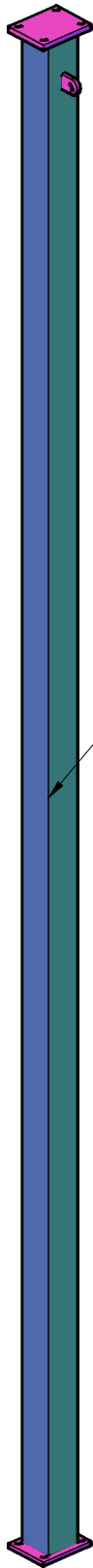
The purpose of this column lateral support is to provide stability to outrigger columns.

Currently, as shown in the figure, there are three angles welded at the top of the I-beam and another three welded at the bottom of the I-beam. They don't have to be in this exact configuration. So, if there is an easier way to construct this, please let us know.

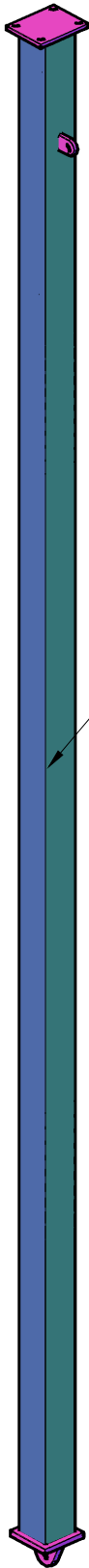


* Ø11/16" hole for 5/8" bolt

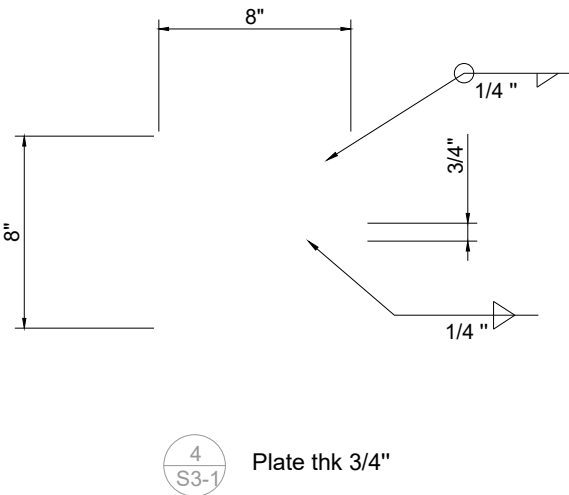
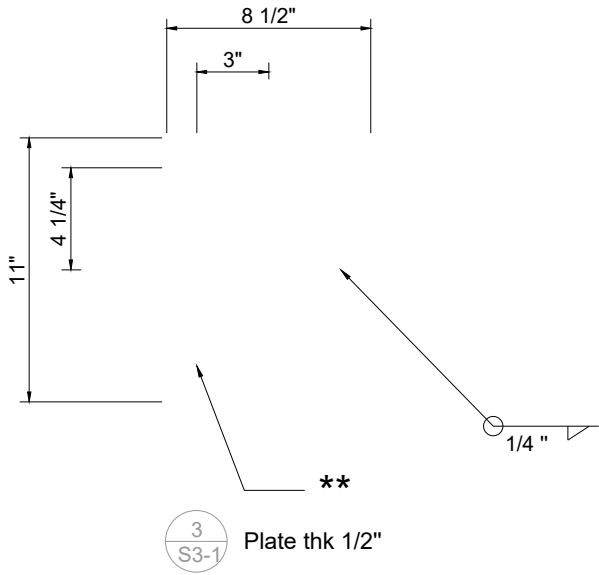
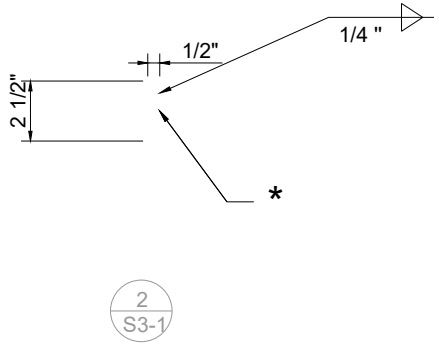
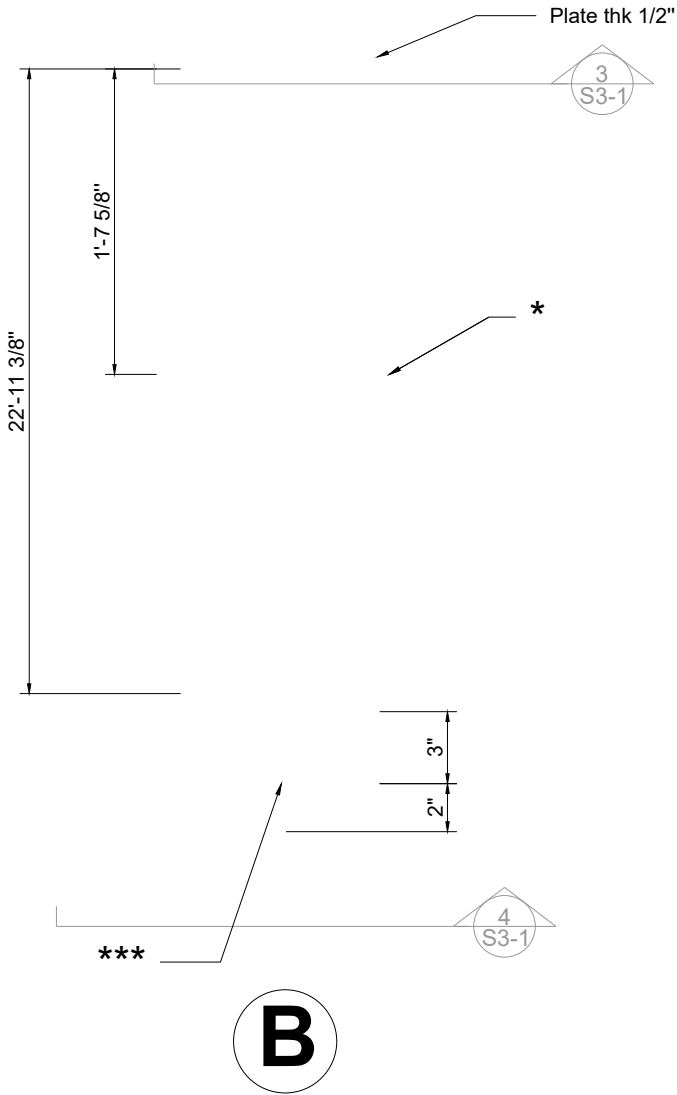
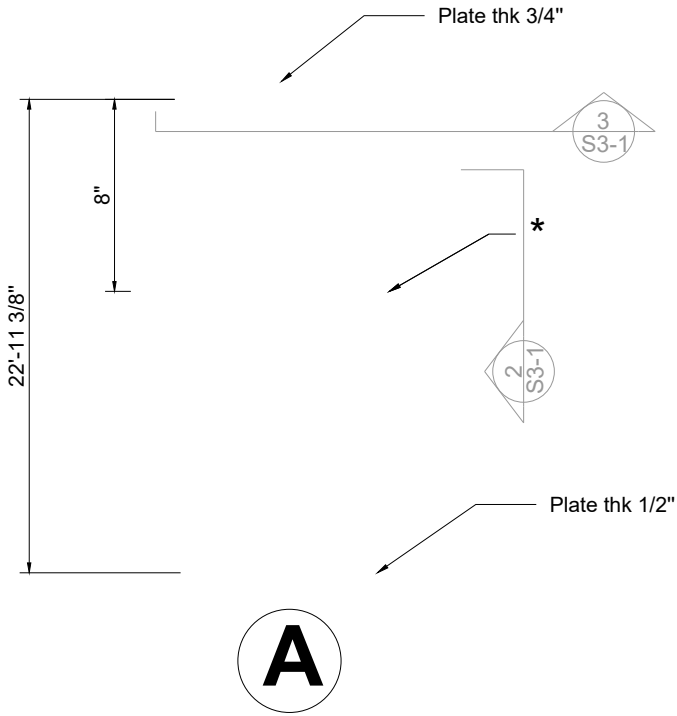
** Drill Ø15/16" hole for 7/8" bolt



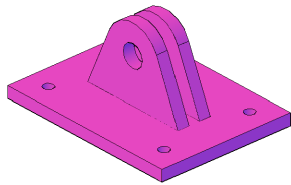
HSS 6X6X3/8
Qty: 4



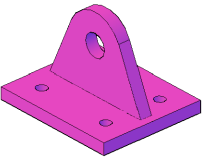
HSS 6X6X3/8
Qty: 4



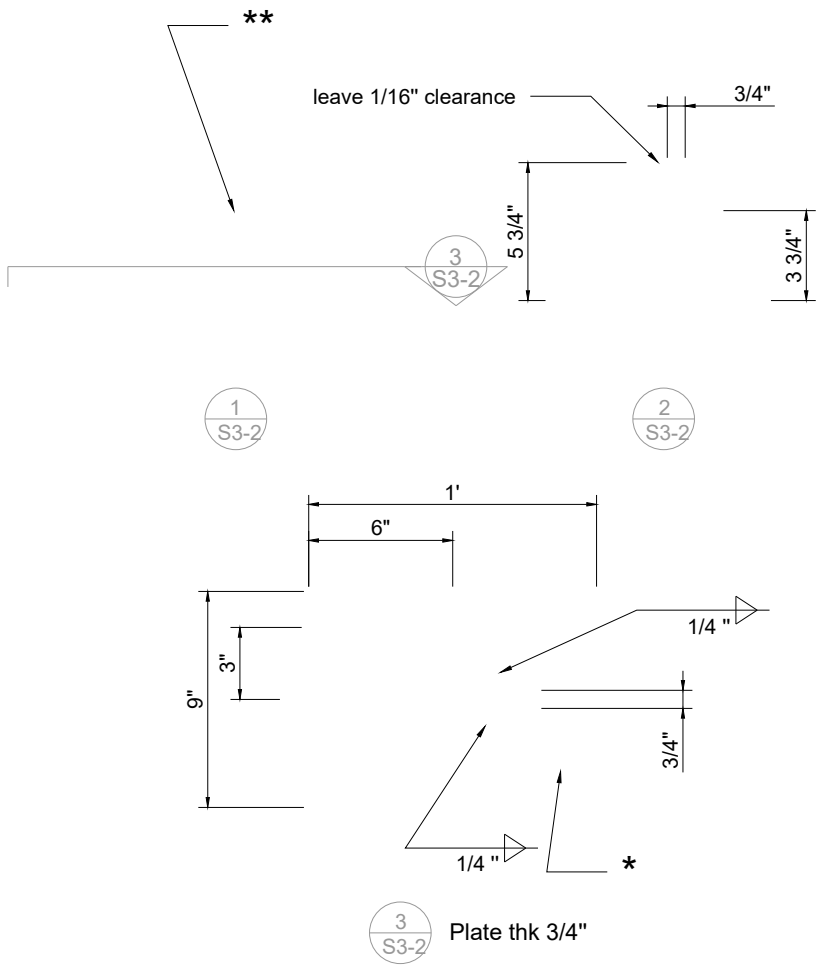
* Ø15/16" hole for 7/8" bolt
** Ø11/16" hole for 5/8" bolts
*** Drill Hole for Ø 1 1/2" pin with tolerance less than 1/32", the less the better



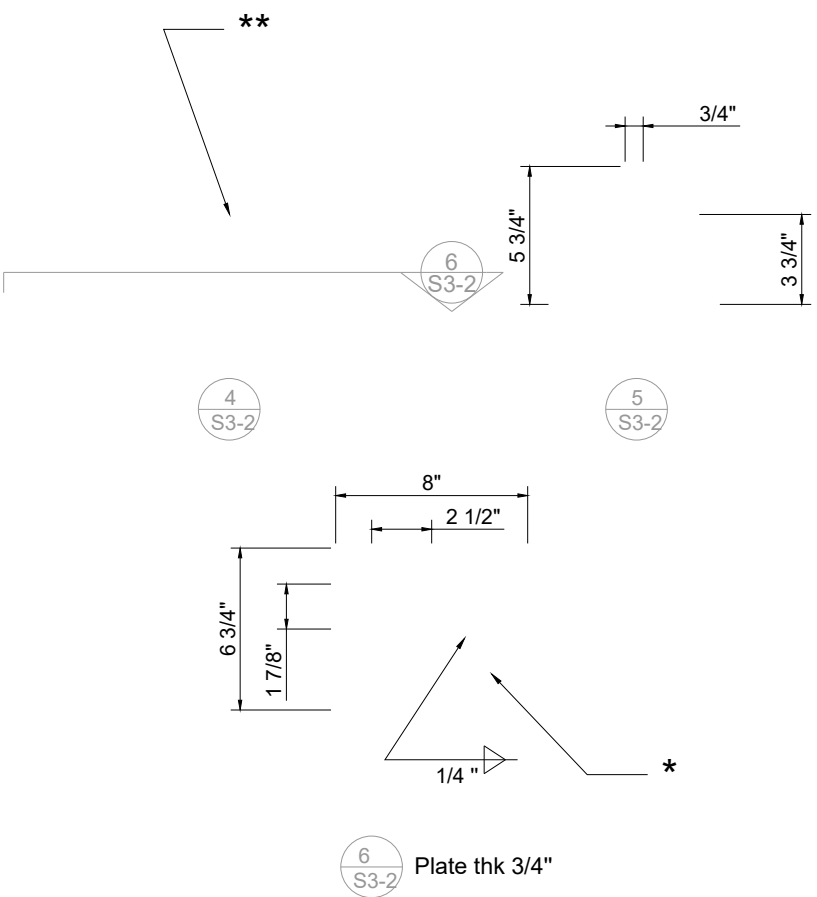
Type I
Qty: 8



Type II
Qty: 4

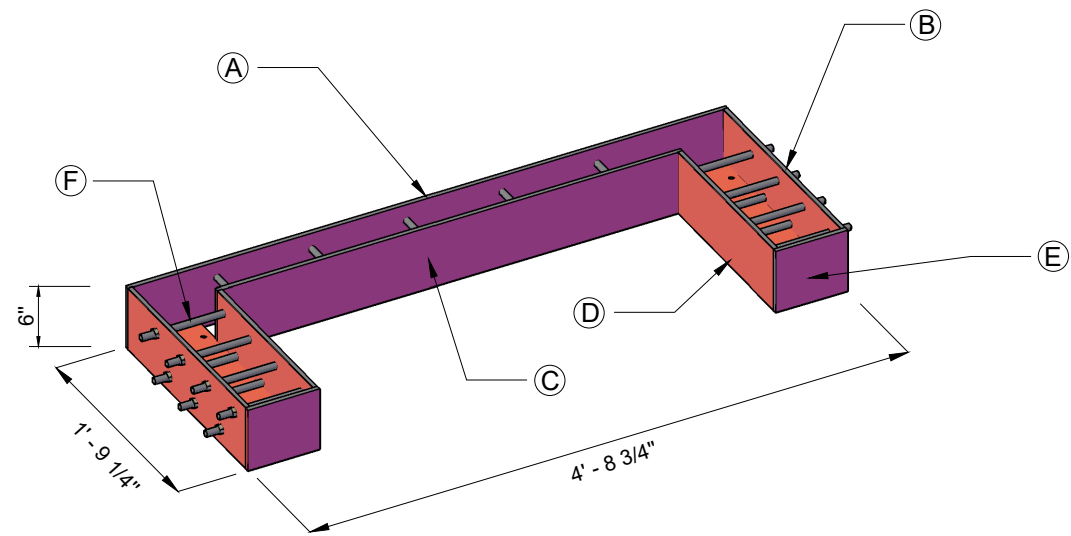


Type I

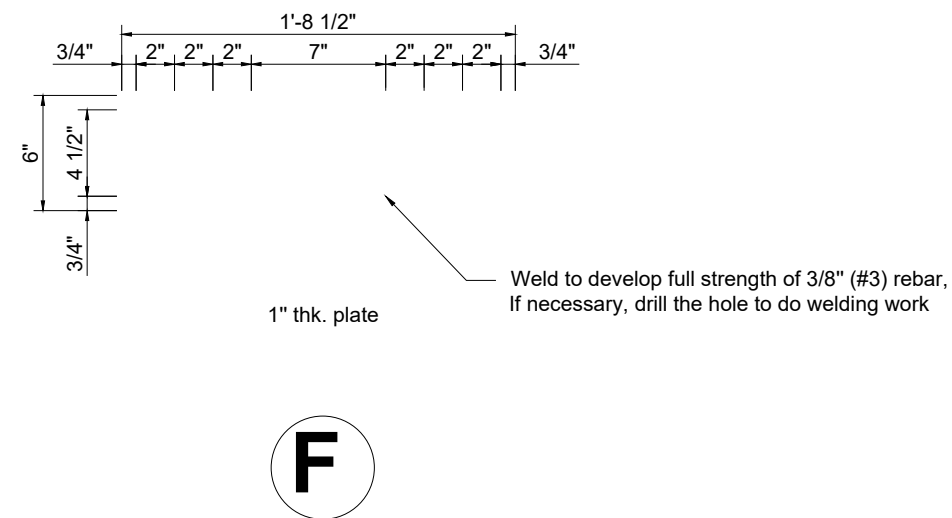
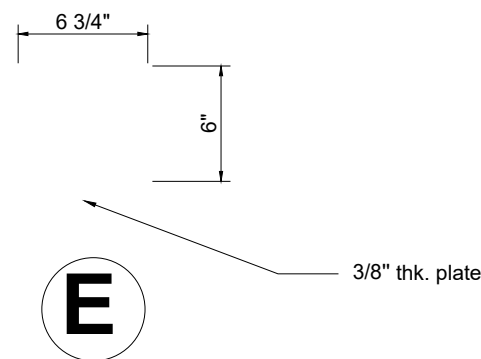
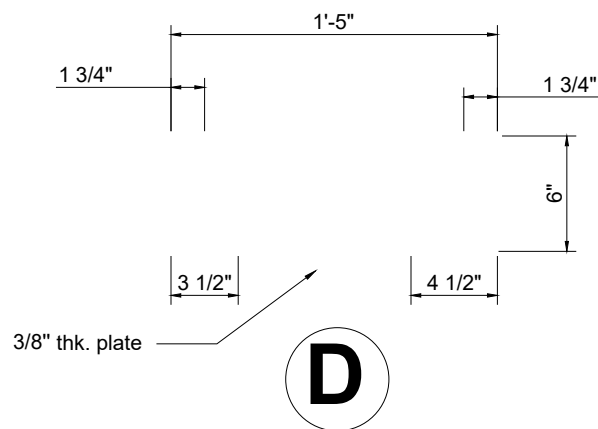
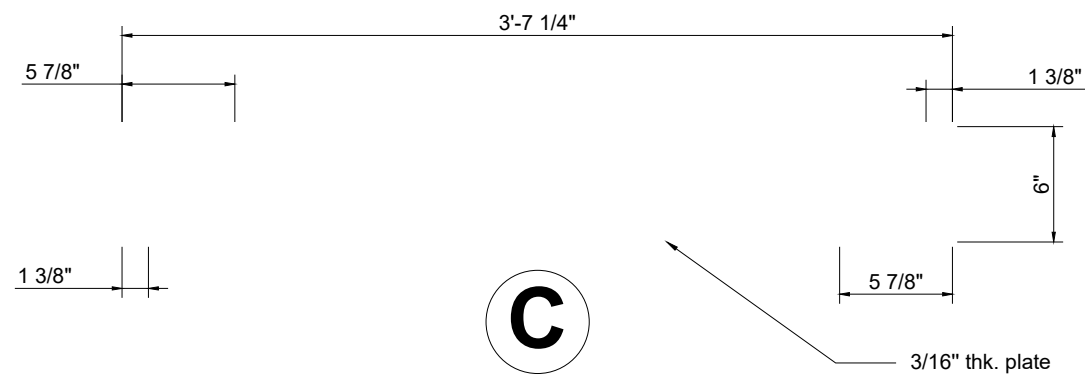
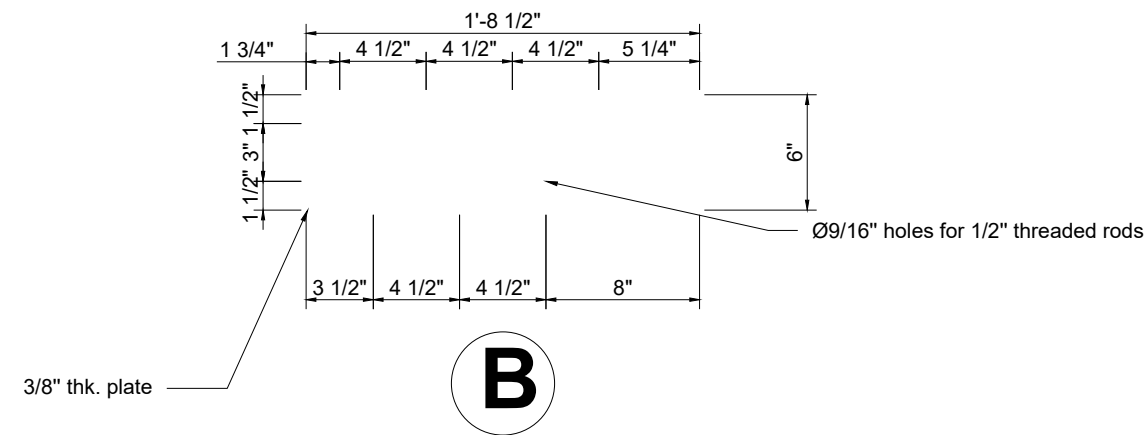
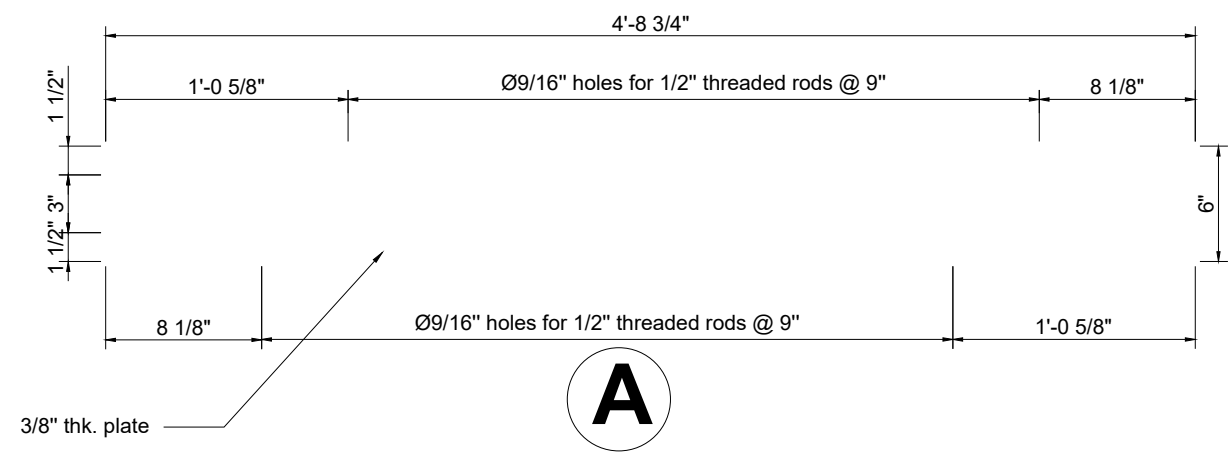


Type II

* Ø11/16" hole for 5/8" threaded rods
** Drill Hole for Ø 1 1/2" pin with tolerance less than 1/32", the less the better



Qty: 2



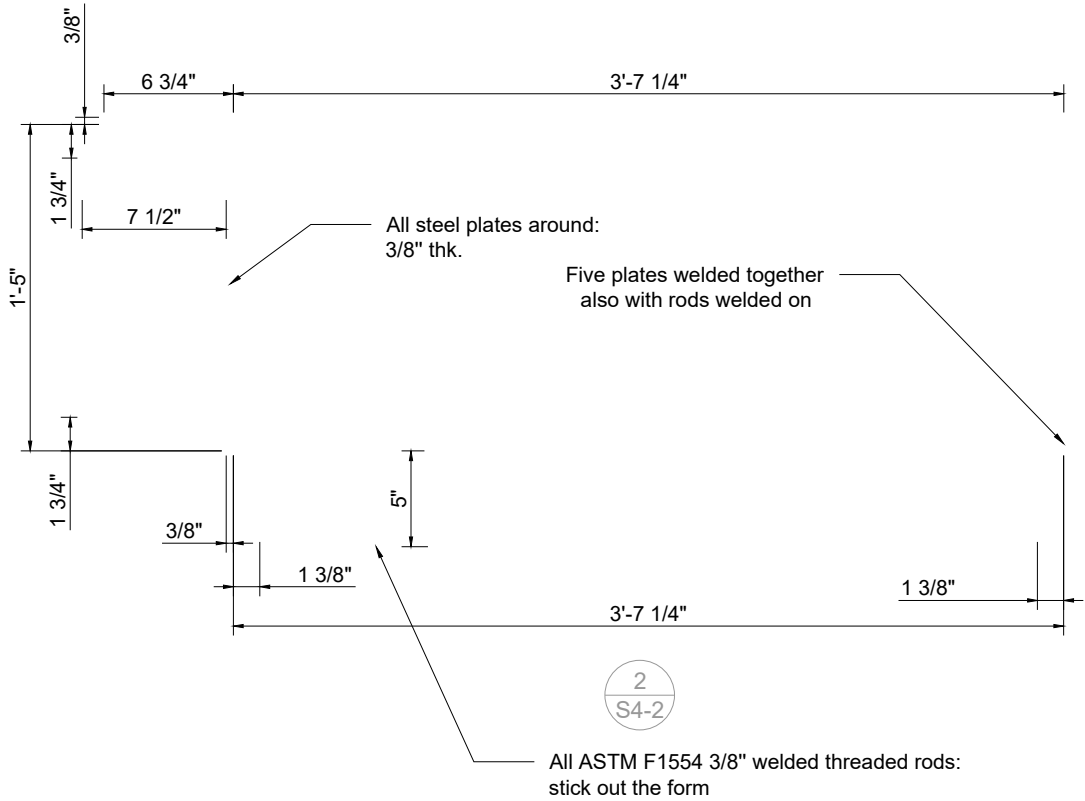
1" thk plate with holes for rebar
is welded to armoring plate, covering the flange from bottom

All threaded rods: welded to plate at one end (interior),
and bolted at the another end (assemble before casting concrete)

All steel plates around:
3/8" thk.

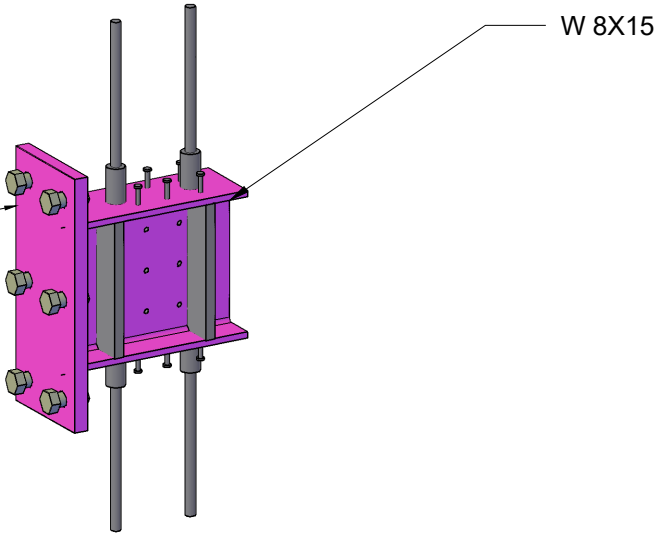
1
S4-2

Five 3/8" thk. steel plates
welded together to be a frame

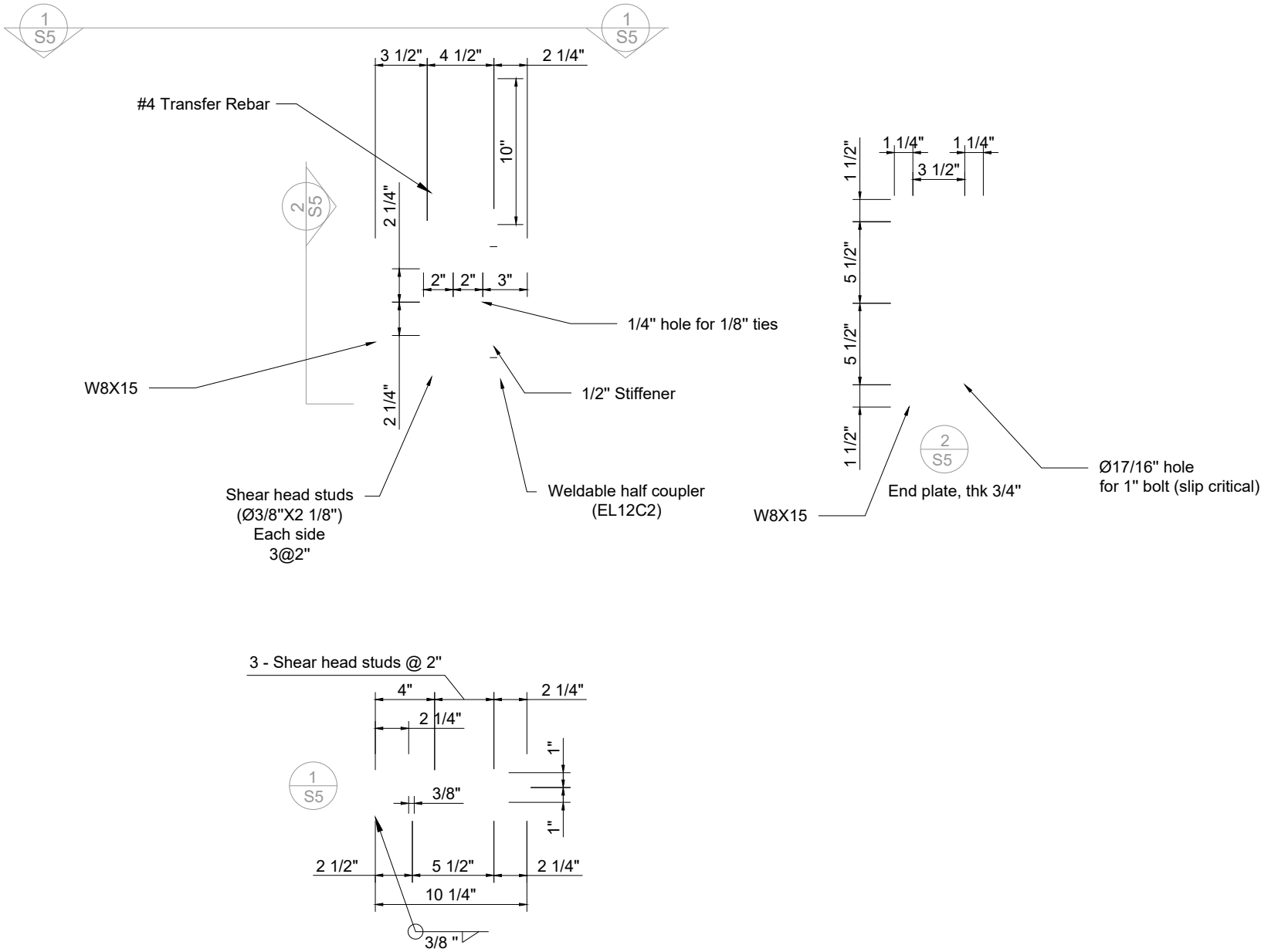


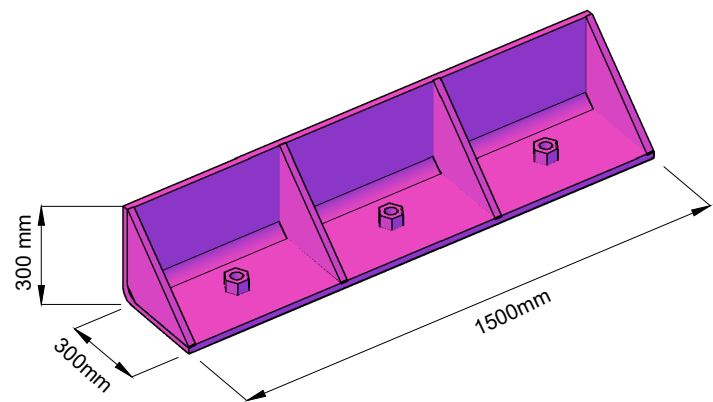
Notes for GTS:

Based on the conversation with Rob, we need steel components to connect the beam to the form work. Please add as you see fit.

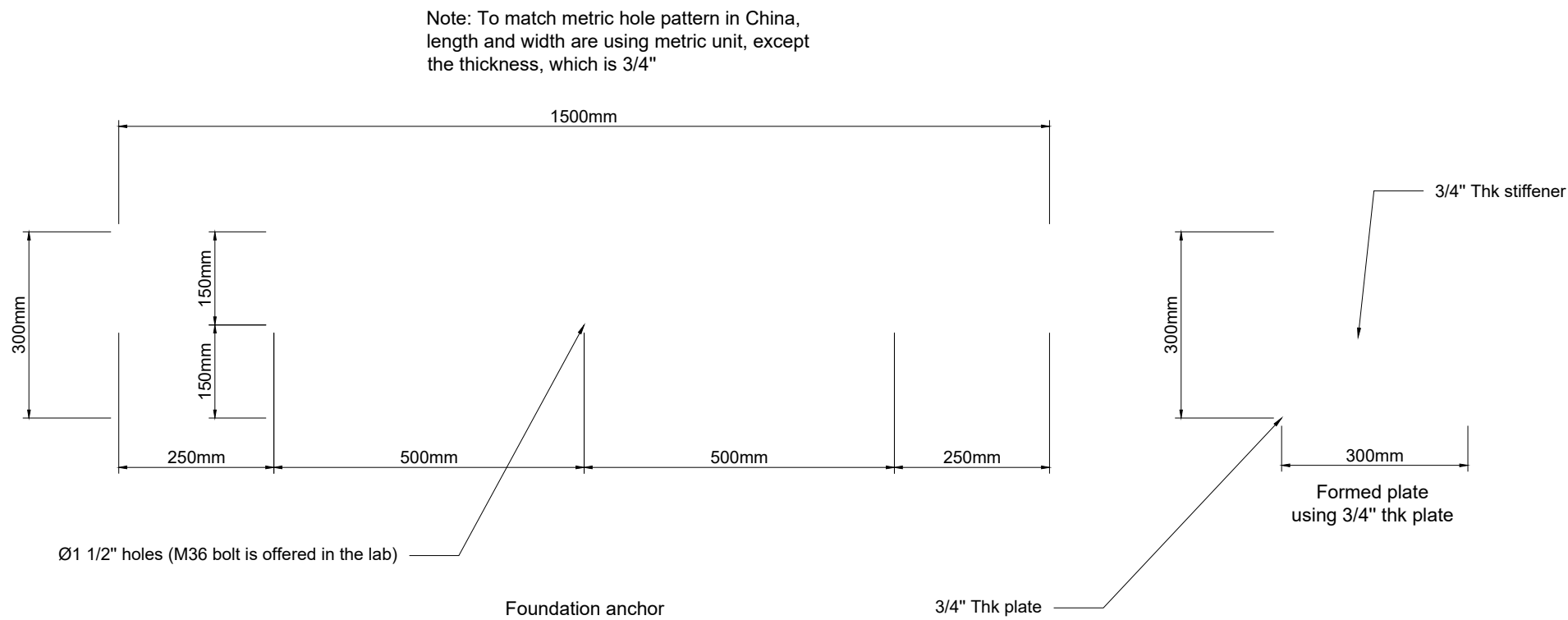


Qty: 32

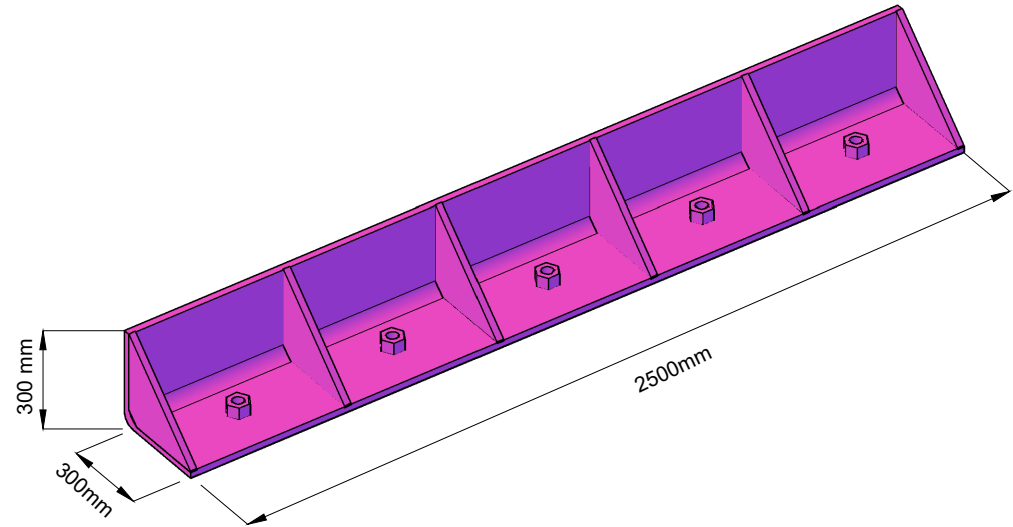




Type I
Qty: 6

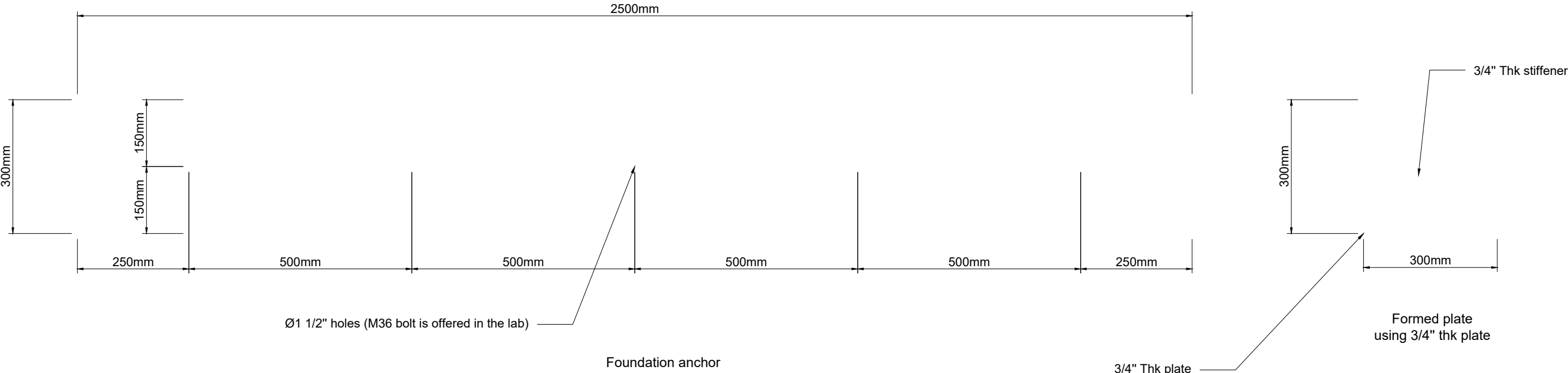


Qty. 4

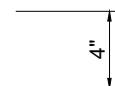


Type II
Qty: 6

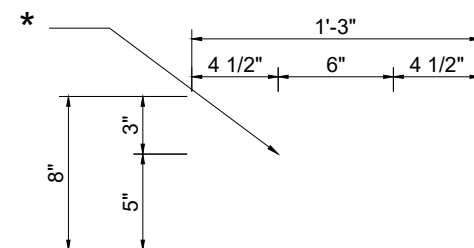
Note: To match metric hole pattern in China, length and width are using metric unit, except the thickness, which is 3/4"



Qty. 2



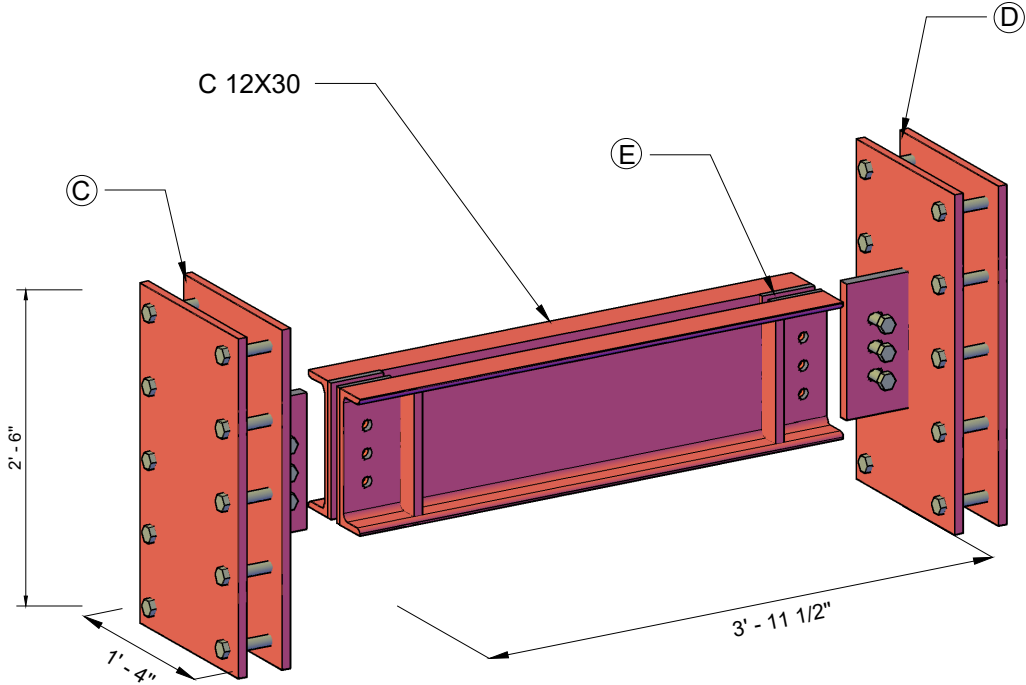
A



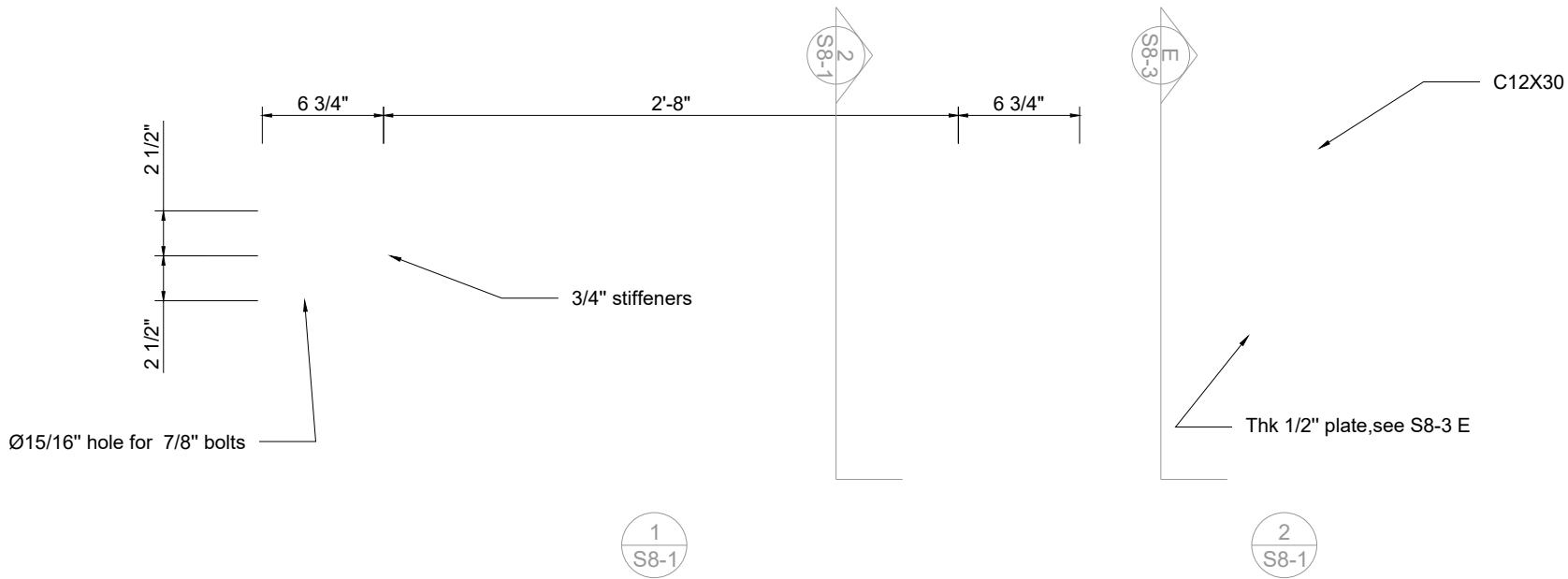
B

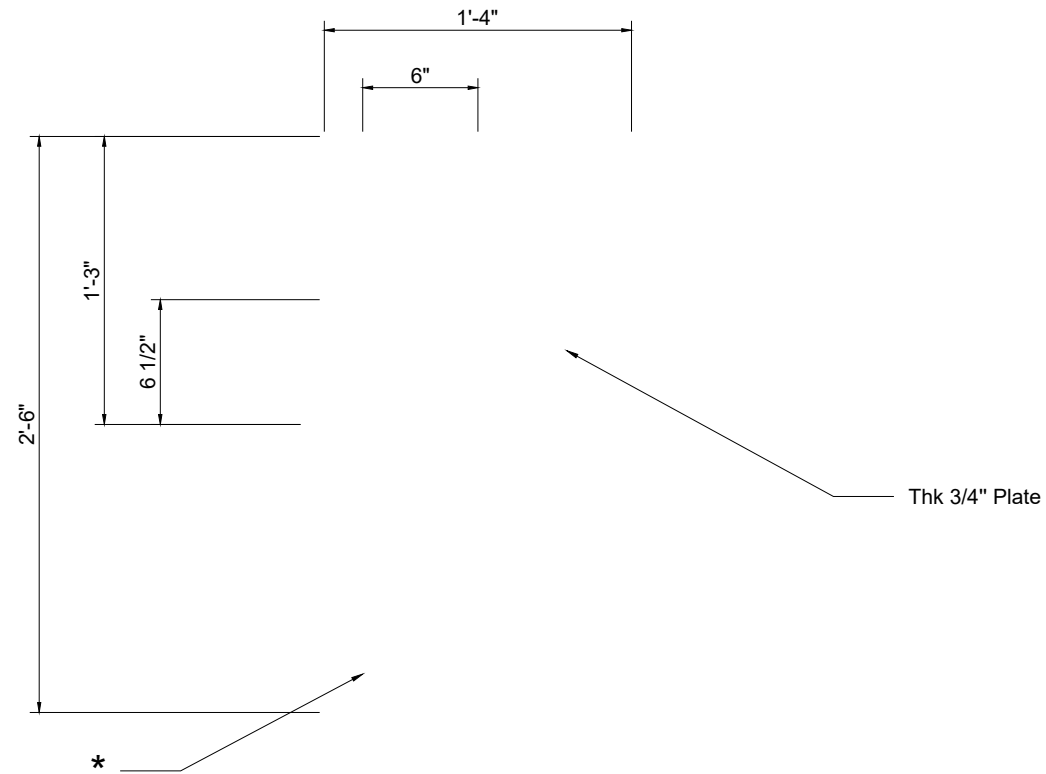
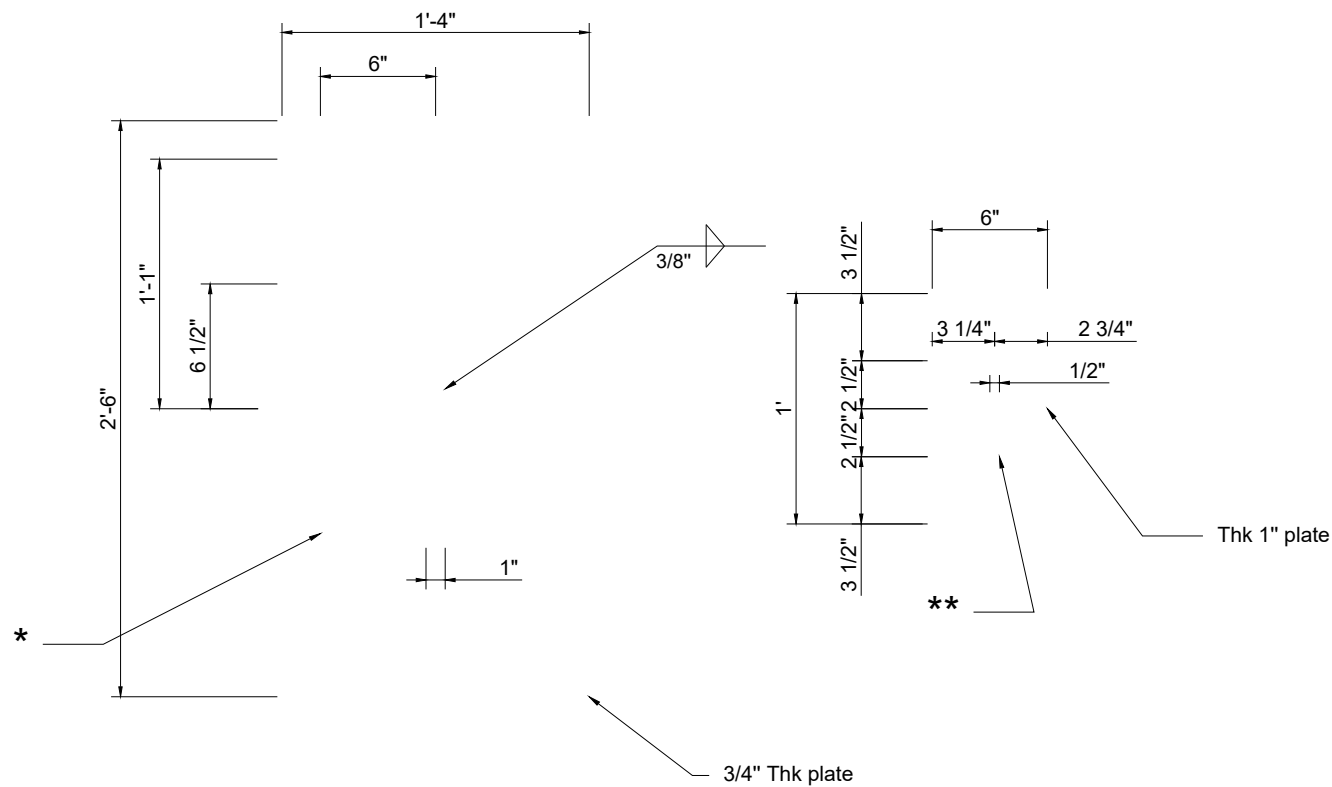


* Ø15/16" hole for 7/8" bolts
** Ø13/16" hole for #6 rebar



Type I
Qty: 1

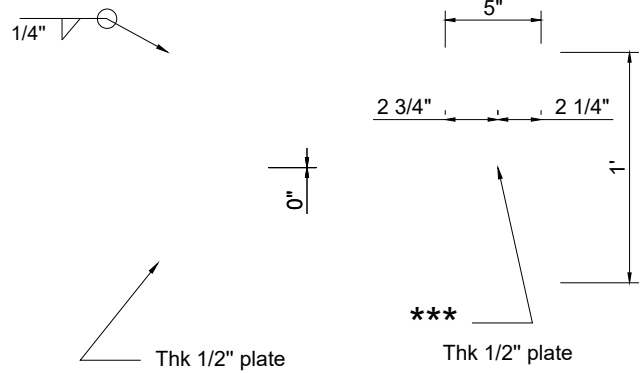
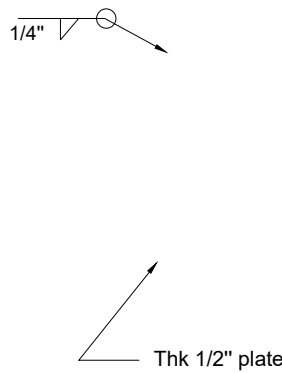




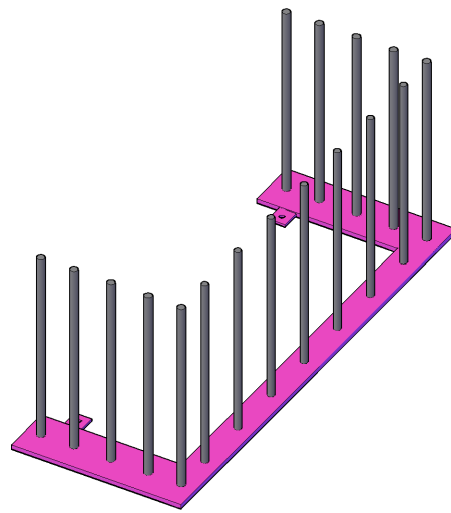
****Notes:**
If possible, use A325M bolts instead of rods

* Ø15/16" hole for 7/8" threaded rods

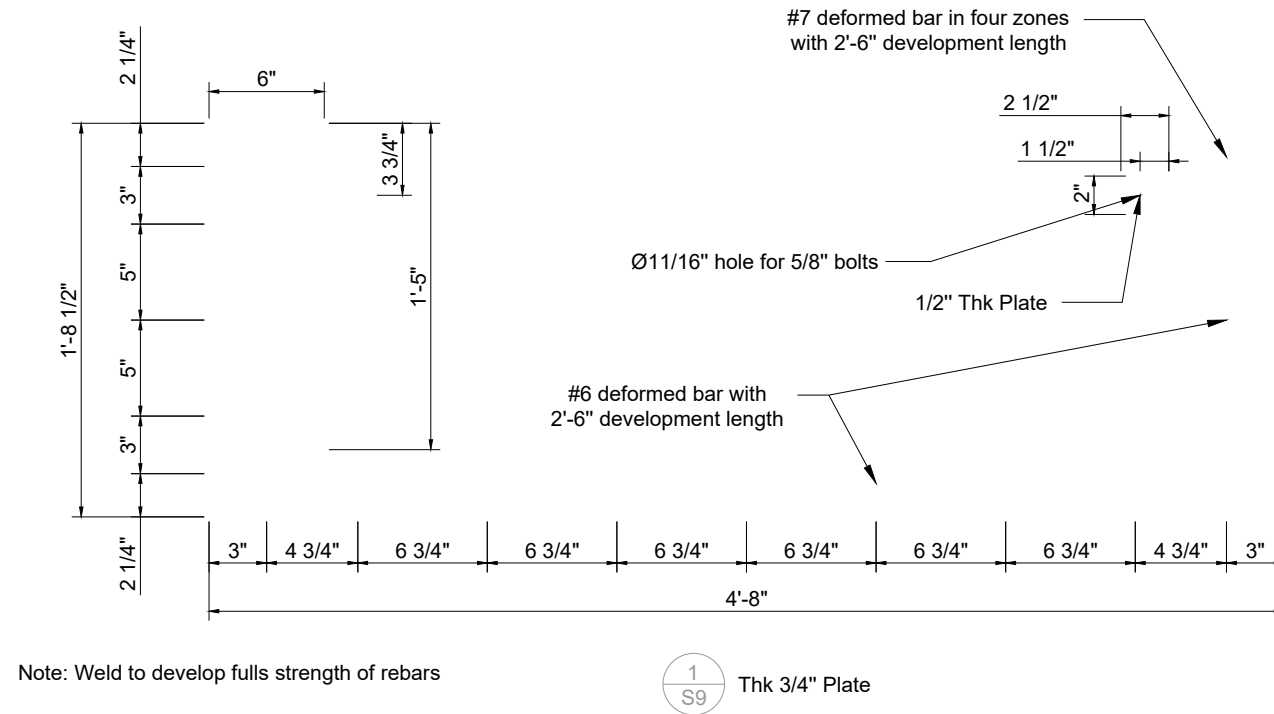
** Slot with Ø1/16" clearance for 7/8" bolt



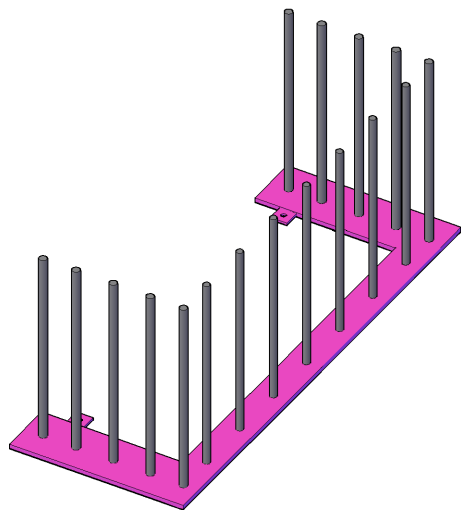
*** Ø15/16" hole for 7/8" bolt



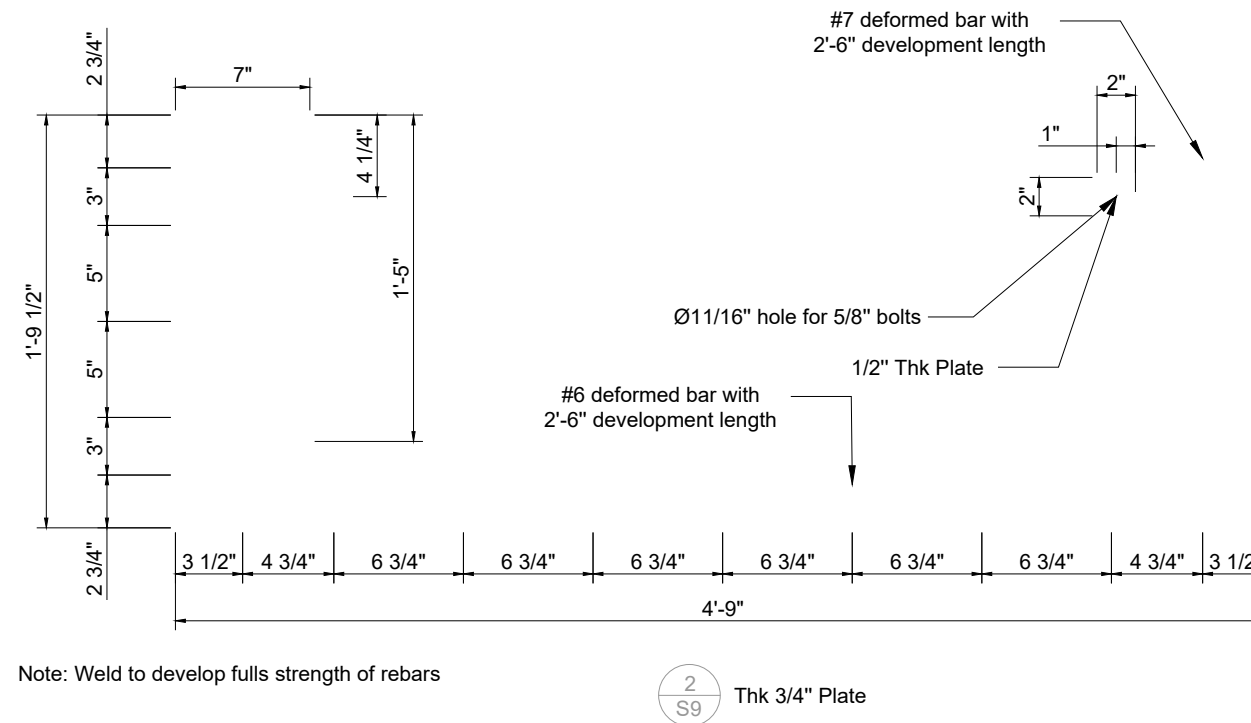
Wall to wall connection plate - Type I
3/4" Thk plate
Qty: 2
Unit Weight: 0.38kN/0.083kips



Qty. 1



Wall to wall connection plate - Type II
3/4" Thk plate
Qty: 2
Unit Weight: 0.44kN/0.098kips



Qty. 1

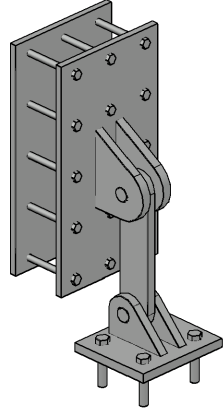


**** Notes:*
If possible, use A325M bolts instead of rods

* Ø11/16" hole for 5/8" threaded rods

** Ø1 1/16" holes for 1" all-threaded rods

*** Drill Hole for Ø 1 1/2" pin with tolerance less than 1/32", the less the better



UNDER REVISION



SHEET NO.

S12

DRAWN BY

Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE	2023/10/08

8 #3

#2 @ 1 3/4 " V.E.F
#2 @ 3 1/2 " H.E.F

#2 @ 4 1/2 " V.E.F
#2 @ 4 1/2 " H.E.F

8 #3



ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Wall to Wall Connection

SHEET NO.

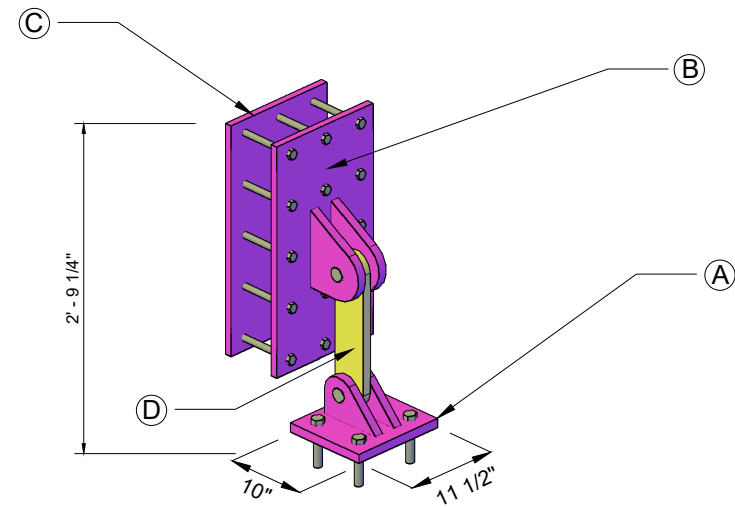
S9

DRAWN BY

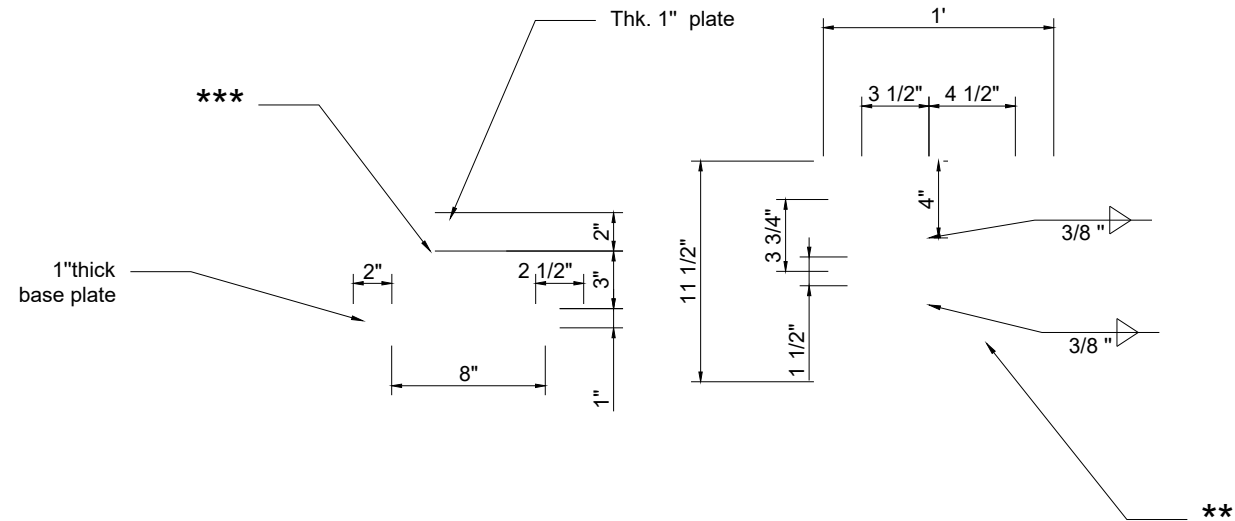
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE

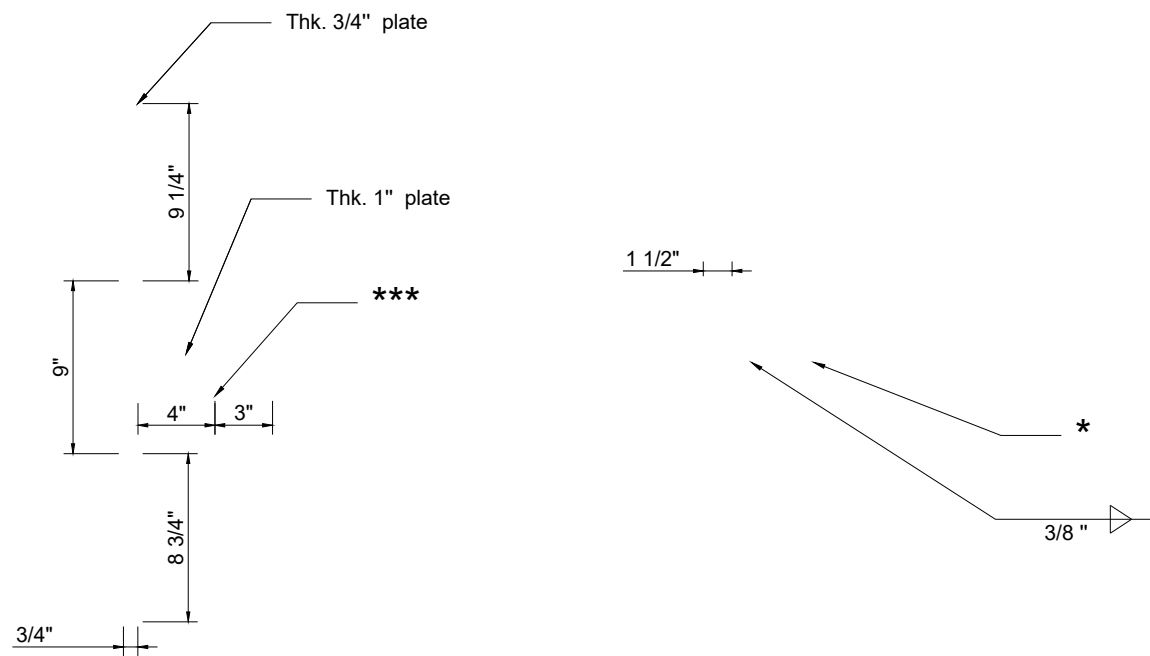
2023/10/08



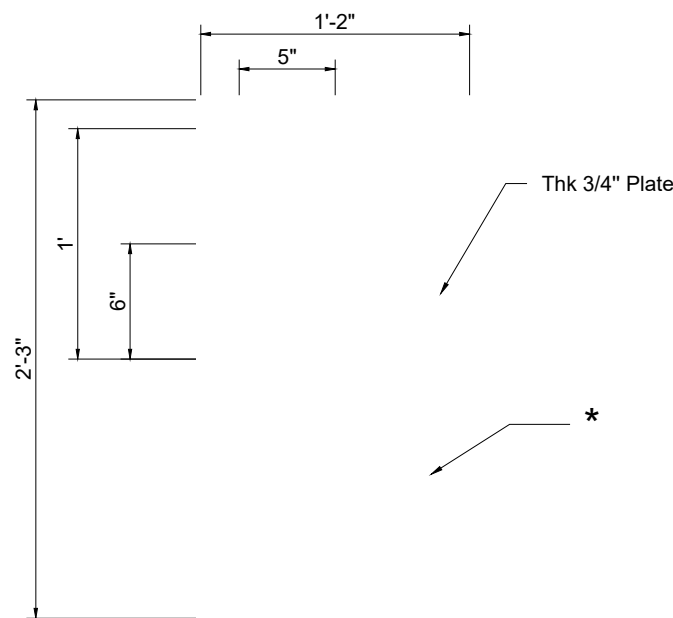
Qty: 4



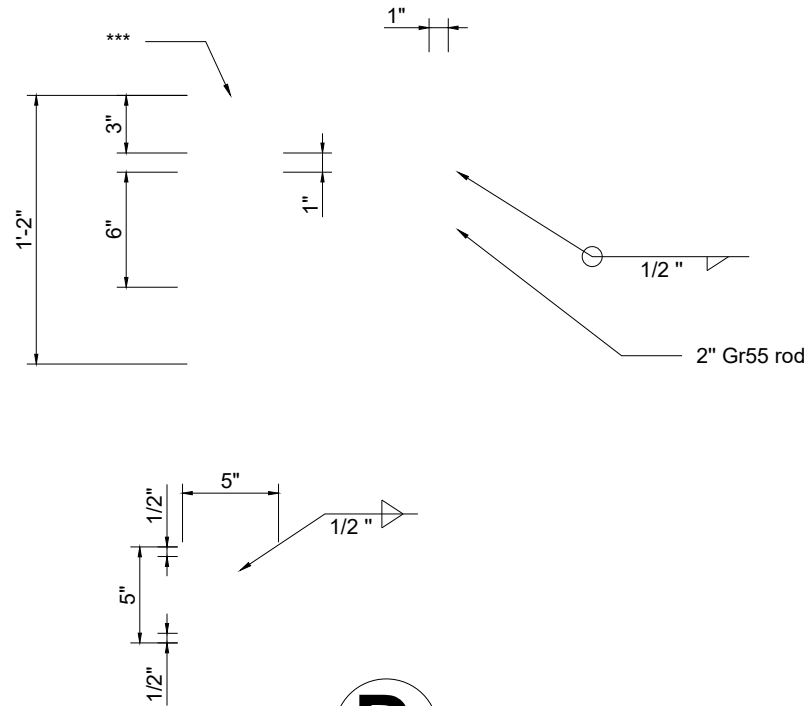
A



B



C



D

*** Notes:
If possible, use A325M bolts instead of rods
* Ø11/16" hole for 5/8" threaded rods
** Ø1 1/16" holes for 1" all-threaded rods
*** Drill Hole for Ø 1 1/2" pin with tolerance less than 1/32", the less the better

ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Base Fuse Connection

SHEET NO.

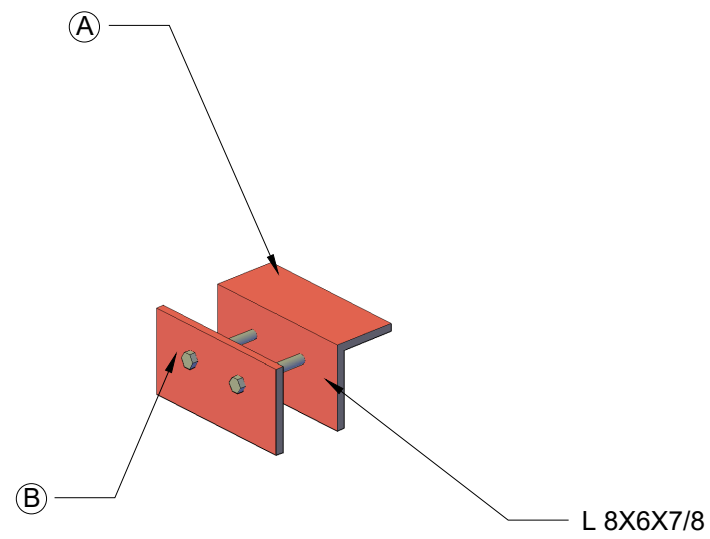
S10-R1

DRAWN BY

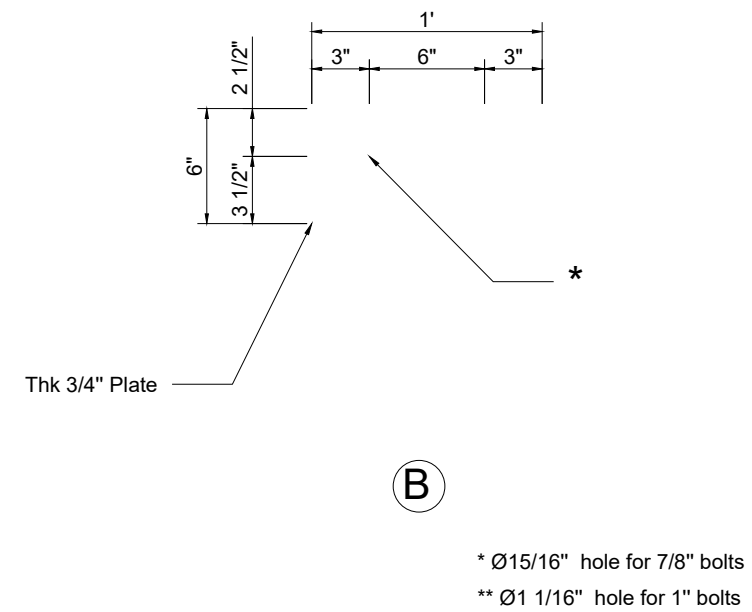
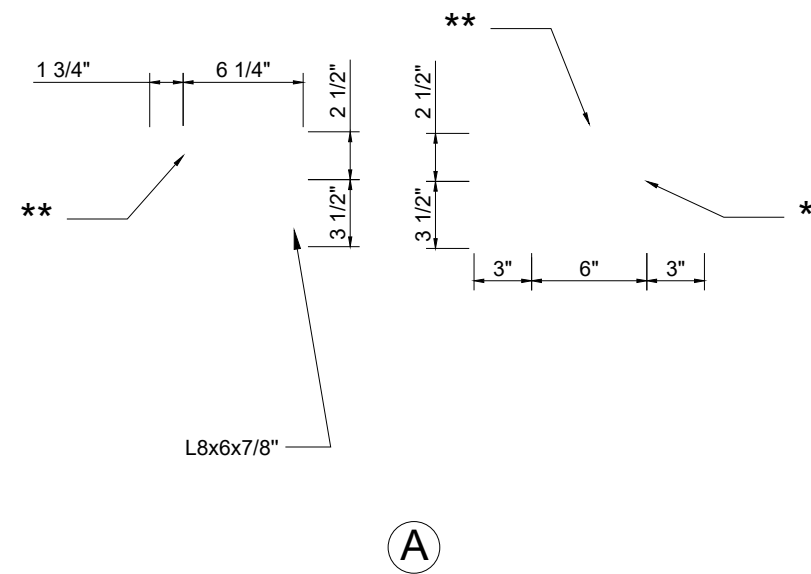
Tianyang Qiao
Hengchao Xu
Lisa Tobber

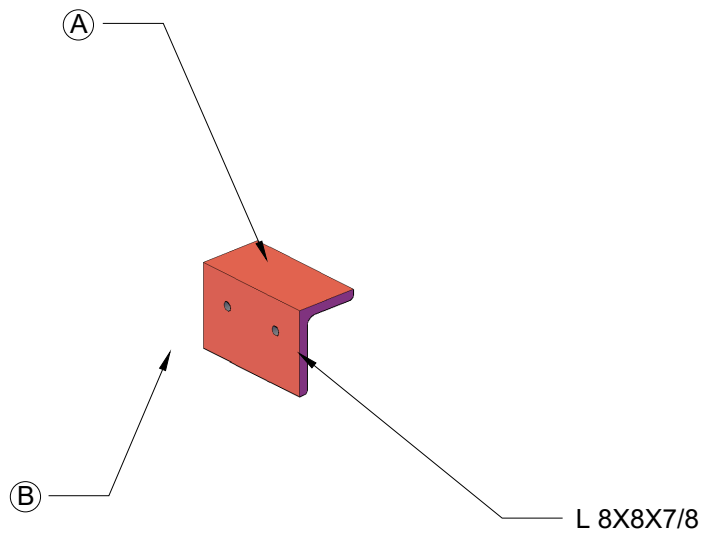
DATE

2018/11/07

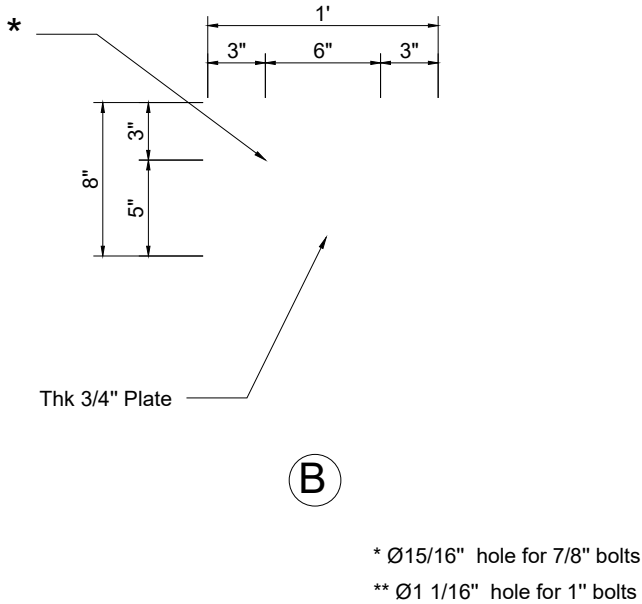
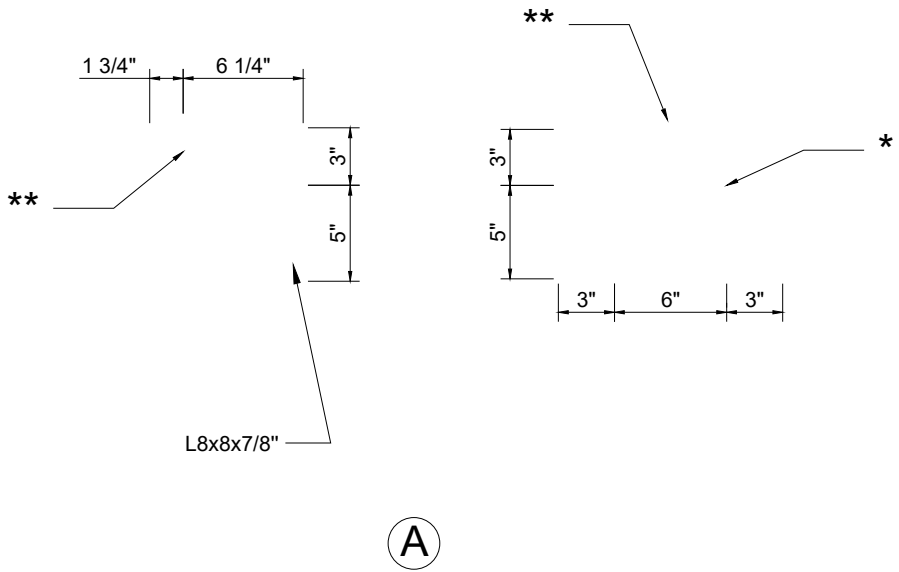


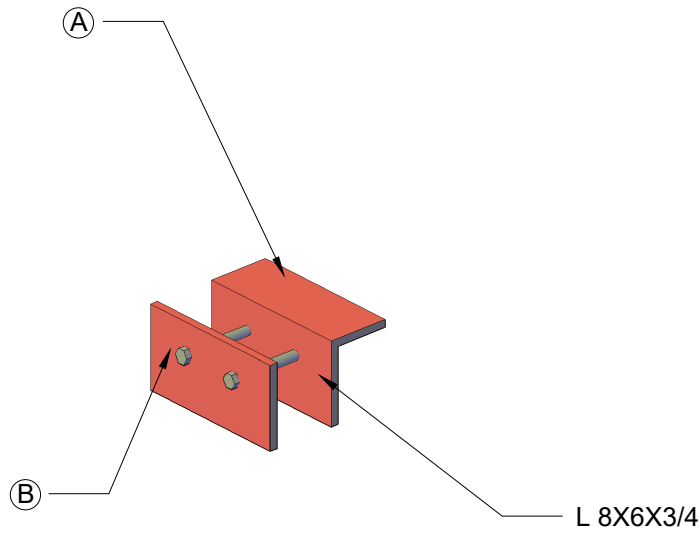
Qyt: 32



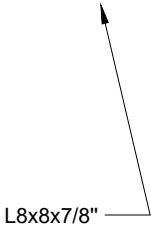


Qty: 32

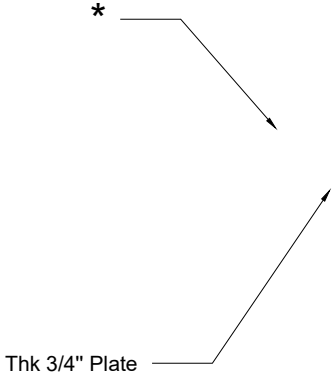
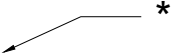




Qty: 32



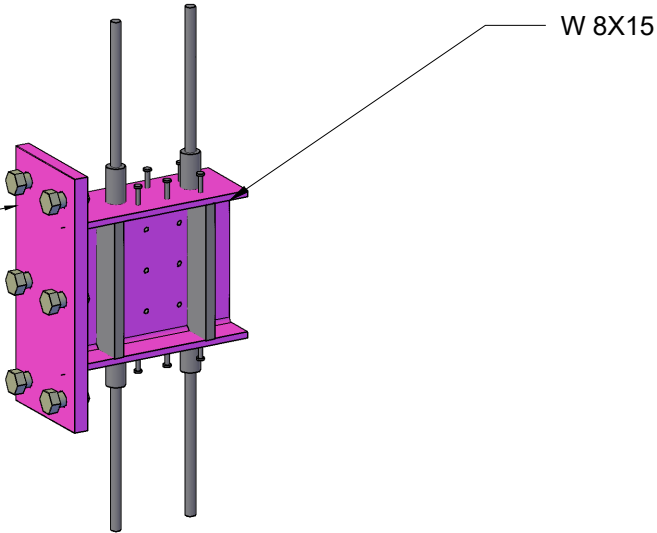
①



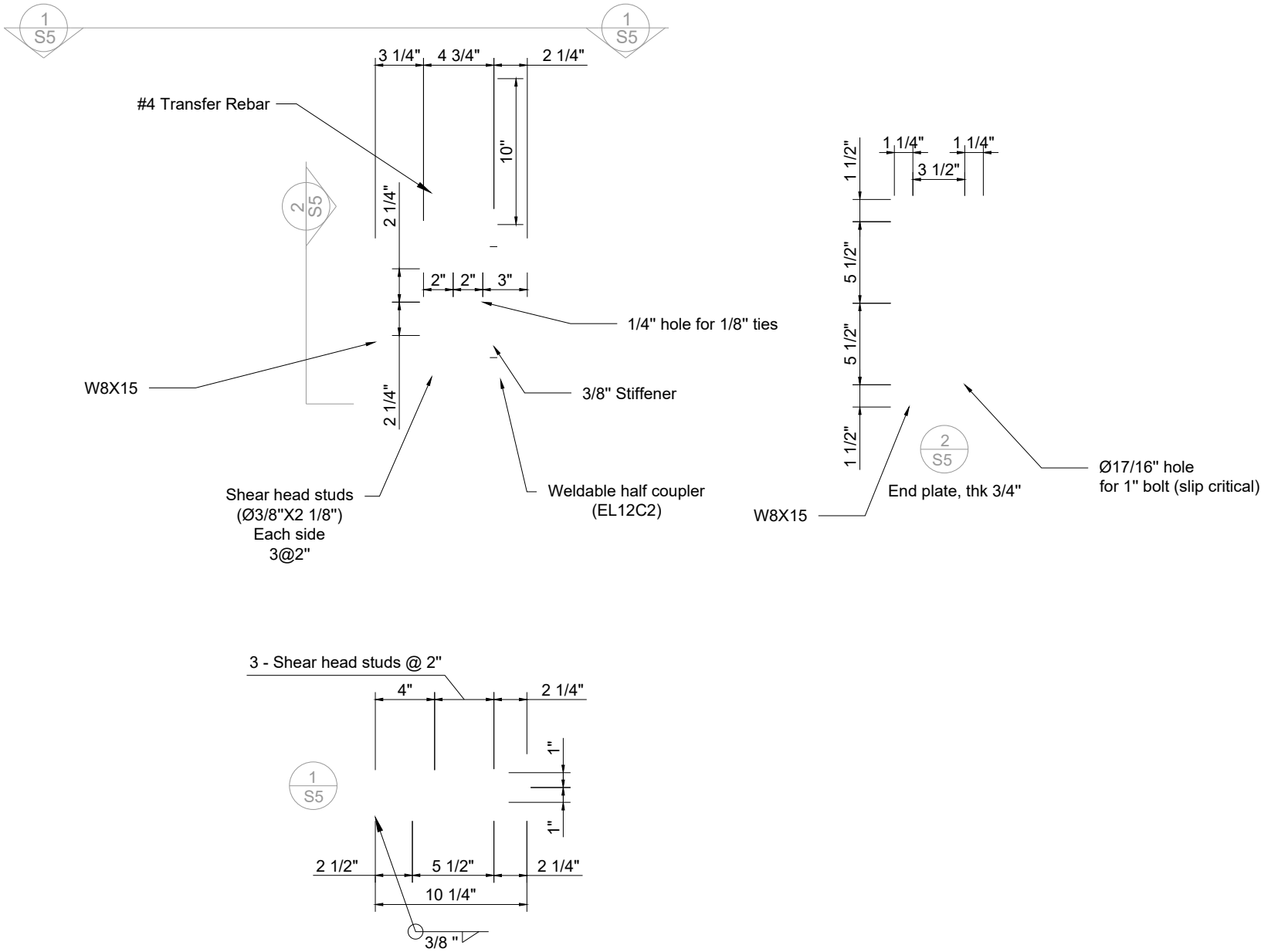
* Ø15/16" hole for 7/8" bolts

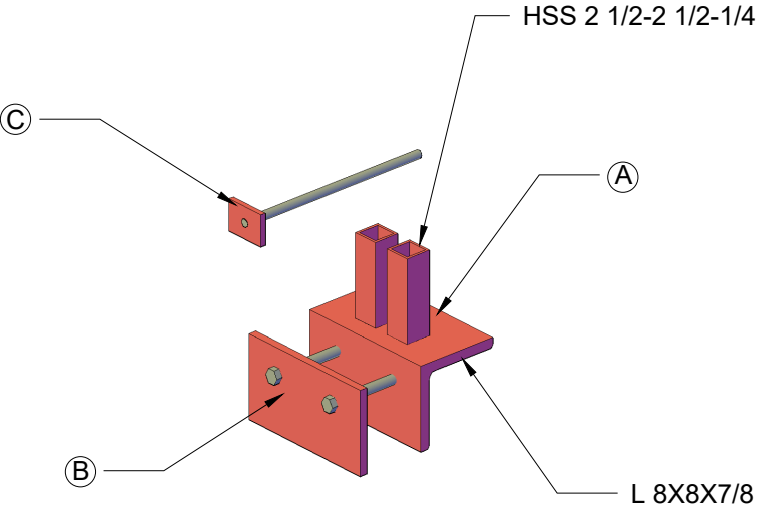
Notes for GTS:

Based on the conversation with Rob, we need steel components to connect the beam to the form work. Please add as you see fit.

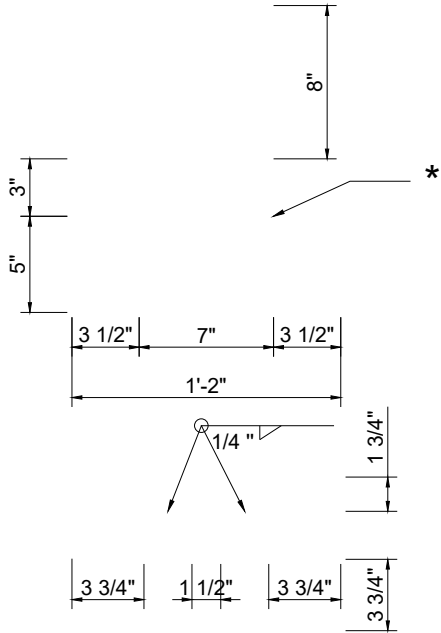


Qyt: 32

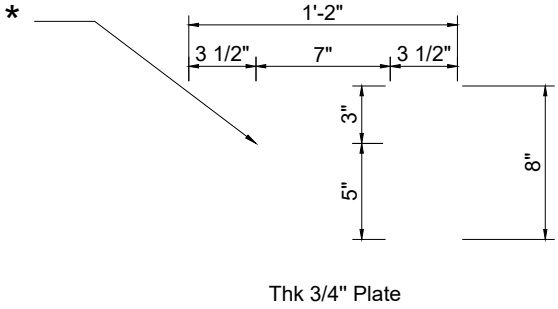




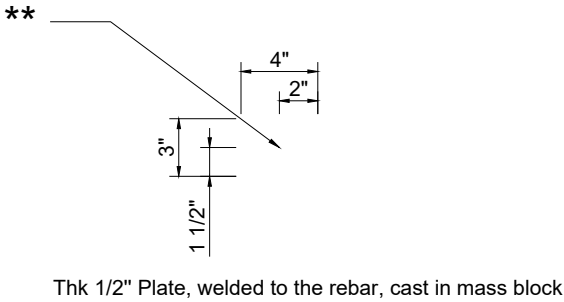
Qty: 32



A



B



C

* Ø15/16" hole for 7/8" bolts
** #6 rebar welded on, drill a hole if necessary

ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

**Mass Block Support
& Rebar End Zone Plate**

SHEET NO.

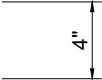
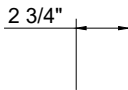
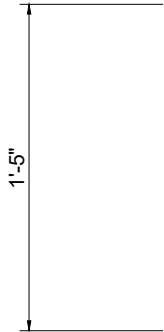
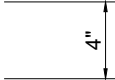
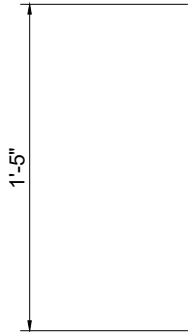
S7-R3

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Lisa Tobber

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Mass Block Support
& Rebar End Zone Plate

SHEET NO.

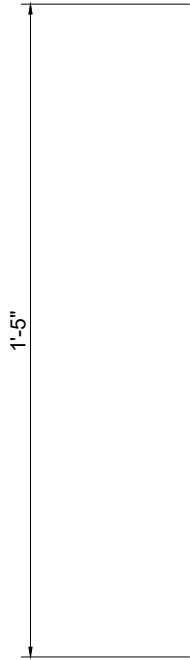
S7

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Lisa Tobber

SHEET NO.

S7

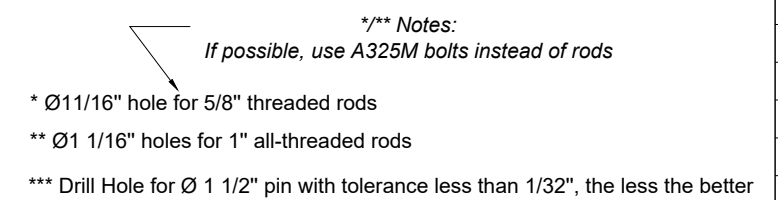
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

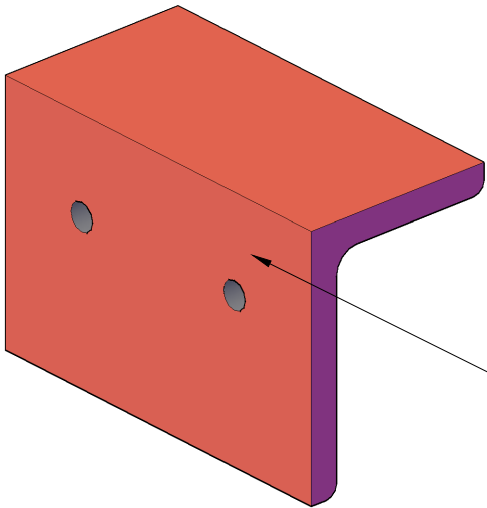
Mass Block Support
& Rebar End Zone Plate



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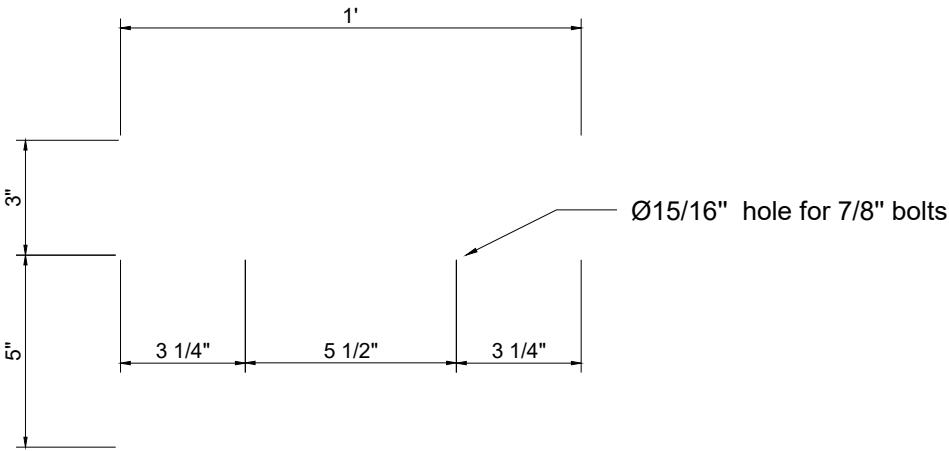
Smart Structures





Qty: 32

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Mass Block Support

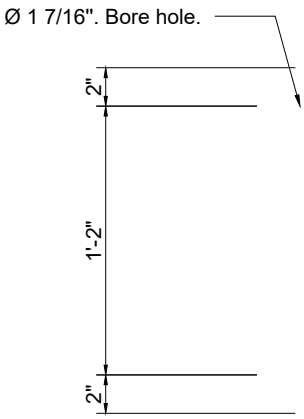
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M109R1

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Qyt: 8

M121



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Smart Structures

ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Base Fuse Connection

SHEET NO.

M121R1

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